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Type A: Manual or drip lubrication
Type B: Bath or disc lubrication
Type C: Oil stream lubrication

TABLE 7-16 Horsepower Ratings—Single-Strand Roller Chain No. 80

No. of teeth	1,000-in pitch							Rotational speed of small sprocket, rev/min																		
	10	25	50	75	88	100	200	300	400	500	600	700	800	900	1000	1200	1400	1600	1800	2000	2500	3000	3500	4000	4500	
11	0.44	1.06	2.07	3.05	3.56	4.03	7.83	11.56	15.23	18.87	22.48	26.07	27.41	22.97	19.61	14.92	11.84	9.69	8.12	6.83	4.96	3.77	3.00	2.45	0.00	
12	0.48	1.16	2.26	3.33	3.88	4.39	8.54	12.61	16.82	20.59	24.53	28.44	31.23	26.17	22.35	17.00	13.49	11.04	9.25	7.90	5.65	4.30	3.41	2.79	0.00	
13	0.52	1.26	2.45	3.61	4.21	4.76	9.26	13.66	18.00	22.31	26.57	30.81	35.02	29.51	25.20	19.17	15.21	12.45	10.43	8.91	6.37	4.85	3.85	3.15		
14	0.56	1.35	2.63	3.89	4.53	5.12	9.97	14.71	19.39	24.02	28.62	33.18	37.72	32.98	28.16	21.42	17.00	13.91	11.66	9.96	7.12	5.42	4.30	3.52		
15	0.60	1.45	2.82	4.16	4.86	5.49	10.68	15.76	20.77	25.74	30.66	35.55	40.41	36.58	31.23	23.76	18.85	15.43	12.93	11.04	7.90	6.01	4.77	0.00		
16	0.64	1.55	3.01	4.44	5.18	5.86	11.39	16.81	22.16	27.45	32.70	37.92	43.11	40.30	34.41	26.17	20.77	17.00	14.25	12.16	8.70	6.62	5.25	0.00		
17	0.68	1.64	3.20	4.72	5.50	6.22	12.10	17.86	23.54	29.17	34.75	40.29	45.80	44.13	37.68	28.66	22.75	18.62	15.60	13.32	9.53	7.25	0.00			
18	0.72	1.74	3.39	5.00	5.83	6.59	12.81	18.91	24.93	30.88	36.79	42.66	48.49	48.08	41.05	31.23	24.78	20.29	17.00	14.51	10.39	7.90	0.00			
19	0.76	1.84	3.57	5.28	6.15	6.95	13.53	19.96	26.31	32.60	38.84	45.03	51.19	52.15	44.52	33.87	26.88	22.00	18.44	15.74	11.26	0.36	0.00			
20	0.80	1.93	3.76	5.55	6.47	7.32	14.24	21.01	27.70	34.32	40.88	47.40	53.88	56.32	48.08	36.58	29.03	23.76	19.91	17.00	12.16	0.00				
21	0.84	2.03	3.95	5.83	6.80	7.69	14.95	22.07	29.08	36.03	42.92	49.77	56.58	60.59	51.73	39.36	31.23	25.56	21.42	18.29	13.09	0.00				
22	0.88	2.13	4.14	6.11	7.12	8.05	15.66	23.12	30.47	37.75	44.97	52.14	59.27	64.97	55.47	42.20	33.49	27.41	22.97	19.61	14.03					
23	0.92	2.22	4.33	6.39	7.45	8.42	16.37	24.17	31.85	39.46	47.01	54.51	61.97	69.38	59.30	45.11	35.80	29.30	24.55	20.97	15.00					
24	0.96	2.32	4.52	6.66	7.77	8.78	17.09	25.22	33.24	41.18	49.06	56.88	64.66	72.40	63.21	48.08	38.16	31.23	26.17	22.35	15.99					
25	1.00	2.42	4.70	6.94	8.09	9.15	17.80	26.27	34.62	42.89	51.10	59.25	67.35	75.42	67.20	51.12	40.57	33.20	27.83	23.76	8.16					
26	1.04	2.51	4.89	7.22	8.42	9.52	18.51	27.32	36.01	44.61	53.14	61.62	70.05	78.43	71.27	54.22	43.02	36.22	29.51	25.20	0.00					
28	1.12	2.71	5.27	7.77	9.06	10.25	19.93	29.42	38.78	48.04	57.23	66.36	75.44	84.47	79.65	60.59	48.08	39.36	32.98	28.16	0.00					
30	1.20	2.90	5.64	8.33	9.71	10.98	21.36	31.52	41.55	51.47	61.32	71.10	80.82	90.50	88.33	67.20	53.33	43.65	36.58	31.23						
32	1.28	3.09	6.02	8.89	10.36	11.71	22.78	33.62	44.32	54.91	65.41	75.84	86.21	96.53	97.31	74.03	58.75	48.08	40.30	5.65						
35	1.40	3.38	6.58	9.72	11.33	12.81	24.92	36.78	48.47	60.05	71.54	82.95	94.29	105.58	111.31	84.68	67.20	55.00	28.15	0.00						
40	1.61	3.87	7.53	11.11	12.95	14.64	28.48	42.03	55.40	68.63	81.76	94.80	107.77			120.67	133.51	103.46	82.10	40.16	0.00					
45	1.81	4.35	8.47	12.49	14.57	16.47	32.04	47.28	62.32	77.21	91.98	106.65	121.24			135.75	150.20	123.45	72.28	0.00						
	Type A				Type B																	Type C				

Type A: Manual or drip lubrication

Type B: Bath or disc lubrication

Type C: Oil stream lubrication

TABLE 7-17 Service Factors for Chain Drives

Load type	Type of driver		
	Hydraulic drive	Electric motor or turbine	Internal combustion engine with mechanical drive
Smooth Agitators; fans; generators; grinders; centrifugal pumps; rotary screens; light, uniformly loaded conveyors	1.0	1.0	1.2
Moderate shock Bucket elevators; machine tools; cranes; heavy conveyors; food mixers and grinders; ball mills; reciprocating pumps; woodworking machinery	1.2	1.3	1.4
Heavy shock Punch presses; hammer mills; boat propellers; crushers; reciprocating conveyors; rolling mills; logging hoists; dredges; printing presses	1.4	1.5	1.7

Lubrication

It is essential that adequate lubrication be provided for chain drives. There are numerous moving parts within the chain, along with the interaction between the chain and the sprocket teeth. The designer must define the lubricant properties and the method of lubrication.

Lubricant Properties. Petroleum-based lubricating oil similar to engine oil is recommended. Its viscosity must enable the oil to flow readily between chain surfaces that move relative to each other while providing adequate lubrication action. The oil should be kept clean and free of moisture. Table 7-18 gives the recommended lubricants for different ambient temperatures.

Method of Lubrication. The American Chain Association recommends three different types of lubrication depending on the speed of operation and the power being transmitted. See Tables 7-14 to 7-16 or manufacturers' catalogs for recommendations. Refer to the following descriptions of the methods and the illustrations in Figure 7-36.

Type A. Manual or drip lubrication: For manual lubrication, oil is applied copiously with a brush or a spout can, at least once every 8 hours of operation. For drip

TABLE 7-18 Recommended Lubricant for Chain Drives

Ambient temperature		Recommended lubricant
°F	°C	
20 to 40	−7 to 5	SAE 20
40 to 100	5 to 38	SAE 30
100 to 120	38 to 49	SAE 40
120 to 140	49 to 60	SAE 50

feed lubrication, oil is fed directly onto the link plates of each chain strand.

Type B. Bath or disc lubrication: The chain cover provides a sump of oil into which the chain dips continuously. Alternatively, a disc or a slinger can be attached to one of the shafts to lift oil to a trough above the lower strand of chain. The trough then delivers a stream of oil to the chain. The chain itself, then, does not need to dip into the oil.

Type C. Oil stream lubrication: An oil pump delivers a continuous stream of oil on the lower part of the chain.

Example Problem 7-5

Figure 7-37 shows a drive for a heavily loaded conveyor for use in the fields of a large commercial produce farm to take heavy containers of potatoes from the field onto trucks that will transport them to the processing plant. The conveyor is to be driven by a gasoline engine delivering 15.0 hp at a speed of 900 rpm. The conveyor pulley speed is to be 230 to 240 rpm. Design the chain drive.

Solution

Objective Design the chain drive.

Given Power transmitted = 15 hp to a heavily loaded produce conveyor
Speed of motor = 900 rpm; output speed range = 230 to 240 rpm

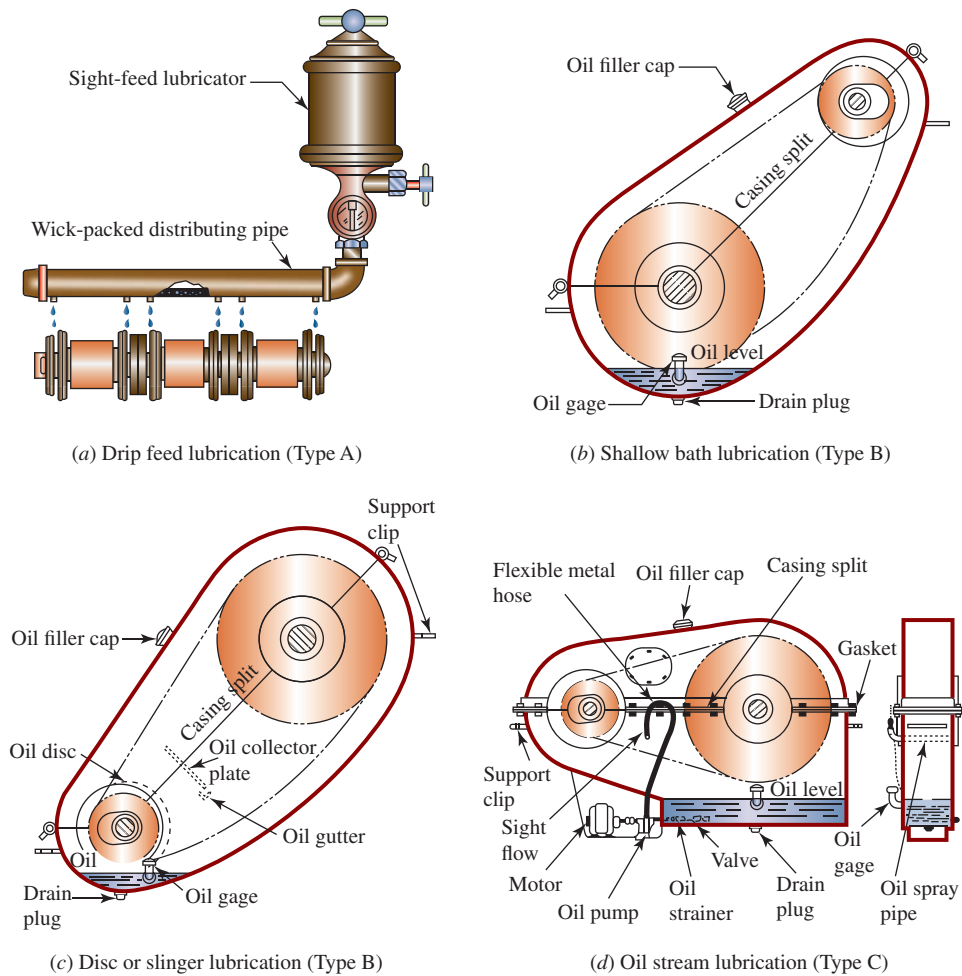


FIGURE 7-36 Lubrication methods

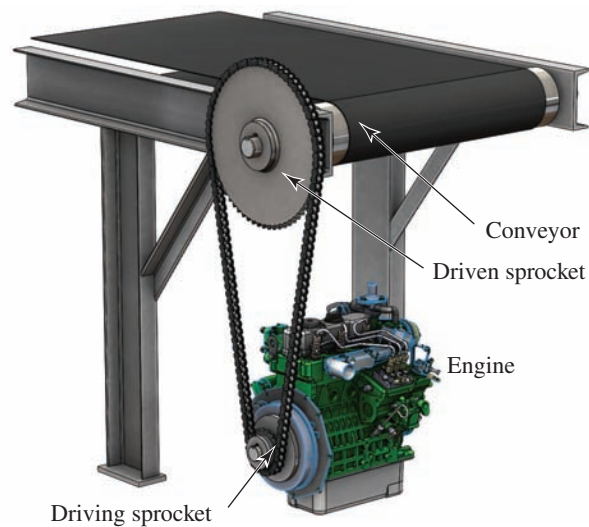


FIGURE 7-37 Chain drive for a heavy-duty conveyor for Example Problem 7-5

Analysis Use the design data presented in this section. The solution procedure is developed within the Results section of the problem solution.

Results **Step 1.** Specify a service factor and compute the design power. From Table 7–17, for moderate shock from the loads applied to the heavy-duty conveyor and a gasoline engine drive, use a service factor of $SF = 1.4$. Then,

$$\text{Design power} = P_{des} = SF(P) = 1.4(15.0 \text{ hp}) = 21.0 \text{ hp}$$

Step 2. Compute the desired ratio. Using the middle of the required range of output speeds, we have

$$\text{Ratio} = (900 \text{ rpm})/(235 \text{ rpm}) = 3.83$$

Step 3. Refer to the tables for power capacity (Tables 7–14 to 7–16), and select the chain pitch. For a single strand, the no. 60 chain with $p = 3/4$ in seems best. A 17-tooth sprocket is rated at 21.96 hp at 900 rpm by interpolation. At this speed, Type B lubrication (oil bath) is required.

Step 4. Compute the required number of teeth on the large sprocket:

$$N_2 = N_1 \times \text{ratio} = 17(3.83) = 65.11$$

Let's use the integer: 65 teeth.

Step 5. Compute the actual expected output speed:

$$n_2 = n_1(N_1/N_2) = 900 \text{ rpm}(17/65) = 235.3 \text{ rpm (Okay! The speed is within the specified range.)}$$

Step 6. Compute the pitch diameters of the sprockets using Equation (7–20):

$$PD_1 = \frac{p}{\sin(180^\circ/N_1)} = \frac{0.75 \text{ in}}{\sin(180^\circ/17)} = 4.082 \text{ in}$$

$$PD_2 = \frac{p}{\sin(180^\circ/N_2)} = \frac{0.75 \text{ in}}{\sin(180^\circ/65)} = 15.524 \text{ in}$$

Step 7. Specify the nominal center distance. Let's use the middle of the recommended range, 40 pitches.

Step 8. Compute the required chain length in pitches from Equation (7–18):

$$L_c = 2CD + \frac{N_2 + N_1}{2} + \frac{(N_2 - N_1)^2}{4\pi^2 CD}$$

$$L_c = 2(40) + \frac{65 + 17}{2} + \frac{(65 - 17)^2}{4\pi^2(40)} = 122.5 \text{ pitches}$$

Step 9. Specify an integral number of pitches for the chain length, and compute the actual theoretical center distance. Let's use 122 pitches, an even number. Then, from Equation (7–19),

$$CD = \frac{1}{4} \left[L_c - \frac{N_2 + N_1}{2} + \sqrt{\left[L_c - \frac{N_2 + N_1}{2} \right]^2 - \frac{8(N_2 - N_1)^2}{4\pi^2}} \right]$$

$$CD = \frac{1}{4} \left[122 - \frac{65 + 17}{2} + \sqrt{\left[122 - \frac{65 + 17}{2} \right]^2 - \frac{8(65 - 17)^2}{4\pi^2}} \right]$$

$$CD = 39.766 \text{ pitches} = 39.766(0.75 \text{ in}) = 29.825 \text{ in}$$

Step 10. Compute the angle of wrap of the chain for each sprocket using Equations (7–21) and (7–22). Note that the minimum angle of wrap should be 120° .

For the small sprocket,

$$\theta_1 = 180^\circ - 2\sin^{-1}[(PD_2 - PD_1)/2CD]$$

$$\theta_1 = 180^\circ - 2\sin^{-1}[(15.524 - 4.082)/(2(29.825))] = 158^\circ$$

Because this is greater than 120° , it is acceptable.
For the larger sprocket,

$$\theta_2 = 180^\circ + 2\sin^{-1}[(PD_2 - PD_1)/2CD]$$

$$\theta_2 = 180^\circ + 2\sin^{-1}[(15.524 - 4.082)/(2(29.825))] = 202^\circ$$

Comments **Summary of Design**

Figure 7–38(a) shows a sketch of the design to scale.

Pitch: No. 60 chain, 3/4-in pitch

Length: 122 pitches = $122(0.75) = 91.50$ in

Center distance: $CD = 29.825$ in (maximum)

Sprockets: Single-strand, no. 60, 3/4-in pitch

Small: 17 teeth, $PD = 4.082$ in

Large: 65 teeth, $PD = 15.524$ in

Type B lubrication is required. The large sprocket can dip into an oil bath.

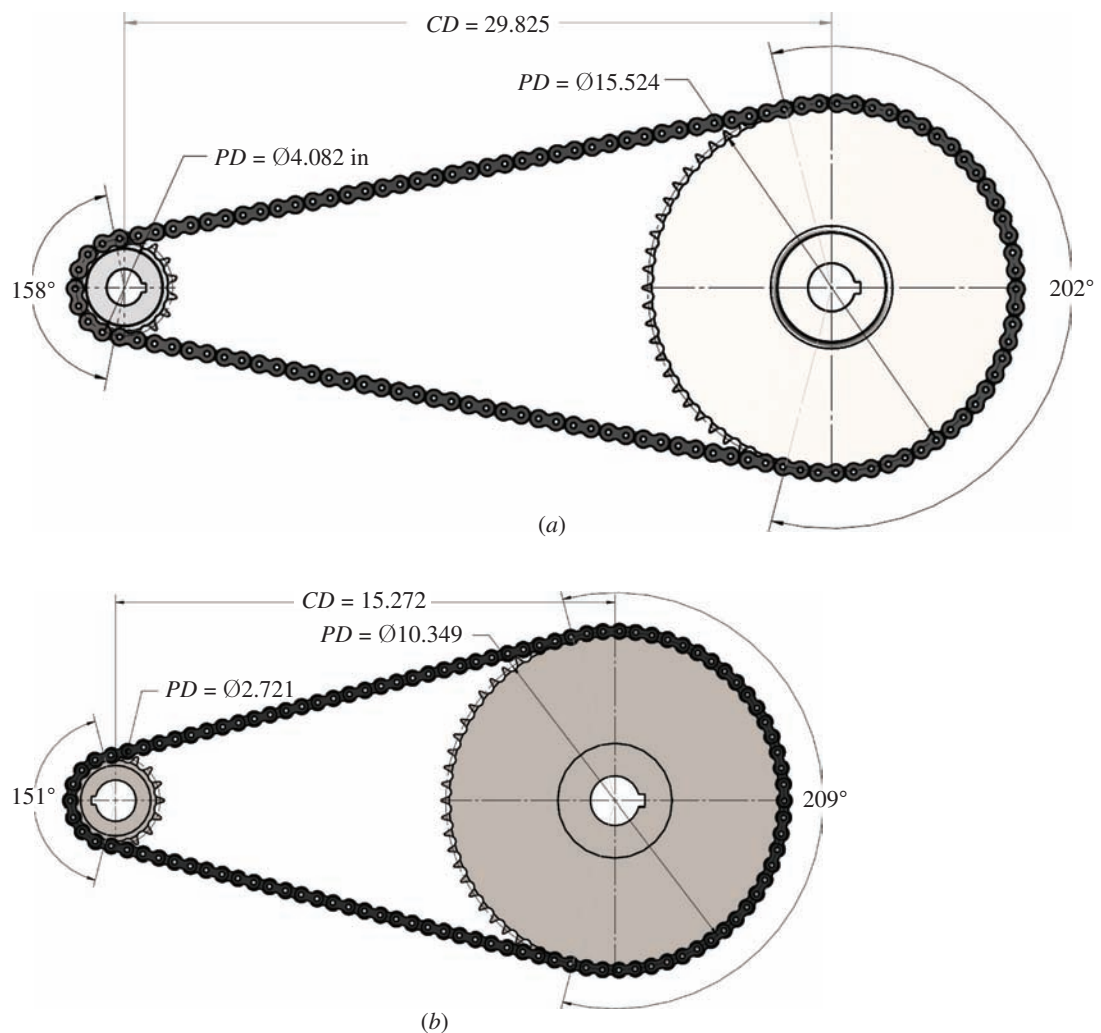


FIGURE 7–38 Scale drawings of layouts for chain drives for Example Problems 7–5 and 7–6

Example Problem 7-6

Create an alternate design for the conditions of Example Problem 7-5 to produce a smaller drive.

Solution

Objective Design a smaller chain drive for the application in Example Problem 7-5.

Given Power transmitted = 15 hp to a conveyor
Speed of motor = 900 rpm; output speed range = 230 to 240 rpm

Analysis Use a multistrand design to permit a smaller-pitch chain to be used to transmit the same design power (21.0 hp) at the same speed (900 rpm). Use the design data presented in this section. The solution procedure is developed within the Results section of the problem solution.

Results Let's try a four-strand chain for which the power capacity factor is 3.3. Then the required power per strand is

$$P = 21.0/3.3 = 6.36 \text{ hp}$$

From Table 7-14, we find that a no. 40 chain (1/2-in pitch) with a 17-tooth sprocket will be satisfactory. Type B lubrication, oil bath, can be used.

For the required large sprocket,

$$N_2 = N_1 \times \text{ratio} = 17(3.83) = 65.11$$

Let's use $N_2 = 65$ teeth.

The sprocket diameters are

$$PD_1 = \frac{p}{\sin(180^\circ/N_1)} = \frac{0.500 \text{ in}}{\sin(180^\circ/17)} = 2.721 \text{ in}$$

$$PD_2 = \frac{p}{\sin(180^\circ/N_2)} = \frac{0.500 \text{ in}}{\sin(180^\circ/65)} = 10.349 \text{ in}$$

For the center distance, let's try the minimum recommended: $C = 30$ pitches.

$$30(0.50 \text{ in}) = 15.0 \text{ in}$$

The chain length is

$$L_c = 2(30) + \frac{65 + 17}{2} + \frac{(65 - 17)^2}{4\pi^2(30)} = 102.9 \text{ pitches}$$

Specify the integer length, $L_c = 104$ pitches = $104(0.50) = 52.0$ in. The actual maximum center distance is

$$CD = \frac{1}{4} \left[104 - \frac{65 + 17}{2} + \sqrt{\left[104 - \frac{65 + 17}{2} \right]^2 - \frac{8(65 - 17)^2}{4\pi^2}} \right]$$

$$CD = 30.54 \text{ pitches} = 30.54(0.50) = 15.272 \text{ in}$$

Compute the angle of wrap of the chain for each sprocket using Equations (7-21) and (7-22). Note that the minimum angle of wrap should be 120° .

For the small sprocket,

$$\theta_1 = 180^\circ - 2\sin^{-1}[(PD_2 - PD_1)/2C]$$

$$\theta_1 = 180^\circ - 2\sin^{-1}[(10.349 - 2.721)/(2(15.272))] = 151.1^\circ$$

Because this is greater than 120° , it is acceptable.

For the larger sprocket,

$$\theta_2 = 180^\circ + 2\sin^{-1}[(PD_2 - PD_1)/2C]$$

$$\theta_2 = 180^\circ + 2\sin^{-1}[(10.349 - 2.721)/(2(15.272))] = 208.9^\circ$$

Comments Summary

Figure 7-38(b) shows the new design to the same scale as the first design. The space reduction is significant.

Chain: No. 40, 1/2-in pitch, four-strand, 104 pitches, 52.0 in length

Sprockets: No. 40–4 (four strands), 1/2-in pitch

Small: 17 teeth, $PD_1 = 2.721$ in

Large: 65 teeth, $PD_2 = 10.349$ in

Maximum center distance: 15.272 in

Type B lubrication (oil bath)

Spreadsheet for Chain Design

Figure 7–39 shows a spreadsheet that assists in the design of chain drives using the procedure developed in this section.

The user enters data shown in italics in the gray-shaded cells. Refer to Tables 7–11 to 7–17 for required data. Results for Example Problem 7–6 are shown in the figure.

CHAIN DRIVE DESIGN						
Initial Input Data:		Example Problem 7–6—Multiple strands				
	Application:	<i>Coal conveyor</i>				
	Drive/type:	<i>Engine-mechanical drive</i>				
	Driven machine:	<i>Heavily loaded conveyor</i>				
	Power input:	15 hp				
	Service factor:	1.4	Table 7–17			
	Input speed:	900 rpm				
	Desired output speed:	235 rpm				
Computed Data:						
	Design power:	21 hp				
	Speed ratio:	3.83				
Design Decisions—Chain Type and Teeth Numbers:						
	Number of strands:	4	1	2	3	4
	Strand factor:	3.3	1.0	1.7	2.5	3.3
	Required power per strand:	6.36 hp				
	Chain number:	40	Table 7–14, 7–15, or 7–16			
	Chain pitch:	0.5 in				
	Number of teeth—driver sprocket:	17				
	Computed no. of teeth—driven sprocket:	65.11				
	Enter: Chosen number of teeth:	65				
Computed Data:						
	Actual output speed:	235.4 rpm				
	Pitch diameter—driver sprocket:	2.721 in				
	Pitch diameter—driven sprocket:	10.349 in				
Center Distance, Chain Length, and Angle of Wrap:						
	Enter: Nominal center distance:	30 pitches	30 to 50 pitches recommended			
	Computed nominal chain length:	102.9 pitches				
	Enter: Specified no. of pitches:	104 pitches	Even number recommended			
	Actual chain length:	52.00 in				
	Computed actual center distance:	30.545 pitches				
	Actual center distance:	15.272 in				
	Angle of wrap—driver sprocket:	151.1°	Should be greater than 120°			
	Angle of wrap—driven sprocket:	208.9°				

FIGURE 7–39 Spreadsheet for chain design

7-7 WIRE ROPE

Application of Wire Rope

Wire rope is used in many industries such as oil and gas, cranes and lifting, construction, mining, military, and shipping. Many types of machines and structures require wire rope for lifting, moving, and stabilizing bodies and it is used in both dynamic and static systems. Examples of dynamic systems entailing wire ropes are material handling, elevators, shovels, and heavy lifting equipment for mechanical power transmission shown in Figure 7-40. Static wire ropes are used to support towers and create suspension bridges.

Due to wire ropes having a variety of functions, there is a complete range of wire rope sizes, steel grades, finishes, and construction types available. A balance of wire rope properties such as strength, abrasion resistance, crush resistance, bending fatigue resistance, and corrosion resistance are used to optimize the selection of wire rope. Understanding the requirements of these applications and wire rope design will optimize wire rope selection leading to optimal performance. Wire ropes must be properly selected for each application while being periodically inspected and maintained during the life of its operation. This will ensure that the wire rope's service life is longer and safer.

Wire Rope Construction

The design of a typical wire rope is shown in Figure 7-41. It is made from three components: wires, strands, and the core. The various combinations of wire arrangement and size form a strand. These strands are helically laid over the center or core to form the wire rope and most wire ropes are constructed with 6, 7, or 8 strands. The center of the wire rope is called a core and is made from steel or fiber and its function is to provide support for the strands and keep the strands properly positioned as the rope flexes and bends under loaded conditions. Two commonly used core designations are a *fiber core* (FC) and an *independent wire rope core* (IWRC). Fiber cores are manufactured from natural or synthetic fibers and the independent wire rope core is made from steel.



FIGURE 7-40 Overhead crane

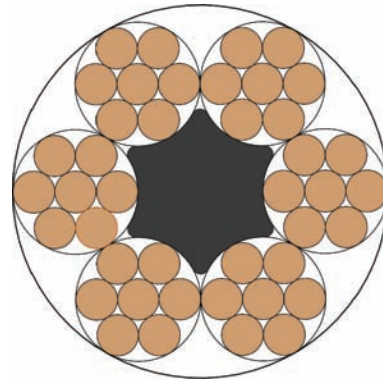


FIGURE 7-41 Wire rope construction

A fiber core provides excellent flexibility while the independent wire rope core provides crush resistance and increased strength. The wire, strands, and core components of the wire rope continuously interact with each other during operation.

The nominal wire rope diameters available are shown in Table 7-19. The wire rope diameter, d , is measured at the extreme outer limits of the strand as shown in Figure 7-42. See Internet sites 17-19 for wire rope producers.

Wire Rope Classification

Wire ropes are grouped into standard classifications based on the number of strands and the number of wires per strand, as shown in Table 7-20. A classification

TABLE 7-19 Nominal Wire Rope Diameter

inches	mm	inches	mm
1/4	6.5	2 1/8	54
5/16	8	2 1/4	58
3/8	9.5	2 3/8	60
7/16	11.5	2 1/2	64
1/2	13	2 5/8	67
9/16	14.5	2 3/4	71
5/8	16	2 7/8	74
3/4	19	3	77
7/8	22	3 1/8	80
1	26	3 1/4	83
1 1/8	29	3 3/8	87
1 1/4	32	3 1/2	90
1 3/8	35	3 3/4	96
1 1/2	38	4	103
1 5/8	42	4 1/4	109
1 3/4	45	4 1/2	115
1 7/8	48	4 3/4	122
2	52	5	128

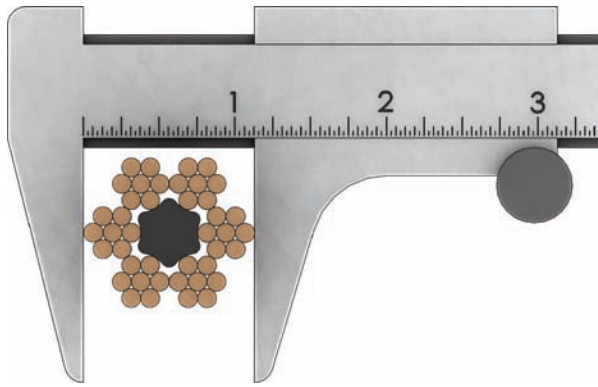


FIGURE 7-42 Correct method of measurement of a wire rope

TABLE 7-20 Wire Rope Classification

Classification	Wires per strand	Maximum number of outer wires in strand
6×7	7 through 15	9
6×19	16 through 26	12
6×36	27 through 49	18
6×61	50 through 74	24

*Classifications are the same in 7 and 8 strand wire ropes

of 6×19 has a wire count of 16 through 26 wires per strand. Wire ropes within a classification have the same strength, weight per foot, and cost.

Basic Strand Construction

Table 7-21 shows the various strands that are constructed from different combinations of wire arrangements and diameters. *Warrington*, *Seale*, and *Filler Wire* are basic strand constructions discussed in this table. The strand design affects the physical characteristics of the wire rope's resistance to abrasion and fatigue. A strand made up of a large number of smaller wires will be less abrasion resistant and more fatigue resistant compared to a strand made up of small number of large wires which will be more abrasion resistant and less fatigue resistant.

The direction of lay of individual wires within the strand, and the direction of lay of the strand in the wire rope are described next. Figure 7-43 shows the different combinations of how the wires and strands are wound within a wire rope. The terms *Right Lay* and *Left Lay* describe the direction the strands are wound in the wire rope. Figures 7-43(a) and (c) show the strands wound with Right Lay. When looking at the wire rope, the strands are angled to the right as shown. Figures 7-43(b) and (d) show the strands wound with Left Lay. This

TABLE 7-21 Strand Construction

Single layer		Single wire in the center with six wires of the same diameter
Seale		Equal number of wires in each layer All wires in layer are the same diameter Large outer wires rest in the valley between the small inner wires
Filler wire		Inner layer having half the number of wires as outer layer Smaller filler wires equal in number to the inner layer are laid in the valleys of the inner layer
Warrington		One diameter of wire in the inner layer Two diameters of wire alternating large and small in the outer layer The large outer layer wires rest in the valleys The smaller wires rest on the crowns of the inner layer
Combined pattern		Strand is formed using two or more of the above constructions

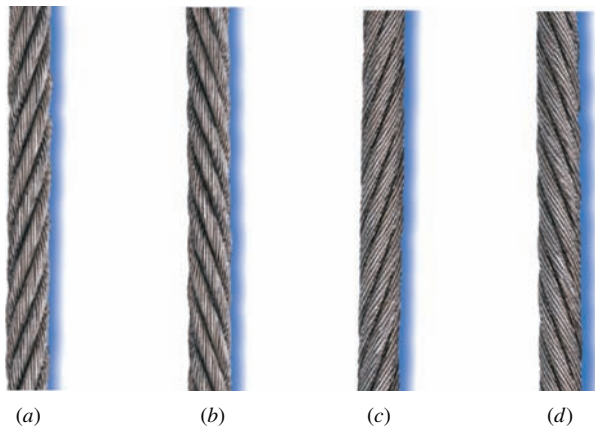


FIGURE 7-43 Types of lay of wire rope

(a) Right Lay – Regular Lay, (b) Left Lay – Regular Lay,
(c) Right Lay – Lang Lay, (d) Left Lay – Lang Lay

shows the strands angled to the left as they are wound around the center core of the wire rope.

The terms *Regular Lay* and *Lang Lay* describe the direction the individual wires are wound in the strand.

- In Regular Lay, the individual wires are wound (lay) in the opposite direction of the strand lay.
- In Lang Lay wire rope, the individual wires have the same lay direction as the strands.

Let's look at the wire rope in Figure 7-43(a) which has *Right Lay – Regular Lay*. The term Right Lay tells us the strand lay is angled to the right in the wire rope. The term Regular Lay tells us the individual wires lay in the opposite direction of the strands, so if the strand is Right Lay the individual wires will lay to the left as shown in Figure 7-44(a). As these strands are wound with Right Lay to form the wire rope, the individual wires in the strands appear to run parallel with the rope length. Figure 7-44(b) shows one strand with Right Lay – Regular Lay and Figure 7-43(a) shows the complete wire rope.

Next, let's look at a wire rope that has *Right Lay – Lang Lay* shown in Figure 7-43(c). The strands in this wire rope also have Right Lay, so the strands will be angled to the right as the previous wire rope. Since this wire rope is Lang Lay, the individual wires are wound in the same direction as the strand. Therefore the individual

wires will lay right since the strand is Right Lay as shown in Figure 7-45(a). The individual wires will appear to run perpendicular to the direction of the wire rope as the strands are wound with Right Lay. Figure 7-45(b) shows one strand wound with Right Lay and the complete wire rope is shown in Figure 7-43(c).

Lang Lay wire ropes are more flexible than Regular Lay wire ropes. As the wire rope bends over a sheave, the individual wires in Lang Lay are not bent as sharply as Regular Lay. The individual wires in Regular Lay lie in parallel with the wire rope axis, and the radius of curvature is equal to the radius of the sheave plus the diameter of the wire rope. The individual wires in a Lang Lay wire rope lie at an angle to the wire rope axis and have a larger radius of curvature. This reduces the stresses in the Lang Lay wire rope compared to the Regular Lay wire rope. Lang Lay will have a higher fatigue resistance due to the lower bending stress. Lang Lay also has a higher abrasion resistance than Regular Lay wire rope, since the wear is distributed differently in each of these types of lays. In Lang Lay, the wear is distributed over a longer distance of the wire, whereas in Regular Lay the wear is concentrated over a shorter distance in the wire. This gives Lang Lay a longer period before the individual wires will be worn and break due to bending stresses compared to Regular Lay. Regular Lay is more resistant to crushing which tends to distort the shape of the wire rope. Regular Lay also tends to be more stable. A stable wire rope is resistant to kinking, does not tangle when relaxed, and spools smoothly on and off the drum.

The distance a strand makes as it is wound one complete revolution around the core of the wire rope is called *Lay Length* and is shown in Figure 7-46.

The selection of the proper lay in the winding of wire rope on a smooth or grooved drum is illustrated in Figures 7-47(a) and (b). Right Lay wire rope is used when winding from the left to the right on a drum rotating clockwise (looking from the left end of the drum). Left Lay wire rope is used when winding from the right to the left on a drum rotating clockwise (looking from the left end of the drum). Another way to look at it is if the drum is left hand grooved, use a Right Lay wire rope. A drum that has a right hand groove will use Left Lay wire rope. The correct lay of rope will keep the winding tight if the load is released.

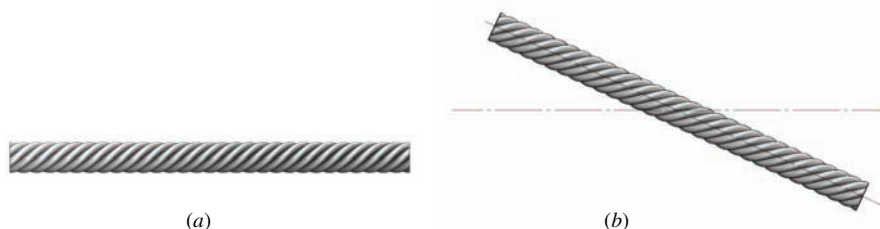


FIGURE 7-44 Right Lay – Regular Lay wire rope

(a) One strand shown straight with Regular Lay—individual wires lay left, opposite of strand lay
(b) One strand shown with Right Lay—individual wires run parallel with length of wire rope

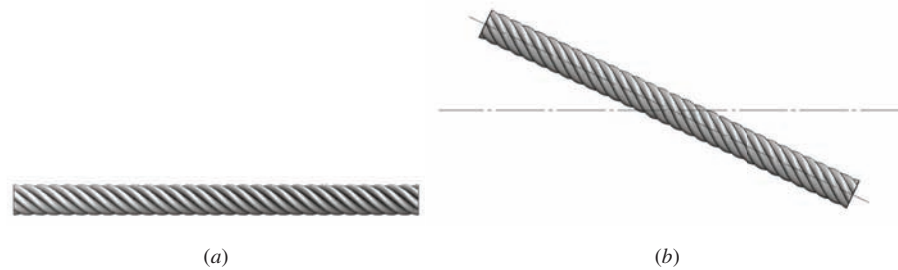


FIGURE 7-45 Right Lay – Lang Lay wire rope

- (a) One strand shown straight with Lang lay—individual wires lay right, same direction as strand lay
 (b) One strand shown with Right Lay—individual wires run perpendicular to the length of wire rope

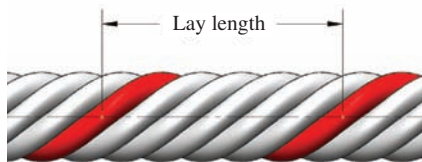


FIGURE 7-46 Lay length

A non-rotating wire rope is designed to be counter-balanced with little tendency to twist in either direction. This is accomplished by having one layer of strands wound with Right Lay and a second layer wound with Left Lay.

Sheave and Drum Design

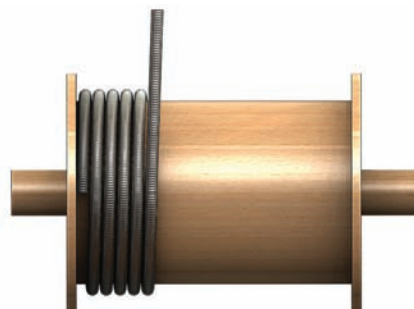
Sheaves, drums, and rollers must be properly designed to optimize the service life of the wire rope. The wire rope strands and wires move relative to one another, as the wire rope travels over a sheave or drum. As the wire rope bends over a sheave, the inside wires would travel a shorter distance than the outside wires.

This distance is compensated by the relative movement of the wires. The wire rope is subjected to bending stresses as it travels over a sheave or drum. This cyclic bending stress will cause fatigue, and eventually failure, of the wire rope. The magnitude of the stresses are related to the load, sheave diameter, and wire rope diameter.

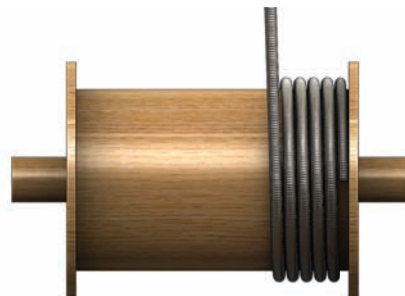
A suggested and minimum diameter (D) of the sheave or drum is selected based on the wire rope diameter (d) and the construction of the wire rope. This diameter is called the *tread diameter* (D) and is where the wire rope rides on the sheave or grooved drum. The D/d ratio measures how tightly the wire rope bends over a sheave. The smaller the ratio, the tighter the bend in the wire rope. The suggested and minimum D/d ratios based on wire rope construction are shown in Table 7-22. Figure 7-48

TABLE 7-22 Sheave and Drum Diameter Factors

Construction	Suggested D/d ratio	Minimum D/d ratio
6×7	72	42
6×19 Seale	51	34
6×21 Filler Wire	45	30
6×25 Filler Wire	39	26
6×31 Warrington Seale	39	26
6×36 Warrington Seale	35	23
6×41 Seale Filler Wire	20	20
6×41 Warrington Seale	32	21
6×42 Tiller	21	14
8×19 Seale	41	27
8×25 Filler Wire	32	21



(a) Right Lay wire rope used on a smooth drum



(b) Left Lay wire rope used on a smooth drum

FIGURE 7-47 Proper lay of wire rope on a smooth drum

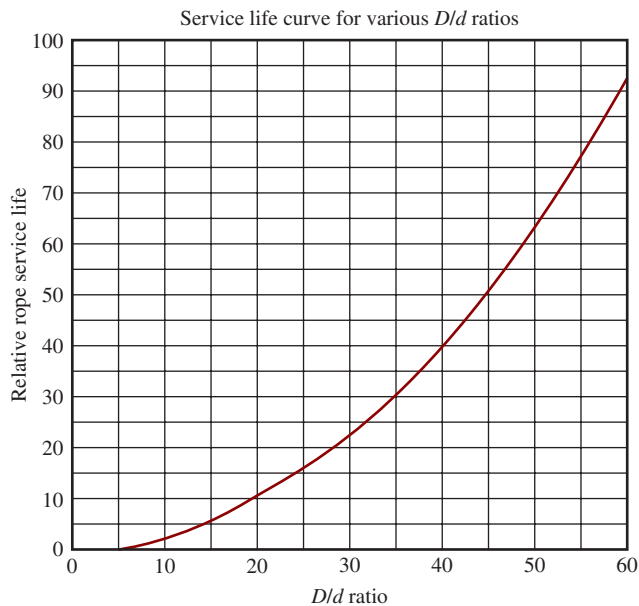


FIGURE 7-48 Service life curve

shows the effects of the D/d ratio on the service life of the wire rope. A larger D/d ratio will extend the life of the wire rope and the sheave and should be based on the practical limit of the machine application, along with cost and weight limitations. Bending a wire rope over a sheave with a small diameter will result in excessive wear in the wire rope and sheave. This is caused by excessive bending and straightening of the wire rope as it enters and exits the sheave. Reverse bending of the wire rope will accelerate wire rope fatigue and should be avoided if possible.

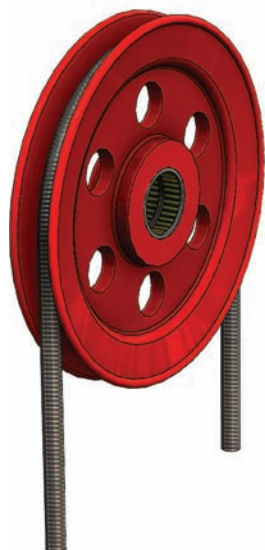
Sheave and Drum Design and Dimensions

As shown in Figure 7-49(a) a typical sheave design will have a hub and bore that will be used to mount a bearing

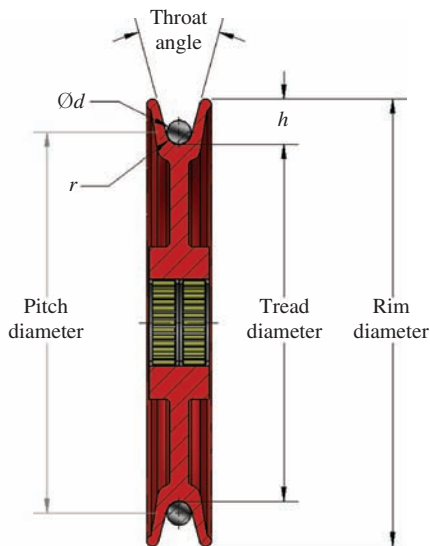
element. The bearing element can be a ball bearing, roller bearing, or a bronze bushing and is selected based on rotational speed, loading, and operating conditions of the sheave. The sheave has a groove designed in the rim that the wire rope will ride in. The web supports this rim and may have lightening holes in order to decrease the inertia of the sheave. Figure 7-49(b) shows the dimensional specifications and terminology of the sheave. The tread diameter (D) is based on the suggested minimum D/d ratio. The sheave throat angle should be between 35° and 45° . The groove depth of a sheave should be between

$$h_{\min} = 1.50 \times d \text{ and } h_{\max} = 1.75 \times d \quad (7-23)$$

The throat angle, along with the groove depth (h), allows the wire rope to enter and exit the sheave with minimum wear on the wire rope and sheave. The proper groove radius (r) of a sheave will give the maximum support to the wire rope. If the groove radius is tight or small, it will pinch the wire rope and increase the groove pressure along this contact point. If the groove radius is oversized, the wire rope will not be properly supported and will tend to flatten out. This will cause the wire rope to be unbalanced and will lead to wire rope crushing and early fatigue. The correct groove radius and angle of contact will improve wire rope support and maximize the wire rope's lifespan. The groove radius for new sheaves and sheaves that are in operation are given in Table 7-23. The groove radius is shown for the different wire rope diameters. A groove gage is used to measure the size and contour of the groove for both new and used sheaves. The gage should make an angle of contact of 150° with the groove and verify the radius. Scheduled maintenance includes periodic inspections of the sheave groove radius. If the groove radius measures smaller than the value listed as "Worn" in Table 7-23, the sheave groove should be re-machined to the "New" radius dimension.



(a) Sheave with roller bearing



(b) Sheave dimensional specifications

FIGURE 7-49 Design of a roller bearing wire rope sheave

TABLE 7-23 Recommended Sheave and Drum Groove Radius

Nominal wire rope diameter		Groove radius			
		New		Worn	
		inches	mm	inches	mm
1/4	6.5	0.137	3.48	0.129	3.28
3/8	9.5	0.201	5.11	0.190	4.83
1/2	13	0.271	6.88	0.256	6.50
5/8	16	0.334	8.48	0.320	8.13
3/4	19	0.401	10.19	0.380	9.65
7/8	22	0.468	11.89	0.440	11.18
1	26	0.543	13.79	0.513	13.03
1 1/4	32	0.669	16.99	0.639	16.23
1 1/2	38	0.803	20.40	0.759	19.28
1 3/4	45	0.939	23.85	0.897	22.78
2	52	1.070	27.18	1.019	25.88
2 1/2	64	1.338	33.99	1.279	32.49
3	77	1.607	40.82	1.538	39.07
3 1/2	90	1.869	47.47	1.794	45.57
4	103	2.139	54.33	2.050	52.07
4 1/2	115	2.396	60.86	2.298	58.37
5	128	2.663	67.64	2.557	64.95

Figure 7-50(a) shows the design of a helically grooved drum. The drum will have end flanges and journal ends that will be mounted in bearing units such as a pillow block to allow the drum to rotate. The drum will also require a means of securing the wire rope to the drum. This could be a clamping device that is bolted to the drum. Figure 7-50(b) shows the dimensional specifications of the grooved drum. The groove radius (r) must be proper for the wire rope diameter used. The pitch (p) of the helical groove is the distance from one groove to

the next adjacent groove along the pitch diameter of the wire rope. The pitch distance should range between

$$\text{Pitch} = p = 2.065 \times r \text{ and } p = 2.18 \times r \quad (7-24)$$

The minimum groove depth (h) is the distance from the drum diameter to the tread diameter (D) on a helically grooved drum. This distance is given by

$$h = 0.374 \times d \quad (7-25)$$

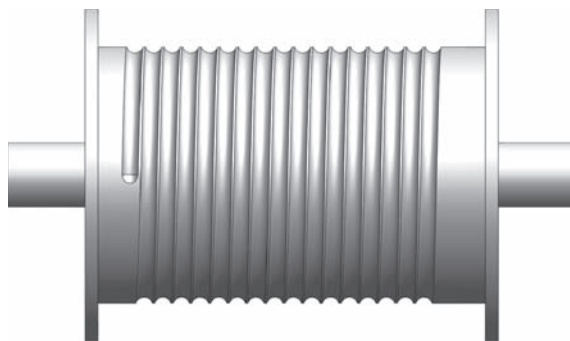
The *fleet angle* is the included angle between the wire rope which extends from the sheave to the drum and the perpendicular line to the axis of the drum. It is desirable to have a small fleet angle to reduce wear between the wire rope and flange of the sheave as the wire rope wraps the drum. The minimum fleet angle should be $\frac{1}{2}^\circ$ to prevent the wire rope from piling up on the drum. The maximum fleet angle should be $1\frac{1}{2}^\circ$ on a smooth drum and 2° on a grooved drum. Figure 7-51 shows the right and left fleet angles between a sheave and a smooth drum.

Wire Material and Grades

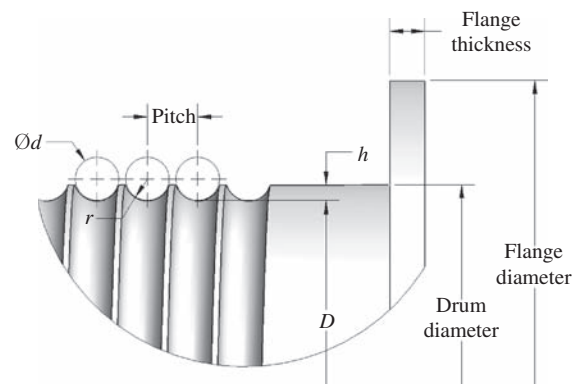
The most common wire material is an uncoated (bright) high-carbon steel. The properties of steel allow it to be drawn into wires with high strength, and good wear and fatigue resistance. Stainless steel or galvanized steel wire are used in applications with corrosive environments.

Plow steel is a term used for wire rope grades. Each different plow steel grade has improved tensile strength as shown in Table 7-24. *Improved plow steel* (IPS) is the most commonly used grade for wire rope, while *Extra improved plow steel* (XIP) is used for special applications.

The hardness of a wire rope can range from 42 to 50 Rc and will wear a sheave or drum if made out of a material that is too soft. A recommended sheave groove hardness should be 250–300 Brinell for a steel material. If a sheave is too soft, the wire rope will score and corrugate the groove of the sheave. A new wire rope will not track in a corrugated sheave and will cause excessive wear in the wire rope.



(a) Grooved drum with flanges and journal ends



(b) Grooved drum dimensional specifications

FIGURE 7-50 Design details for a grooved wire rope drum

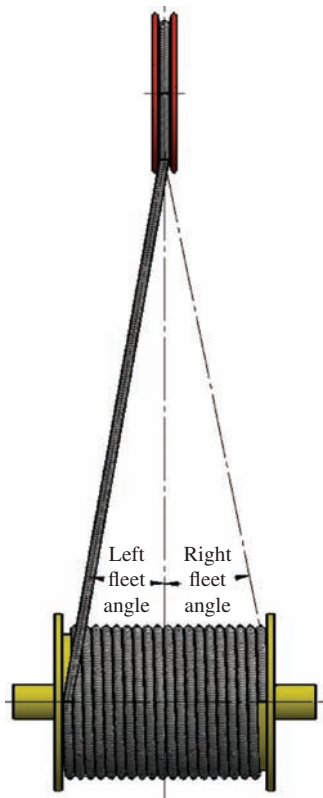


FIGURE 7-51 Fleet angle

TABLE 7-24 Grades of Wire Rope

Grade	Tensile strength
Plow Steel	1570 N/mm ²
Improved Plow Steel	1770 N/mm ²
Extra Improved Plow Steel	1960 N/mm ²

Wire Rope Selection

Wire rope construction refers to the arrangement of wires, strands, and the type of core used. The construction produces specific operating characteristics of the wire rope that will allow it to meet the performance requirements of its application. The attributes of a wire rope include strength, flexibility, resistance to bending fatigue, abrasion resistance, and resistance to crushing. A compromise between these attributes may have to be made in order to select the best wire rope for a specific application.

A wire rope's *strength* is shown as a minimum breaking force and is measured in tons (2000 lb/ton). Its minimum breaking force is determined by placing a new rope in a pull testing machine that applies a constantly increasing force until the wire rope breaks. The strength of the wire rope can be increased by increasing the diameter of the wire rope, using a better grade of steel, using a steel core construction, or increasing the cross-sectional steel content of the wire rope.

Flexibility relates to the wire rope's ability to bend around a sheave or drum. Some applications require the use of more sheaves where the wire rope will have to make a lot of bends or smaller-diameter sheaves where the wire rope will have to make a tighter bend. Flexibility can be increased by increasing the quantity of smaller-diameter wires in a strand. Some wire rope constructions are more flexible than others. The same diameter wire rope that has a strand with 37 smaller-diameter wires instead of 26 larger-diameter wires would improve the flexibility of the wire rope. Using a fiber core instead of a steel core will also improve the flexibility of the wire rope.

Fatigue resistance is required when wire ropes are subjected to repeated bending over sheaves and drums. This phenomenon involves metal fatigue of the individual wires that make up the wire rope. This is similar to taking a paper clip and bending it back and forth until it breaks. The same occurrence takes place in the individual wires as they are repeatedly bent around sheaves and drums. Sharper bends and reverse bending will accelerate the rate of fatigue of the wire rope. Using a larger number of smaller individual wires, while keeping the overall diameter of the wire rope the same, will increase the fatigue resistance of the wire rope. The smaller individual wires have a greater ability to bend than a larger wire. As discussed previously, the bending fatigue can also be reduced by having a large D/d ratio.

Abrasion resistance refers to the actual wearing of material away from the individual wires in a wire rope. The removal of material creates a destructive condition and weakens the wire rope which will lead to eventual failure. Abrasion can occur when the wire rope is dragged through a gritty material or across a stationary object. Abrasion also happens to individual wires within the wire rope as the wires rub against each other when the wire rope is bending and unbending or being loaded and unloaded. Abrasion will also manifest when the wire rope is being wound or unwound from a drum and the wire rope will have rope-to-rope contact. Proper sheave alignment, groove diameter, fleet angle, and drum winding will minimize abrasion of the wire rope. These should be checked during regular scheduled maintenance outages. Large-diameter wires will improve the wire rope's resistance to abrasion.

Crushing resistance is the ability of the wire rope to maintain its cross-sectional shape while under an external pressure. IWRC core, six strand, and regular lay wire rope construction will improve crush resistance. A crushed or distorted wire rope will not allow the individual wires to move relative to one another and operate properly. An IRWC core has a higher crush resistance compared to a fiber core.

As was discussed earlier, using a small number of larger-diameter wires will improve abrasion resistance, but a lower resistance to bending fatigue while using a large number of small-diameter wires will improve the resistance to bending fatigue and reduce the resistance

to abrasion. A fiber core is more flexible than an IWRC core but has less crush resistance and strength. Lang lay has a higher abrasion resistance, fatigue resistance, and is more flexible than a regular lay wire rope. Regular lay wire rope is more resistant to crushing and is more stable than Lang lay. Based on the best suitable wire rope design for the application, a compromise between these characteristics of the wire ropes will need to be made.

Several characteristics of the design application should be considered:

- Speed of operation
- Acceleration and deceleration
- Length of rope
- Number, size, and location of sheaves and drums
- Conditions of the environment
- Human and property safety concerns

These design characteristics will be used to select of the best wire rope for the application.

Design Factors and Working Loads

Industry standards and regulations require minimum design factors to be applied when using wire rope. The design factor is applied to the minimum breaking force of the wire rope. A minimum design factor (SF) of 5 is used in overhead cranes, gantry cranes, and overhead hoists. The maximum working load of a wire rope can then be determined in the following equation.

$$\text{Maximum working load} = \text{Minimum breaking force}/SF$$

This allowable working load is the load the wire rope is expected to carry not only due to static loading, but also loading resulting from accelerations and shock loading conditions on a system. Shock loads can be significantly higher than static loads. Shock loading can occur when there is a sudden change or jerking movement of the load and should be avoided.

The technical data are the same for both 6×19 and 6×36 classification of wire rope and are shown in Table 7–25. The minimum breaking force is given for

TABLE 7–25 6×19 and 6×36 Classes Technical Data

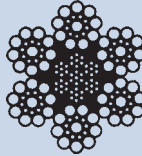
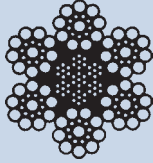
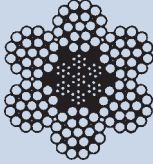
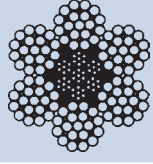
Fiber core 				IWRC 		
Diameter	Weight per foot	Min breaking force		Weight per foot	Min breaking force	
		IPS	XIP		IPS	XIP
in	lb/ft	tons	tons	lb/ft	tons	tons
1/4	0.105	2.74	3.02	0.116	2.94	3.4
5/16	0.164	4.26	4.69	0.18	4.58	5.27
3/8	0.236	6.1	6.72	0.26	6.56	7.55
7/16	0.32	8.27	9.1	0.35	8.89	10.2
1/2	0.42	10.7	11.8	0.46	11.5	13.3
9/16	0.53	13.5	14.9	0.59	14.5	16.8
5/8	0.66	16.7	18.3	0.72	17.9	20.6
3/4	0.95	23.8	26.2	1.04	25.6	29.4
7/8	1.29	32.2	35.4	1.42	34.6	39.8
1	1.68	41.8	46	1.85	44.9	51.7
1 1/8	2.13	52.6	57.8	2.34	56.5	65
1 1/4	2.63	64.6	71.1	2.89	69.4	79.9
1 3/8	3.18	77.7	85.5	3.5	83.5	96
1 1/2	3.78	92	101	4.16	98.9	114
1 5/8	4.44	107	118	4.88	115	132
1 3/4	5.15	124	137	5.67	133	153
1 7/8	5.91	141	156	6.5	152	174
2	6.72	160	176	7.39	172	198
2 1/8	7.59	179	197	8.35	192	221
2 1/4	8.51	200	220	9.36	215	247

TABLE 7-26 Properties of Standard 6×19 and 6×36 Wire Ropes

6×19S (Seale)		Good resistance to abrasion and crushing Fatigue resistance is less than 6×25
6×25F (Filler Wire)		Higher resistance to fatigue than 6×19 class Best combination of flexibility and wear resistance for 6×19 class Filler wire provides support and stability to the strand
6×26 (Warrington Seale)		High resistance to crushing Good wear resistance Good Flexibility
6×36 (Warrington Seale)		Provides good fatigue resistance without having wires that are too small Most flexible due to large number of small wires Susceptible to crushing, use IWRC to minimize

both improved plow steel and extra improved plow steel with a fiber core and an IWRC core. The weight per foot is shown for both fiber and IWRC cores.

The 6×19 steel core and fiber core classifications of wire rope are the most widely used. The combination of flexibility and wear resistance makes it suitable for diverse types of machinery and equipment. Compared to the 6×19 classification, the 6×36 classification of wire rope is more flexible but less abrasion resistant. Table

7-26 gives the operating properties of some 6×19 and 6×36 wire ropes.

The nomenclature of a wire rope defines the length, size (diameter), direction of lay, grade of rope, finish, construction, and type of core. An example of a complete wire rope designation is

500 ft × 5/8 in diameter 6×19 Filler Wire – Right Lay–Left Lang Improved Plow Steel IWRC

Example Problem
7-7

Recommend the diameter for a grooved drum on which a wire rope with a diameter of 16 mm is to be wound as part of a winch system. The construction of the rope is to be 6×36 Warrington Seale type.

Solution

Given 6×36 Warrington Seale type wire rope with a diameter of $d = 16$ mm

Results Table 7-22 lists the suggested D/d ratio for this size and type wire rope to be 35, where the value of D is the tread diameter of the drum. Then,

$$D = 35 d = 35 \cdot 16 \text{ mm} = 560 \text{ mm}$$

The tread diameter, D , is the measurement between the lowest point of the groove on the top and bottom of the drum, as shown in Figure 7-50(b).

The minimum groove depth, h , can be calculated from Equation (7-20).

$$h = 0.374 \cdot d = 0.374 \cdot 16 \text{ mm} = 6 \text{ mm}$$

Therefore, the drum diameter is

$$\text{Drum diameter} = D + 2 \cdot h = 560 \text{ mm} + 2 \cdot 6 \text{ mm} = 572 \text{ mm}$$

Example Problem 7–8 Recommend the allowable working load for a wire rope having 6×19 construction, a diameter of 1.25 in, and made from XIP steel to be used in a crane application.

Solution Given Wire rope, 6×19 construction, $D = 1.25$ in

Results Table 7–25 lists the minimum breaking load for 1.25-in diameter wire rope made from XIP (Extra improved plow steel) to be 71.1 tons. A minimum service factor (SF) of 5 is recommended. Then,

Maximum working load = Breaking force/ $SF = 71.1$ tons/5 = 14.22 tons

Converting to pounds force gives,

$14.22 \text{ tons} \times 2000 \text{ lb/ton} = 28\,440 \text{ lb}$

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INTERNET SITES RELATED TO BELT DRIVES AND CHAIN DRIVES

1. **American Chain Association.** A national trade organization for companies providing products for the chain drive industry. Publishes standards and design aids for designing, applying, and maintaining chain drives and engineering chain conveyor systems.
2. **Dayco Products LLC.** Manufacturer of Dayco industrial and automotive belt drive systems of the poly-v and timing belt designs
3. **Baldor-Dodge.** Manufacturer of numerous power transmission components, including V-belt and synchronous belt drive systems.
4. **Regal Beloit Americas, Inc.** Manufacturer of a variety of power transmission products under several brand names, including Browning V-belt drives and Morse roller-chain drives.
5. **Gates Rubber Company.** Rubber products for the automotive and industrial markets, including V-belt drives and synchronous belt drives.
6. **Grainger Industrial Supply.** A distributor of a wide array of industrial products, including belts, sheaves, chain, and sprockets for power transmission.
7. **International Organization for Standardization.** The premier organization for establishing and promulgating technical standards for worldwide implementation. Search on *Standards* and the subject for which you are seeking ISO standards.
8. **Martin Sprocket and Gear Company.** Manufacturer of a wide range of mechanical power transmission products, including chain sprockets, V-belt sheaves, synchronous belt sprockets, and bushings.

9. **Maryland Metrics.** U.S. based company that supplies metric power transmission products rated by the ISO, BS, and DIN standards.
10. **Power Transmission.** A comprehensive website for companies providing products for the power transmission industry, many of which supply belt and chain drive systems.
11. **Ensinger Precision Components.** Producer of plastic injection-molded mechanical drive components, including plastic chain, sprockets, and synchronous belt pulleys.
12. **Rexnord Corporation.** Manufacturer of power transmission and conveying components, including roller chain drives and engineered chain drive systems.
13. **SAE International.** The Society of Automotive Engineers, the engineering society for advancing mobility on land or sea, in air or space. Offers standards on V-belts, synchronous belts, pulleys, and drives for automotive applications.
14. **SDP/SI.** The Stock Drive Products/Sterling Instruments Company distributes a wide array of mechanical drive components with a heavy emphasis on small, precision mechanical drive components, including synchronous belts, pulleys, chain, and sprockets. Both U.S. Customary and Metric styles of components are included.
15. **Wippermann Company.** Manufacturer of a wide range of chain drive products based in Germany; wide selection of German DIN standard chains and sprockets.
16. **T. B. Wood's Sons Company.** Manufacturer of many mechanical drives products, including V-belt drives, synchronous belt drives, and adjustable speed drives.
17. **Bridon.** Manufacturer of wire rope for the crane, oilfield, surface mining, underground mining, and related industries.
18. **Union - A WireCo(R) WorldGroup Brand.** Manufacturer of wire rope for crane, oil and gas, surface mining, logging, slings, and related products. The site includes a Technical Reference section providing guidance on installation, operation, maintenance, inspection and other topics.
19. **Wirerope Works, Inc.** Manufacturer of Bethlehem Wire Rope(R) for elevators, oilfield and marine applications, logging, mining, ski lifts, bridge suspension, and other applications.

PROBLEMS

V-Belt Drives

1. Specify the standard 3V belt length (from Table 7-2) that would be applied to two sheaves with pitch diameters of 5.25 in and 13.95 in with a center distance of no more than 24.0 in.
 2. For the standard belt specified in Problem 1, compute the actual center distance that would result.
 3. For the standard belt specified in Problem 1, compute the angle of wrap on both of the sheaves.
 4. Specify the standard 5V belt length (from Table 7-2) that would be applied to two sheaves with pitch diameters of 8.4 in and 27.7 in with a center distance of no more than 60.0 in.
 5. For the standard belt specified in Problem 4, compute the actual center distance that would result.
 6. For the standard belt specified in Problem 4, compute the angle of wrap on both of the sheaves.
 7. Specify the standard 8V belt length (from Table 7-2) that would be applied to two sheaves with pitch diameters of 13.8 in and 94.8 in with a center distance of no more than 144 in.
 8. For the standard belt specified in Problem 7, compute the actual center distance that would result.
 9. For the standard belt specified in Problem 7, compute the angle of wrap on both of the sheaves.
 10. If the small sheave of Problem 1 is rotating at 1750 rpm, compute the linear speed of the belt.
 11. If the small sheave of Problem 4 is rotating at 1160 rpm, compute the linear speed of the belt.
 12. If the small sheave of Problem 7 is rotating at 870 rpm, compute the linear speed of the belt.
 13. For the belt drive from Problems 1 and 10, compute the rated power, considering corrections for speed ratio, belt length, and angle of wrap.
 14. For the belt drive from Problems 4 and 11, compute the rated power, considering corrections for speed ratio, belt length, and angle of wrap.
 15. For the belt drive from Problems 7 and 12, compute the rated power, considering corrections for speed ratio, belt length, and angle of wrap.
 16. Describe a standard 15N belt cross section. To what size belt (inches) would it be closest?
 17. Describe a standard 17A belt cross section. To what size belt (inches) would it be closest?
- For Problems 18-22 (Table 7-27), design a V-belt drive. Specify the belt size, the sheave sizes, the number of belts, the actual output speed, and the center distance.

Roller Chain

23. Describe a standard roller chain, no. 140.
24. Describe a standard roller chain, no. 60.
25. Specify a suitable standard chain to exert a static pulling force of 1250 lb.
26. Roller chain is used in a hydraulic forklift truck to elevate the forks. If two strands support the load equally, which size would you specify for a design load of 5000 lb?
27. List three typical failure modes of roller chain.
28. Determine the power rating of a no. 60 chain, single-strand, operating on a 20-tooth sprocket at 750 rpm. Describe the preferred method of lubrication. The chain connects a hydraulic drive with a meat grinder.
29. For the data of Problem 28, what would be the rating for three strands?
30. Determine the power rating of a no. 40 chain, single-strand, operating on a 12-tooth sprocket at 860 rpm. Describe the preferred method of lubrication. The small

TABLE 7-27 V-Belt Drive Design Problems

Problem number	Driver type	Driven machine	Service (h/day)	Input speed (rpm)	Input power		Nominal output speed (rpm)
					(hp)	(kW)	
18.	AC motor (HT)	Hammer mill	8	870	25	18.6	310
19.	AC motor (NT)	Fan	22	1750	5	3.73	725
20.	6-cylinder engine	Heavy conveyor	16	1500	40	29.8	550
21.	DC motor (compound)	Milling machine	16	1250	20	14.9	695
22.	AC motor (HT)	Rock crusher	8	870	100	74.6	625

Note: NT indicates a normal-torque electric motor. HT indicates a high-torque electric motor.

TABLE 7-28 Chain Drive Design Problems

Problem number	Driver type	Driven machine	Input speed (rpm)	Input power		Nominal output speed (rpm)
				(hp)	(kW)	
38.	AC motor	Hammer mill	310	25	18.6	160
39.	AC motor	Agitator	750	5	3.73	325
40.	6-cylinder engine	Heavy conveyor	500	40	29.8	250
41.	Steam turbine	Centrifugal pump	2200	20	14.9	775
42.	Hydraulic drive	Rock crusher	625	100	74.6	225

sprocket is applied to the shaft of an electric motor. The output is to a coal conveyor.

31. For the data of Problem 30, what would be the rating for four strands?
32. Determine the power rating of a no. 80 chain, single-strand, operating on a 32-tooth sprocket at 1160 rpm. Describe the preferred method of lubrication. The input is an internal combustion engine, and the output is to a fluid agitator.
33. For the data of Problem 32, what would be the rating for two strands?
34. Specify the required length of no. 60 chain to mount on sprockets having 15 and 50 teeth with a center distance of no more than 36 in.
35. For the chain specified in Problem 34, compute the actual center distance.
36. Specify the required length of no. 40 chain to mount on sprockets having 11 and 45 teeth with a center distance of no more than 24 in.
37. For the chain specified in Problem 36, compute the actual center distance.

For Problems 38–42 (Table 7–28), design a roller chain drive. Specify the chain size, the sizes and number of teeth in the sprockets, the number of chain pitches, and the center distance.

Synchronous Belts

For any of the sets of problem data in Tables 7–27 and 7–28, use manufacturers' catalogs to specify the belt size and sprocket sizes for synchronous belts. See Internet sites 3–6, 8, 9, 11, 14, and 16 for manufacturers of synchronous belts and sprockets.

43. A synchronous belt drive is to have an input sprocket rotational speed of 800 rpm and an output sprocket speed of 600 rpm.
 - (a) List the sprocket combinations that could be used and list the pitch diameters for each sprocket.
 - (b) For each sprocket combination from (a) and using a 1440-8MGT belt, compute the center distance and the belt linear velocity.
44. A synchronous belt drive is to have an input sprocket rotational speed of 1200 rpm and an output sprocket speed of 600 rpm.
 - (a) List the sprocket combinations with an 8-mm pitch that could be used and list the pitch diameters for each sprocket.
 - (b) For each sprocket combination from (a) and using a 1800-8MGT-30 belt, compute the center distance and the belt linear velocity.
 - (c) For each sprocket, list the taper-lock bushing required and the minimum and maximum bores.

KINEMATICS OF GEARS

The Big Picture

You Are the Designer

- 8–1 Objectives of This Chapter
- 8–2 Spur Gear Styles
- 8–3 Spur Gear Geometry-Involute-Tooth Form
- 8–4 Spur Gear Nomenclature and Gear-Tooth Features
- 8–5 Interference between Mating Spur Gear Teeth
- 8–6 Internal Gear Geometry
- 8–7 Helical Gear Geometry
- 8–8 Bevel Gear Geometry
- 8–9 Types of Wormgearing
- 8–10 Geometry of Worms and Wormgears
- 8–11 Gear Manufacturing
- 8–12 Gear Quality
- 8–13 Velocity Ratio and Gear Trains
- 8–14 Devising Gear Trains

THE BIG PICTURE

Kinematics of Gears

Discussion Map

- Gears are toothed, cylindrical wheels used for transmitting motion and power from one rotating shaft to another.
- Most gear drives cause a change in the speed of the output gear relative to the input gear.
- Some of the most common types of gears are *spur gears*, *helical gears*, *bevel gears*, and *worm/wormgear sets*.

This chapter will help you learn about the features of different kinds of gears, the kinematics of a pair of gears operating together, and the operation of gear trains having more than two gears.

Gears are toothed, cylindrical wheels used for transmitting motion and power from one rotating shaft to another. The teeth of a driving gear mesh accurately in the spaces between teeth on the driven gear as shown in Figure 8–1. The driving teeth push on the driven teeth, exerting a force perpendicular to the radius of the gear. Thus, a torque is transmitted, and because the gear is rotating, power is also transmitted.

Speed Reduction Ratio. Often gears are employed to produce a change in the speed of rotation of the

Discover

Identify at least two machines or devices that employ gears. Describe the operation of the machines or devices and the appearance of the gears.

driven gear relative to the driving gear. In Figure 8–1, if the smaller top gear, called a *pinion*, is driving the larger lower gear, simply called the *gear*, the larger gear will rotate more slowly. The amount of speed reduction is dependent on the ratio of the number of teeth in the pinion to the number of teeth in the gear according to this relationship:

$$n_p/n_G = N_G/N_p \quad (8-1)$$

The basis for this equation will be shown later in this chapter. But to show an example of its application

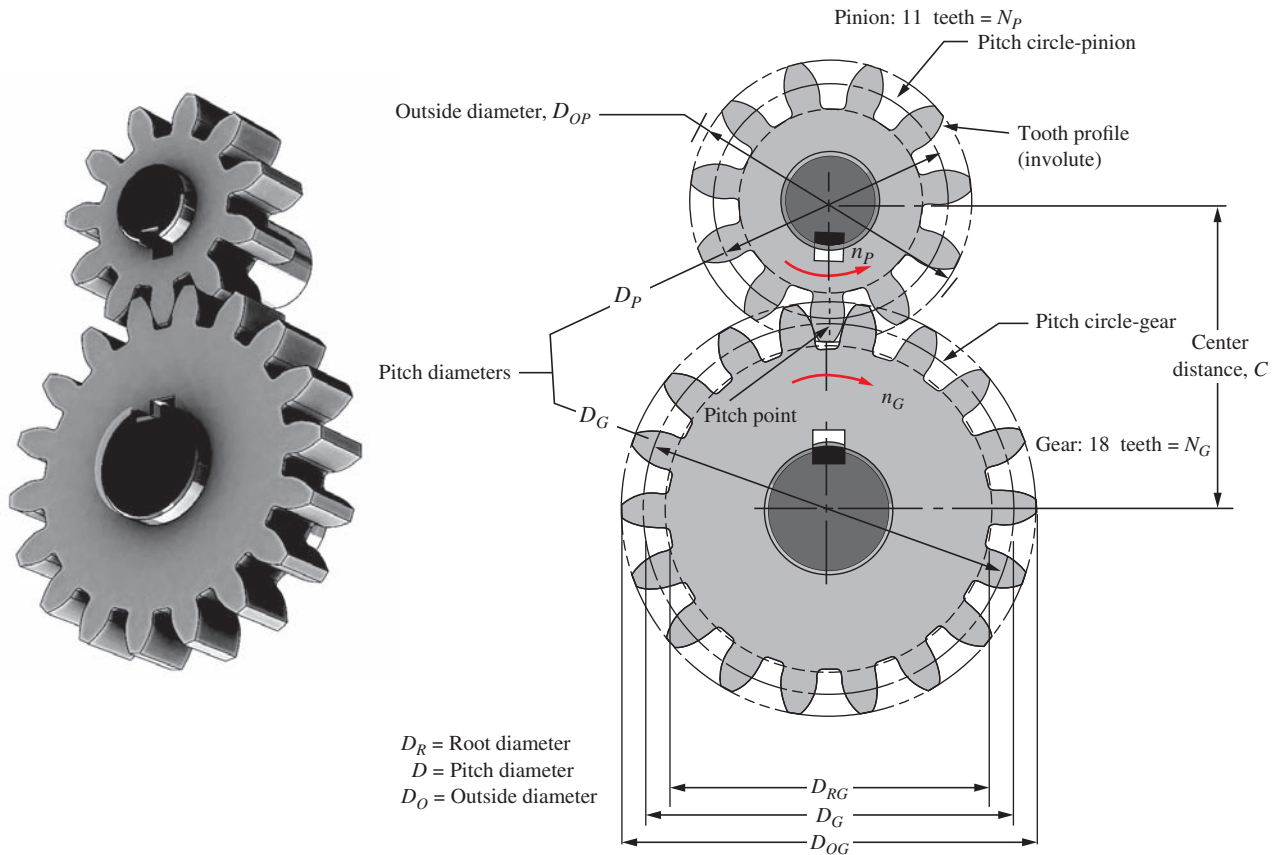


FIGURE 8-1 Pair of spur gears. The pinion drives the gear.

here, consider that the pinion in Figure 8-1 is rotating at 1800 rpm. You can count the number of teeth in the pinion to be 11 and the number of teeth in the gear to be 18. Then we can compute the rotational speed of the gear by solving Equation (8-1) for n_G :

$$n_G = n_P(N_P/N_G) = (1800 \text{ rpm})(11/18) = 1100 \text{ rpm}$$

When there is a reduction in the speed of rotation of the gear, there is a simultaneous proportional *increase* in the torque transmitted to the shaft carrying the gear. More will be said about this later, also.

Kinds of Gears. Several kinds of gears having different tooth geometries are in common use. To acquaint you with the general appearance of some, their basic descriptions are given here. Later we will describe their geometry more completely.

Figure 8-2 shows a photograph of many kinds of gears. Labels indicate the major types of gears that are discussed in this chapter: *spur gears*, *helical gears*, *bevel gears*, and *worm/wormgear sets*. Obviously, the shafts that would carry the gears are not included in this photograph. See References 4, 7, 10, 11-13, and 19 and Internet sites 1, 4, and 5 for more information on gearing.

Spur gears have teeth that are straight and arranged parallel to the axis of the shaft that carries

the gear. The curved shape of the faces of the spur gear teeth have a special geometry called an *involute curve*, described later in this chapter. This shape makes it possible for two gears to operate together with smooth, positive transmission of power. Figure 8-1 also shows the side view of spur gear teeth, and the involute curve shape is evident there. The shafts carrying the gears are parallel.

The teeth of *helical gears* are arranged so that they lie at an angle with respect to the axis of the shaft. The angle, called the *helix angle*, can be virtually any angle. Typical helix angles range from approximately 10° to 30° , but angles up to 45° are practical. The helical teeth operate more smoothly than equivalent spur gear teeth, and stresses are lower. Therefore, a smaller helical gear can be designed for a given power-transmitting capacity as compared with spur gears. One disadvantage of helical gears is that an axial force, called a *thrust force*, is generated in addition to the driving force that acts tangent to the basic cylinder on which the teeth are arranged. The designer must consider the thrust force when selecting bearings that will hold the shaft during operation. Shafts carrying helical gears are typically arranged parallel to each other. However, a special design, called *crossed helical gears*, has 45° helix angles, and their shafts operate 90° to each other.



FIGURE 8-2 A variety of gear types
(Courtesy of Boston Gear, an Altra Industrial Motion Company)

Bevel gears have teeth that are arranged as elements on the surface of a cone. The teeth of straight bevel gears appear to be similar to spur gear teeth, but they are tapered, being wider at the outside and narrower at the top of the cone. Bevel gears typically operate on shafts that are 90° to each other. Indeed, this is often the reason for specifying bevel gears in a drive system. Specially designed bevel gears can operate on shafts that are at some angle other than 90° . When bevel gears are made with teeth that form a helix angle similar to that in helical gears, they are called *spiral bevel gears*. They operate more smoothly than straight bevel gears and can be made smaller for a given power transmission capacity. When both bevel gears in a pair have the same number of teeth, they are called *miter gears* and are used only to change the axes of the shafts to 90° . No speed change occurs.

Now look closely at Figure 8–3 that shows an example of a large, commercially available reducer with three stages that employs a combination of bevel, helical, and spur gears that were just described. See Internet site 6. Seeing them in one unit can help you appreciate the similarities and differences among them. Follow the flow of power through the reducer as outlined here:

1. The input shaft at the left end carries the spiral bevel pinion for the right angle first stage of reduction.
2. The helical pinion behind the output gear of the bevel gear pair drives the large helical output gear of the second stage of reduction.
3. The output shaft from the helical gear pair carries the spur-type sun gear of a planetary gear train whose output shaft drives the final output shaft projecting from the front of the reducer.

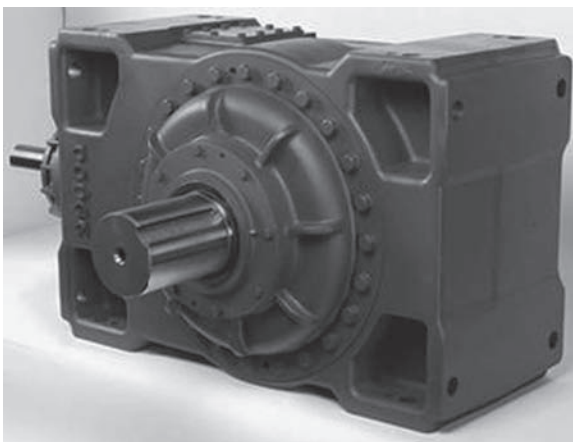
A *rack* is a straight gear that moves linearly instead of rotating. When a circular gear is mated with a rack, as shown toward the right side of Figure 8–2, the

combination is called a *rack and pinion drive*. You may have heard that term applied to the steering mechanism of a car or to a part of other machinery. See Section 8–3 for more discussion about a rack.

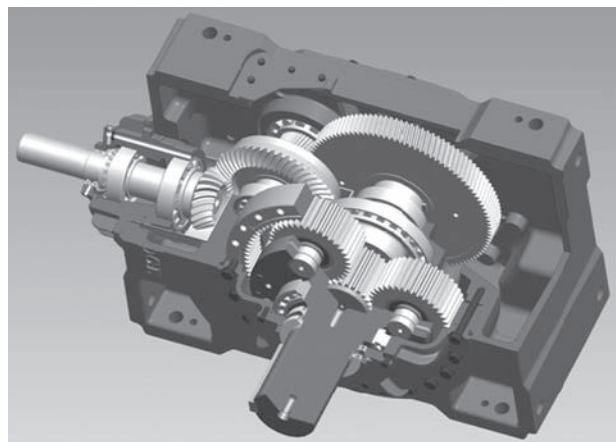
A *worm and its mating wormgear* operate on shafts that are at 90° to each other. They typically accomplish a rather large speed reduction ratio compared with other types of gears. The worm is the driver, and the wormgear is the driven gear. The teeth on the worm appear similar to screw threads, and, indeed, they are often called *threads* rather than *teeth*. The teeth of the wormgear can be straight like spur gear teeth, or they can be helical. Often the shape of the tip of the wormgear teeth is enlarged to partially wrap around the threads of the worm to improve the power transmission capacity of the set. One disadvantage of the worm/wormgear drive is that it has a somewhat lower mechanical efficiency than most other kinds of gears because there is extensive rubbing contact between the surfaces of the worm threads and the sides of the wormgear teeth.

Where Have You Observed Gears? Think of examples where you have seen gears in actual equipment. Describe the operation of the equipment, particularly the power transmission system. Sometimes, of course, the gears and the shafts are enclosed in a housing, making it difficult for you to observe the actual gears. Perhaps you can find a manual for the equipment that shows the drive system. Or look elsewhere in this chapter and in Chapters 9 and 10 for some photographs of commercially available gear reducers. (*Note: If the equipment you are observing is operating, be very careful not to come in contact with any moving parts!*) Try to answer these questions:

- What was the source of the power? An electric motor, a gasoline engine, a steam turbine, a hydraulic motor? Or were the gears operated by hand?



(a) External view of reducer



(b) Cutaway view showing internal components

FIGURE 8–3 Triple reduction gear reducer employing bevel, helical, and spur gears (Baldor/Dodge, Greenville, SC)

- How were the gears arranged together, and how were they attached to the driving source and the driven machine?
- Was there a speed change? Can you determine how much of a change?
- Were there more than two gears in the drive system?
- What types of gears were used? (You should refer to Figure 8–2.)
- What materials were the gears made from?
- How were the gears attached to the shafts that supported them?
- Were the shafts for mating gears aligned parallel to each other, or were they perpendicular to one another?

- How were the shafts themselves supported?
- Was the gear transmission system enclosed in a housing? If so, describe it.

This chapter will help you learn the basic geometries and kinematics of gears and pairs of gears operating together. You will also learn how to analyze gear trains having more than two gears so that you can describe the motion of each gear. Then you will learn how to devise a gear train to produce a given speed reduction ratio. In later chapters, you will learn how to analyze gears for their power transmission capacity and to design gear trains to transmit a given amount of power at a specified ratio of the speed of the input shaft to the speed of the output shaft.

YOU ARE THE DESIGNER

A gear-type speed reducer was described in Chapter 1, and a sketch of the layout of the gears within the reducer was shown in Figure 1–12. You are advised to review that discussion now because it will help you understand how the present chapter on *gear geometry* and *kinematics* fits into the design of the complete speed reducer.

Assume that you are responsible for the design of a speed reducer that will take the power from the shaft of an electric motor rotating at 1750 rpm and deliver it to a machine that is to operate at approximately 292 rpm. You have decided to use gears to transmit the power, and you are proposing a double-reduction speed reducer like the concept sketch shown in Figure 1–12. This chapter will give you the information you need to define the general nature of the gears, including their arrangement and their relative sizes.

The input shaft (shaft 1) is coupled to the motor shaft. The first gear of the gear train is mounted on this shaft and rotates at the same speed as the motor, 1750 rpm. Gear 1 drives the mating gear 2, which is larger, causing the speed of rotation of shaft 2 to be slower than that of shaft 1. But the speed is not yet down to 292 rpm as desired.

The next step is to mount a third gear (gear 3) on shaft 2 and mate it with gear 4 mounted on the output shaft, shaft 3. With proper sizing of all four gears, you should be able to produce an output speed equal or quite close to the desired speed. This process requires knowledge of the concept of *velocity ratio* and the techniques of designing gear trains as presented in this chapter.

But you will also need to specify the appearance of the gears and the geometry of the several features that make up each gear. Whereas the final specification also requires the information from following chapters, you will learn how to recognize common

forms of gears and to compute the dimensions of key features. This will be important when completing the design for strength and wear resistance in later chapters.

Let's say that you have chosen to use spur gears in your design. What design decisions must you make to complete the specification of all four gears? The following list gives some of the important parameters for each gear:

- The number of teeth.
- The form of the teeth.
- The size of the teeth as indicated by the *pitch*.
- The width of the face of the teeth.
- The style and dimensions of the gear blank into which the gear teeth are to be machined.
- The design of the hub for the gear that facilitates its mounting to the shaft.
- The degree of precision of the gear teeth and the corresponding method of manufacture that can produce that precision.
- The means of attaching the gear to its shaft.
- The means of locating the gear axially on the shaft.

To make reliable decisions about these parameters, you must understand the special geometry of spur gears as presented first in this chapter. However, there are other forms of gears that you could choose. Later sections give the special geometry of helical gears, bevel gears, and worm/wormgear sets. The methods of analyzing the forces on these various kinds of gears are described in later chapters, including the stress analysis of the gear teeth and recommendations on material selection to ensure safe operation with long life. ■

8–1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will be able to:

1. Recognize and describe the main features of *spur gears*, *helical gears*, *bevel gears*, and *worm/wormgear sets*.
2. Describe the important operating characteristics of these various types of gears with regard to the

similarities and differences among them and their general advantages and disadvantages.

3. Describe the *involute-tooth form* and discuss its relationship to the *law of gearing*.
4. Describe the basic functions of the American Gear Manufacturers Association (AGMA) and identify pertinent standards developed and published by this organization.

5. Define *velocity ratio* as it pertains to two gears operating together.
6. Specify appropriate numbers of teeth for a mating pair of gears to produce a given velocity ratio.
7. Define *train value* as it pertains to the overall speed ratio between the input and output shafts of a gear-type speed reducer (or speed increaser) that uses more than two gears.

8-2 SPUR GEAR STYLES

Figure 8-4 shows several different styles of commercially available spur gears. When gears are large, the spoked design in Part (a) is often used to save weight. The gear teeth are machined into a relatively thin rim that is held by a set of spokes connecting to the hub. The bore of the hub is typically designed to be a close sliding fit with the shaft that carries the gear. A keyway is usually machined into the bore to allow a key to be inserted for positive transmission of torque. The first illustration does not include a keyway because this gear is sold as a stock item, and the ultimate user finishes the bore to match a given piece of equipment.

The solid hub design in Figure 8-4(b) is typical of smaller spur gears. Here the finished bore with a keyway is visible. The set screw over the keyway allows the locking of the key in place after assembly.

When spur gear teeth are machined into a straight, flat bar, the assembly is called a rack, as shown in Figure 8-4(c). The rack is essentially a spur gear with an infinite radius. In this form, the teeth become straight-sided, rather than the curved, involute form typical of smaller gears.

Gears with diameters between the small solid form [Part (b)] and the larger spoked form [Part (a)] are often

produced with a thinned web as shown in Part (d), again to save weight.

You as a designer may create special designs for gears that you implement into a mechanical device or system. One useful approach is to machine the gear teeth of small pinions directly into the surface of the shaft that carries the gear. This is very often done for the input shaft of gear reducers.

8-3 SPUR GEAR GEOMETRY: INVOLUTE-TOOTH FORM

The most widely used spur gear tooth form is the full-depth involute form. Its characteristic shape is shown in Figure 8-5. See References 10–15 and 18 for more on the kinematics of gearing.

The involute is one of a class of geometric curves called *conjugate curves*. When two such gear teeth are in mesh and rotating, there is a *constant angular velocity* ratio between them: From the moment of initial contact to the moment of disengagement, the speed of the driving gear is in a constant proportion to the speed of the driven gear. The resulting action of the two gears is very smooth. If this were not the case, there would be some speeding up and slowing down during the engagement, with the resulting accelerations causing vibration, noise, and dangerous torsional oscillations in the system.

You can easily visualize an involute curve by taking a cylinder and wrapping a string around its circumference. Tie a pencil to the end of the string. Then start with the pencil tight against the cylinder, and hold the string taut. Move the pencil away from the cylinder while keeping the string taut. The curve that you will draw is an involute. Figure 8-6 is a sketch of the process.

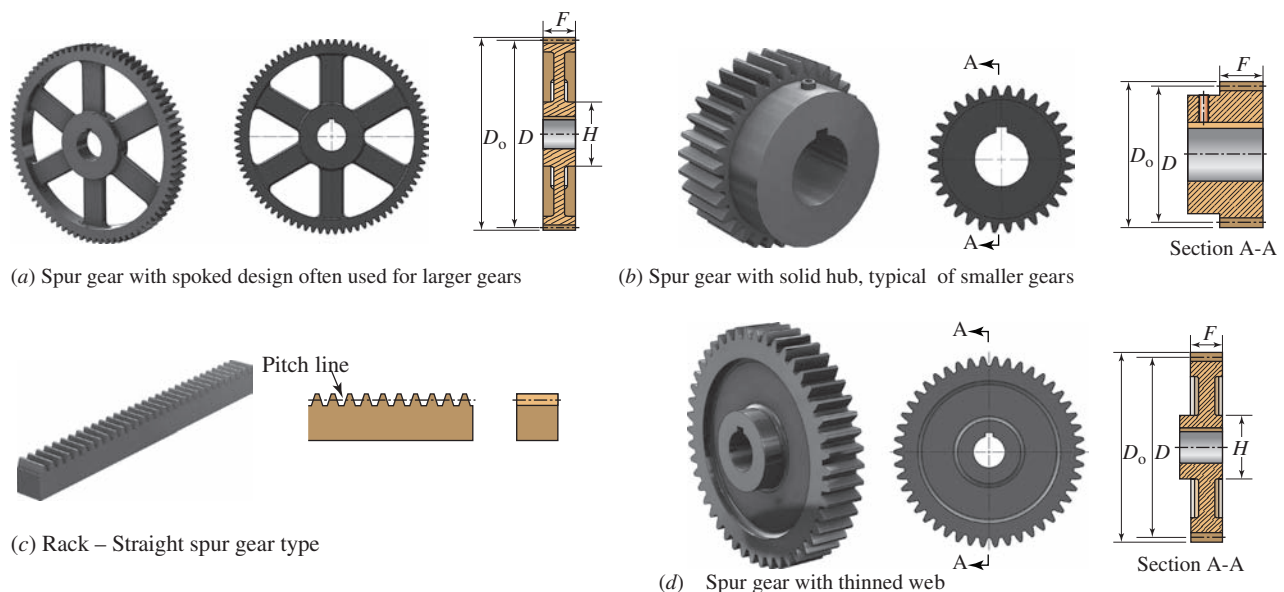


FIGURE 8-4 Examples of spur gears and a rack

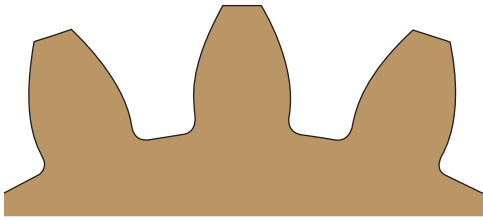


FIGURE 8-5 Involute-tooth form

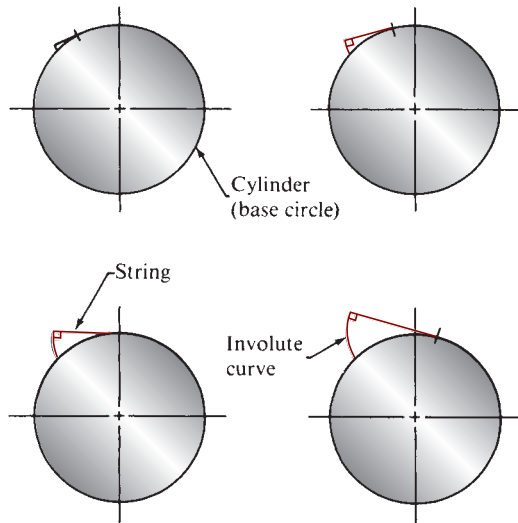


FIGURE 8-6 Graphical presentation of an involute curve

The circle, the end view of the cylinder, is called the *base circle*. Notice that at any position on the curve, the string represents a line tangent to the base circle and, at

the same time, perpendicular to the involute. Drawing another base circle along the same centerline in such a position that the resulting involute is tangent to the first one, as shown in Figure 8-7, demonstrates that at the point of contact, the two lines tangent to the base circles are coincident and will stay in the same position as the base circles rotate. This is what happens when two gear teeth are in mesh.

It is a fundamental principle of *kinematics*, the study of motion, that if the line drawn perpendicular to the surfaces of two rotating bodies at their point of contact always crosses the centerline between the two bodies at the same place, the angular velocity ratio of the two bodies will be constant. This is a statement of the *law of gearing*. As demonstrated here, the gear teeth made in the involute-tooth form obey the law.

Of course, only the part of the gear tooth that actually comes into contact with the mating tooth needs to be in the involute form.

Figure 8-8 shows the gear tooth involute profile. The tooth profile is the curve forming the side of the tooth and is bounded by the major diameter and minor diameters of the gear. The active profile is the portion of the gear tooth involute curve that makes contact with the mating gear. The *start of the active profile* (SAP) corresponds to the lowest point on the active profile. The *true involute form* (TIF) diameter is the diameter above which the tooth profile is a true involute, but the mating gear teeth do not contact. The tip chamfer is a tooth modification used to eliminate the sharp edge of the gear tooth flank and the major diameter of the gear.

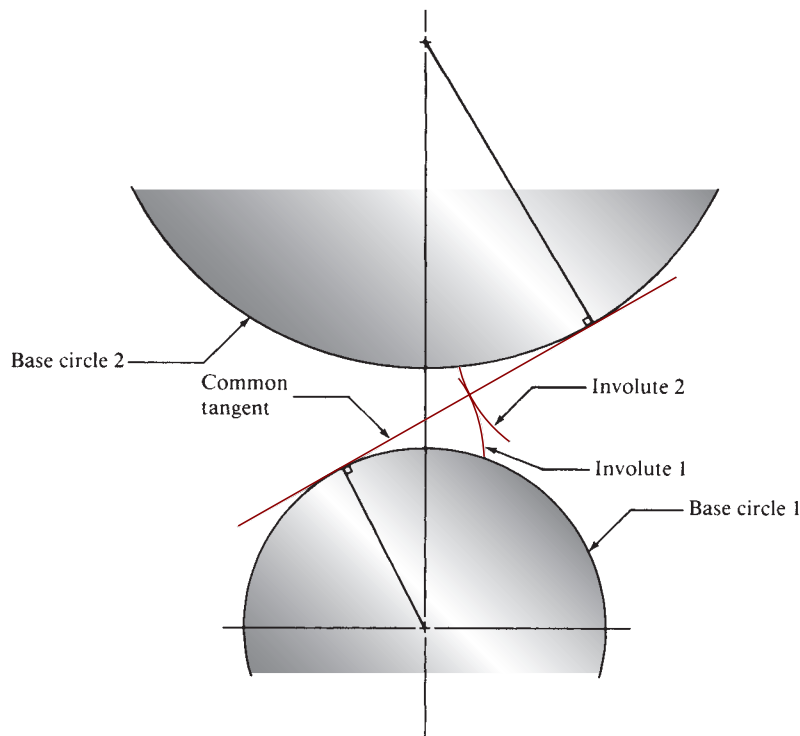


FIGURE 8-7 Mating involutes

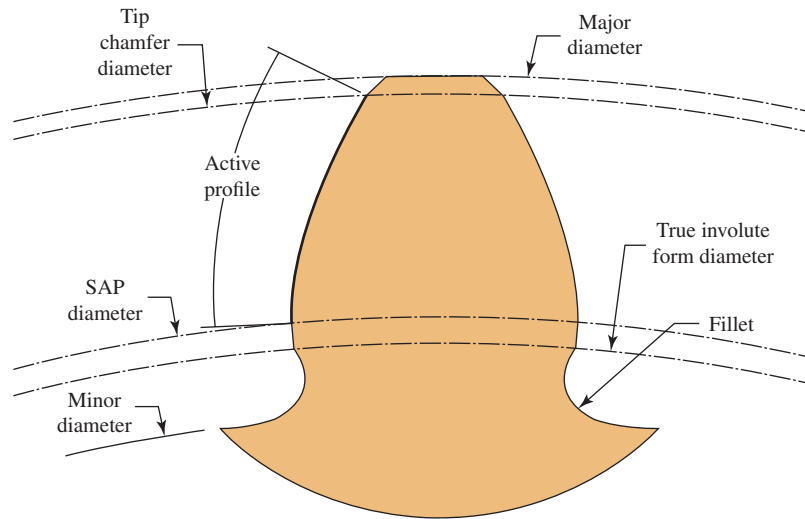


FIGURE 8-8 Gear tooth profile

8-4 SPUR GEAR NOMENCLATURE AND GEAR-TOOTH FEATURES

This section describes several features of individual spur gear teeth, complete gears, and the basic geometry of two mating gears. Terms and symbols used here conform mostly to American Gear Manufacturers Association (AGMA) standards. Because there is variation among the several applicable standards, the primary reference is AGMA 2001-D04 *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*. This standard is the basis for analytical design methods that are described in Chapters 9 and 10 for spur gears and helical gears, respectively. Where appropriate, the terms and symbols used by other AGMA standards and international standards such as ISO, DIN (Germany), and JIS (Japan) are noted. Both the conventional U.S. system of units, called the *Diametral Pitch*

System, and the SI metric system, called the *Metric Module System*, are discussed.

Reference is made to several figures and tables that depict the geometry of interest in the design of gear pairs:

1. Figure 8-1 shows two mating spur gears, indicating the dimensions related to diameters and center distance.
2. Figure 8-9 shows details of spur gear teeth with the many terms used to denote specific parts of the teeth and their relationship with the pitch diameter. These terms are defined later in this section.
3. Figure 8-10 shows two gears in mesh with several important diameters, center distance, and other features. See also Internet sites 7 and 8 for animations of teeth engagement.
4. Figure 8-11 shows how spur gear teeth engage as the gears rotate. Gear 1 rotates clockwise and drives gear 2 that rotates counterclockwise. The teeth on gear 1, labeled A_1 , B_1 , C_1 , and D_1 , contact the teeth on gear 2, labeled A_2 , B_2 , C_2 , and D_2 respectively. The contact between any two teeth remains along the line of action, until the teeth are no longer engaged.
5. Figures 8-12 and 8-13 show various sizes of gear teeth in both the diametral pitch and metric module systems. Both figures are full size, enabling you to compare physical gears to the drawings to gain an appreciation of gear tooth sizes.
6. Table 8-1 is a composite reference tool for identifying the names, symbols, definitions, units, and formulas related to the several features of gear teeth and mating gears.

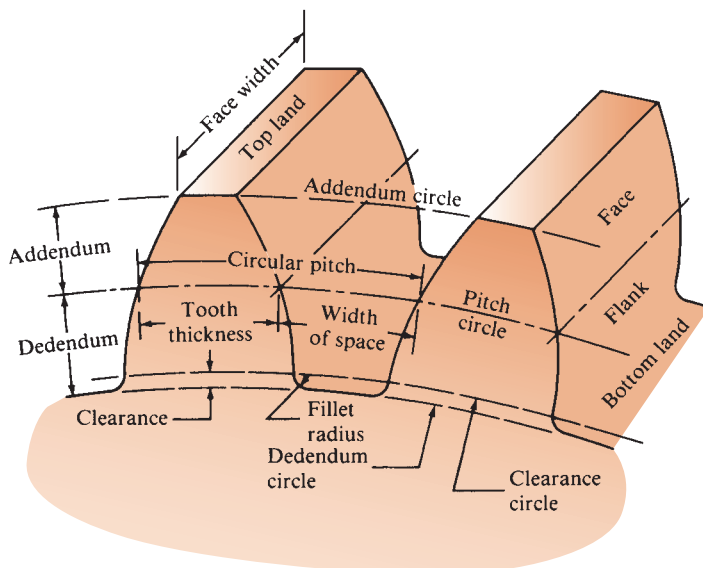


FIGURE 8-9 Spur gear teeth features

A note about accuracy: Gears and gear trains are precision mechanical devices with tolerances on critical dimensions typically in the range of a few ten thousandths of an inch (0.0001 in or about 0.0025 mm). Therefore, it is expected that such dimensions be

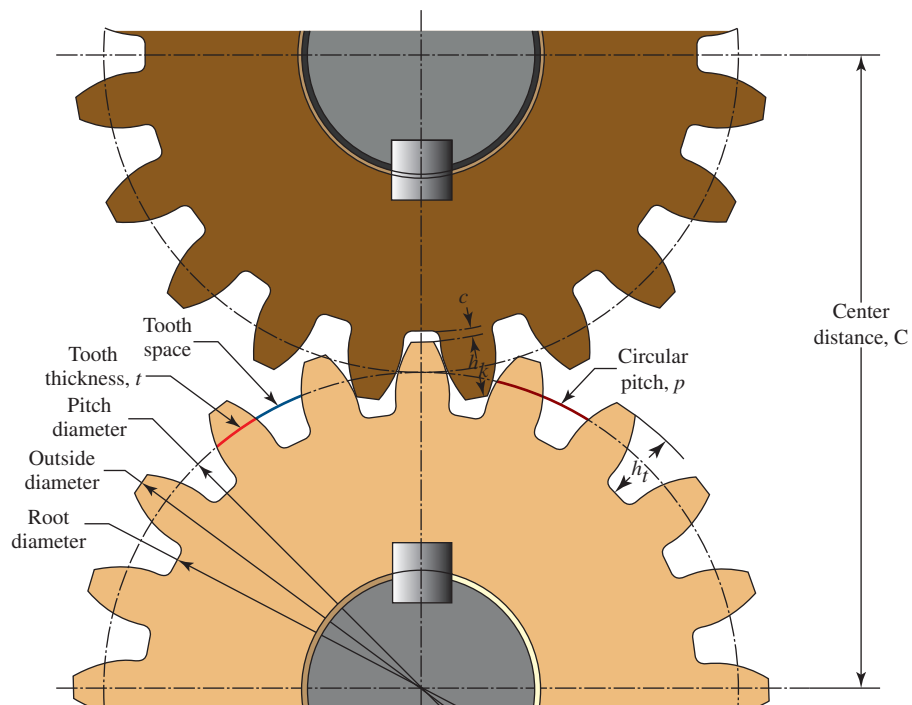
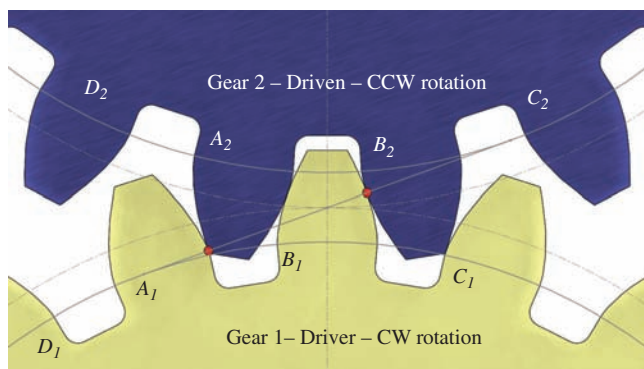
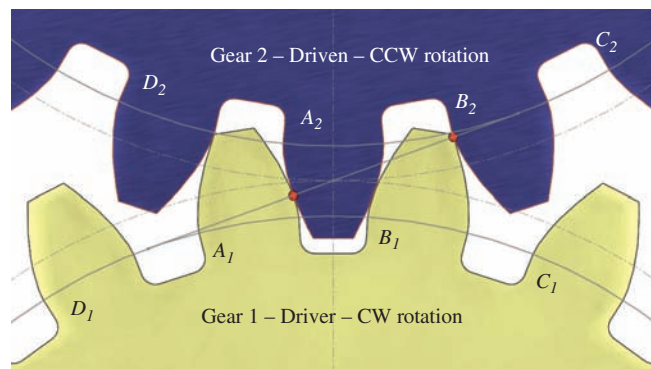


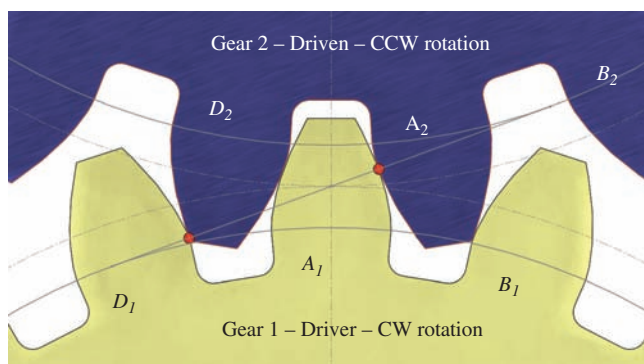
FIGURE 8-10 Details of two meshing spur gears showing several important geometric features.



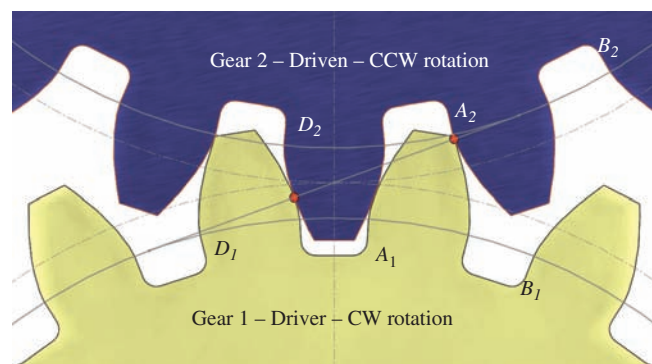
(a) Tooth A_1 on driver beginning engagement with tooth A_2 on driven gear; B_1 fully engaged with B_2 ; C_1 not engaged.



(b) Tooth A_1 fully engaged with A_2 ; B_1 leaving engagement with B_2



(c) Tooth A_1 fully engaged with A_2 ; D_1 beginning engagement with D_2



(d) Tooth D_1 fully engaged with D_2 ; A_1 leaving engagement with A_2

FIGURE 8-11 Cycle of engagement of gear teeth

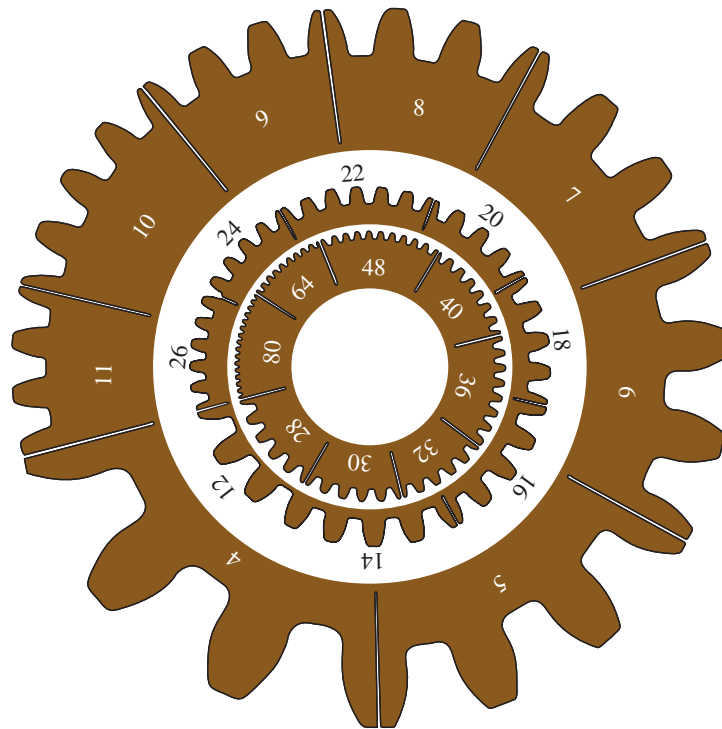


FIGURE 8-12 Gear-tooth size as a function of diametral pitch—actual size

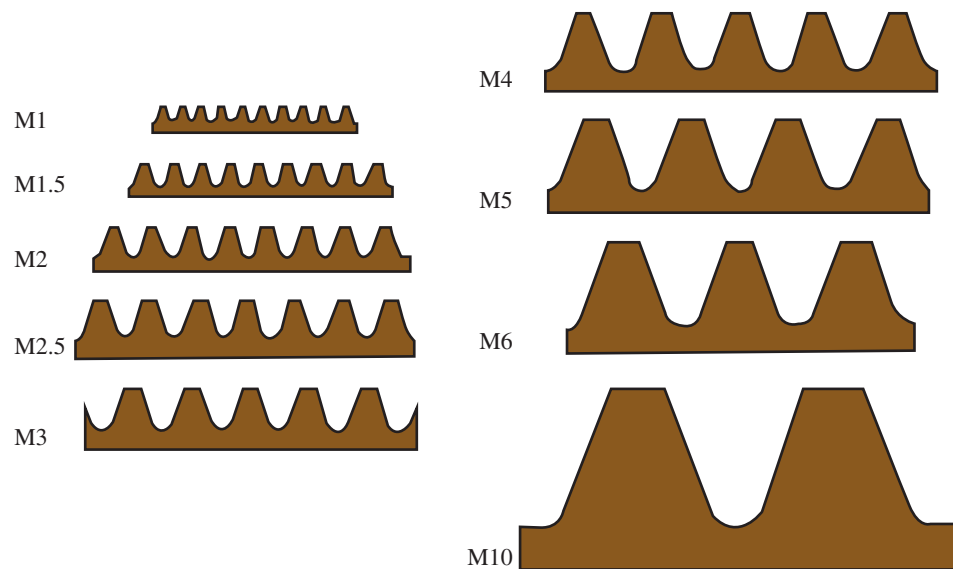


FIGURE 8-13 Selected standard metric modules in rack form—actual size

TABLE 8-1 Gear and Tooth Features, Diameters, Center Distance for a Gear Pair

Formulas						
Number of teeth and Pitches	Symbol	Definition	Typical unit	General formula	U.S. Full-depth involute system	
					Coarse pitch $P_d < 20$ (in)	Fine pitch $P_d \geq 20$ (in)
Number of teeth	N	Integer count of teeth on a gear				
Circular pitch	p	Arc distance between corresponding points on adjacent teeth	in or mm	$p = \pi D/N$		$p = \pi/P_d$
Diametral pitch	P_d	Number of teeth per inch of pitch diameter	in^{-1}	$P_d = N/D$		
Module	m	Pitch diameter divided by number of teeth	mm	$m = D/N$		$m = 25.4/P_d$
Diameters						
Pitch diameter	D	Kinematic characteristic diameter for a gear; Diameter of the pitch circle	in or mm	$D = N/P_d$		$D = mN$
Outside diameter	D_o	Diameter to the outside surface of the gear teeth	in or mm	$D_o = (N + 2)/P_d$		$D_o = m(N + 2)$
Root diameter	D_R	Diameter to the root circle of the gear at the base of the teeth	in or mm	$D_R = D - 2b$		
Gear Tooth Features						
Addendum	a	Radial distance from pitch circle to outside of tooth	in or mm	$a = 1.00/P_d$		$a = 1.00m$
Dedendum	b	Radial distance from pitch circle to bottom of tooth space	in or mm	$b = 1.25/P_d$	$b = 1.20/P_d + 0.002$	$b = 1.25m^1$
Clearance	c	Radial distance from top of fully engaged tooth of mating gear to bottom of tooth space	in or mm	$c = 0.25/P_d$	$c = 0.20/P_d + 0.002$	$c = 0.25m^1$
Whole depth	h_t	Radial distance from top of a tooth to bottom of tooth space	in or mm	$h_t = a + b$	$h_t = 2.20/P_d + 0.002$	$h_t = 2.25m^1$
Working depth	h_k	Radial distance a gear tooth projects into tooth space of mating gear	in or mm	$h_k = a + a = 2a$	$h_k = 2.00/P_d$	$h_k = 2.00m^1$
Tooth thickness	t	Theoretical arc distance equal to 1/2 of circular pitch	in or mm	$t = p/2$	$t = \pi/[2(P_d)]$	$t = \pi m/2$
Face width	F	Width of tooth parallel to axis of gear	in or mm	<i>Design decision</i>	Approximately $12/P_d$	
Pressure angle	ϕ	Angle between the tangent to the pitch circle and the perpendicular to the gear tooth surface	degrees	<i>Design decision</i>	Most common value = 20° Others: $14\ 1/2^\circ$, 25°	
Center Distance	C	Distance from between centerlines of mating gears	in or mm	$C = (D_P + D_G)/2$	$C = (N_P + N_G)/2P_d$	$C = m(N_P + N_G)/2$

Note: ¹Factors in formula for dedendum may vary in metric module system to obtain custom clearance.

reported to at least the nearest ten thousandth of an inch (four decimal places) or the nearest 0.001 mm. Some applications require even more precision. See References 8 and 9 for more on accuracy of gearing. Additional discussion about accuracy is included later in Section 8–12 on Gear Quality.

Pinion and Gear. For two gears in mesh, the smaller gear is called the *pinion* and the larger is called, simply, the *gear*.

Number of Teeth, N . It is essential that there are an integer number of teeth in any gear. This text uses the symbol N for the number of teeth, with N_p for the pinion and N_G for the gear. These subscripts are applied to other gear features as well. Another commonly used symbol for the number of teeth is z , with similar subscripts or simply called z_1 and z_2 .

Pitch. Refer to Figures 8–9 and 8–10. The *pitch* of a gear, in general, is defined as follows:

The pitch of a gear is the arc distance from a point on a tooth at the pitch circle to the corresponding point on the next adjacent tooth, measured along the pitch circle.

The pitch circle is defined next. It is important to note that the pitch of both mating gears must be identical to ensure the smooth engagement of the teeth as the gears rotate. Standard pitches are defined in three different systems, described later in this section.

Pitch Circle and Pitch Diameter. When two gears are in mesh, they behave as if two smooth rollers are rolling on each other without slipping. The surface of each roller defines the *pitch circle* and its diameter is called the *pitch diameter*. The *pitch diameter*, called D in this book, is used as the characteristic size of the gear for calculations of speeds. Note that the pitch diameter for a gear is a theoretical concept and cannot be measured directly. It falls within the gear teeth and is dependent on which standard system for pitch is specified for a particular gear pair. The commonly used units for D are inches (in) for the U.S. system and millimeters (mm) for the SI metric system. Figures 8–1, 8–10, and 8–15 show the pitch diameters on meshing gears.

TABLE 8–2 Standard Circular Pitches (in)

10.0	7.5	5.0
9.5	7.0	4.5
9.0	6.5	4.0
8.5	6.0	3.5
8.0	5.5	

Circular Pitch, p . The pitch corresponding exactly to the basic definition of pitch given above is called the *circular pitch*, p . Some large gears that are made by casting are made to standard sizes of circular pitch such as those listed in Table 8–2. They represent a very small portion of gears in common use. The formula for p comes from dividing the circumference of the pitch circle of the gear into N parts. That is,

✦ **Circular Pitch**

$$p = \pi D / N \quad (8-2)$$

Diametral Pitch, P_d . The most common pitch system in use in the United States at this time is *diametral pitch system*. We use the symbol, P_d , to denote *diametral pitch*. Note that some references use the term DP . The definition of P_d is stated here for either the pinion or the gear and both must be identical.

✦ **Diametral Pitch**

$$P_d = N_p / D_p = N_G / D_G \quad (8-3)$$

Analysis of units shows that P_d has the unit of in^{-1} , but the unit is rarely reported. It is necessary to not confuse the terms *diametral pitch*, P_d , and *pitch diameter*, D . Note that designers often refer to gears in this system as, for example, 8-pitch for $P_d = 8$ and 20-pitch for $P_d = 20$.

In this book, we use only those values of P_d listed in Table 8–3 because they are the most readily available as stock gears and most gear manufacturers have tooling for these sizes. Smaller pitches have larger teeth; larger pitches have smaller teeth. Note that pitches under 20 are called *coarse*, while those 20 and higher are called

TABLE 8–3 Standard Diametral Pitches (teeth/in)

Coarse pitch ($P_d < 20$)				Fine pitch ($P_d \geq 20$)	
1	2	5	12	20	72
1.25	2.5	6	14	24	80
1.5	3	8	16	32	96
1.75	4	10	18	48	120
				64	

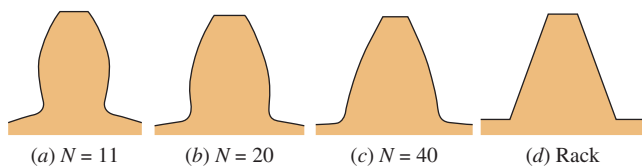


FIGURE 8-14 Involute curve shape for varying numbers of teeth for a diametral pitch of 5

fine. Refer to Figure 8-13 that shows actual sizes of teeth with certain diametral pitches. **Note that not all listed values of P_d are readily available and those such as 7, 9, 11, 22, and 26 should not be specified.**

Number of Teeth Related to Pitch Diameter

The relationship between the pitch diameter, diametral pitch, and the number of teeth is given by Equation 8-3. As the number of teeth increases for a given diametral pitch, the pitch diameter increases. So as the pitch diameter increases, the base circle diameter increases. A larger base circle will decrease the curvature of the involute curve of the gear tooth. Figure 8-14 shows as the number of teeth increases for a given diametral pitch, the radius of the involute curve of the gear tooth will decrease. Inversely, as the number of teeth decrease the radius of the involute curve of the gear tooth will increase. The teeth in this figure are all drawn with a diametral pitch of 5. The gear rack can be thought of as a gear with an infinitely large diameter, where the involute curve of the tooth profile becomes a straight line.

Metric Module, m . The basic definition of the metric module, m , is given here for both the pinion and the gear and they must be identical.

➤ Metric Module

$$m = D_p/N_p = D_G/N_G \quad (8-4)$$

The unit of mm is typically used. Note that smaller values of m denote smaller teeth and vice versa. Some references and vendors' tables use the symbol, M, for module and write M5 for $m = 5$, for example. Figure 8-13 shows actual sizes of nine standard modules, illustrated as the teeth of racks, straight gears with infinite diameters. Many more modules are available and Table 8-4 lists a total of 19 values that cover typical applications featured in this book.

Relation between P_d and m . With globally integrated design and marketing of products and systems, it is likely that the need to convert from one system to another will be needed. Note that the definition of P_d is fundamentally the inverse of the definition of m . That is,

$$m = 1/P_d \quad \text{or} \quad P_d = 1/m$$

However, because different units are employed for each term, a conversion factor of 25.4 mm/in is required, resulting in useful forms of the relationship as

TABLE 8-4 Standard modules

Module (mm)	Equivalent P_d	Closest standard P_d (teeth/in)
0.3	84.667	80
0.4	63.500	64
0.5	50.800	48
0.8	31.750	32
1	25.400	24
1.25	20.320	20
1.5	16.933	16
2	12.700	12
2.5	10.160	10
3	8.466	8
4	6.350	6
5	5.080	5
6	4.233	4
8	3.175	3
10	2.540	2.5
12	2.117	2
16	1.587	1.5
20	1.270	1.25
25	1.016	1

➤ Relation between Module and Diametral Pitch

$$m = 25.4/P_d \quad \text{or} \quad P_d = 25.4/m \quad (8-5)$$

Table 8-4 uses this relationship to compute the equivalent diametral pitch, P_d , for given standard metric modules, m . Note that the conversions do not deliver standard values of P_d such as those listed in Table 8-3. Therefore, we list the nearest standard P_d value in Table 8-4 as an aid to designers who are considering converting a design from one system to the other. We can say, for example, that a module of $m = 1.25$ is a closely similar size to $P_d = 20$. Of course, the design for either system must be completed independently.

Gear Tooth Features. Table 8-1 includes the definitions of several other features of individual teeth that designers must be familiar with. Refer to Figures 8-9 and 8-10 to visualize those features.

Face Width, F . Face width is the width of the gear parallel to the axis of the gear. It is defined by the designer as one of the required *design decisions*. More is said about face width in Chapter 9, where strength of the teeth is considered. For now, we can state that a nominal value for face width is approximately $F \sim 12/P_d$, but a wide range is permitted.

Center Distance, C . One of the most critical dimensions for a gear pair is the *center distance*, defined as the linear distance from the centerline of the pinion to the centerline of the gear as shown in Figure 8–1. The theoretical value is best represented as the sum of the pitch radii of the pinion and the gear. That is,

$$\begin{aligned}\text{Pitch radii:} \quad R_P &= D_P/2 \quad \text{and} \quad R_G = D_G/2 \\ \text{Center distance:} \quad C &= R_P + R_G = D_P/2 + D_G/2 \\ C &= (D_P + D_G)/2\end{aligned}\quad (8-6)$$

Other useful equations for C , recommended for use in this book, are developed here.

Diametral pitch system: From Equation (8–3),

⇒ **Center Distance in terms of N_G , N_P , and P_d**

$$\begin{aligned}D_P &= N_P/P_d \quad \text{and} \quad D_G = N_G/P_d \\ C &= (D_P + D_G)/2 = (N_P/P_d + N_G/P_d)/2 \\ C &= (N_P + N_G)/2P_d\end{aligned}\quad (8-7)$$

Metric module system: From Equation (8–4),

⇒ **Center Distance in terms of N_G , N_P , and m**

$$\begin{aligned}D_P &= mN_P \quad \text{and} \quad D_G = mN_G \\ C &= (D_P + D_G)/2 = (mN_P + mN_G)/2 \\ C &= m(N_P + N_G)/2\end{aligned}\quad (8-8)$$

An important advantage of the final forms of the center distance Equation (8–7) and (8–8) is that all numbers on the right side are typically integers or exact fractional dimensions such as 1.25, 1.5, or 2.5. Therefore, the highest level of accuracy is obtained from using those forms. Conversely, some values for pitch diameters are irrational numbers. For example, a 12-pitch gear with 65 teeth has a pitch diameter of

$$\begin{aligned}D &= N/P_d = 65/12 = 5.416666 \dots \text{ in} \quad \text{and} \\ R &= D/2 = 2.708333 \dots \text{ in}\end{aligned}$$

Depending on rounding, an inaccurate calculation for center distance could result from using these values.

Comments on Gear Tooth Features. The definitions and formulas given in Table 8–1 yield the theoretical values and it is typical for gear designers and manufacturers to modify some features to produce preferred performance characteristics. It is important to realize that some of these practices change the fundamental geometry of the gears and they may result in weaker tooth shapes, vibration during operation, and/or increased noise. Some examples are given below:

1. **Addendum modification:** The upper part of the flank of a gear tooth is the first to engage its mating tooth and it penetrates most deeply into the tooth space of the mating gear. Some applications benefit from relieving the true surface of the tooth or

from shortening the addendum to promote smooth engagement or to avoid damage.

2. **Dedendum modification:** Some designers prefer greater clearance between the bottom of the tooth space and the fully engaged mating tooth to facilitate lubrication. Using a longer dedendum dimension provides this capability. Plastic gears may also employ dedendum modification to accommodate thermal expansion and/or swelling because of moisture absorption.
3. **Center distance modification:** Some combinations of gear tooth geometry result in very small clearances between teeth and possibly interference between the top of the engaging tooth and the lower part of the flank of the mating tooth. These conditions should be avoided by proper design decisions to avoid the interference and this topic is discussed in Chapter 9. However, some designers remove the interference by expanding the center distance slightly from its theoretical dimension.
4. **Tooth thickness modification:** A tooth thickness defined exactly as $\frac{1}{2}$ of the circular pitch will result in a tight fit of a tooth in the tooth space of the mating gear, possibly causing binding or precluding entry of lubrication at the contact point on the teeth. Using a smaller tooth thickness provides space for lubrication and facilitates assembly. The space created is called *backlash* and it is discussed next.

- **Backlash:** To provide backlash, the cutter generating the gear teeth can be fed more deeply into the gear blank than the theoretical value on either or both of the mating gears. Alternatively, backlash can be created by adjusting the center distance to a larger value than the theoretical value.

The magnitude of backlash depends on the desired precision of the gear pair and on the size and the pitch of the gears. It is actually a design decision, balancing cost of production with desired performance. The American Gear Manufacturers Association (AGMA) provides recommendations for backlash in their standards. (See Reference 2.) Table 8–5 lists examples of recommended ranges for several values of pitch.

Pressure Angle

The pressure angle is the angle between the tangent to the pitch circles and the line drawn normal (perpendicular) to the surface of the gear tooth (see Figure 8–15).

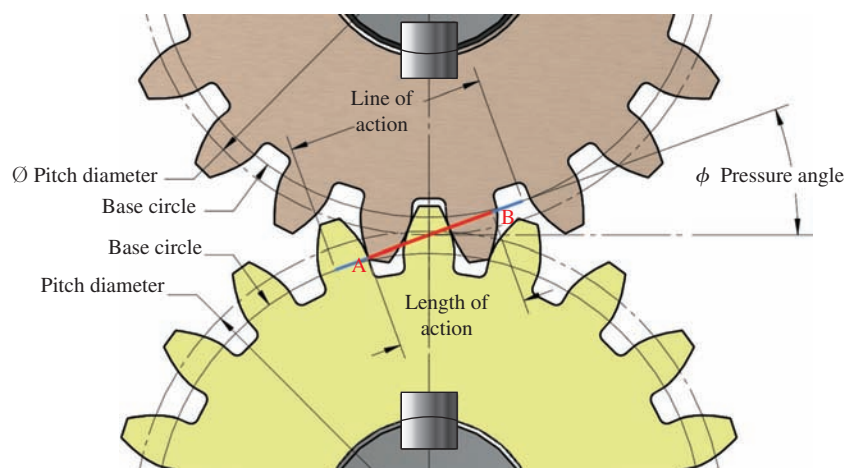
The normal line is sometimes referred to as the *line of action*. When two gear teeth are in mesh and are transmitting power, the force transferred from the driver to the driven gear tooth acts in a direction along the line of action. The *length of action* is the distance AB along

TABLE 8-5 Recommended Minimum Backlash for Coarse Pitch Gears**A. Diametral pitch system (backlash in inches)**

P_d	Center distance, C (in)				
	2	4	8	16	32
18	0.005	0.006			
12	0.006	0.007	0.009		
8	0.007	0.008	0.010	0.014	
5		0.010	0.012	0.016	
3		0.014	0.016	0.020	0.028
2			0.021	0.025	0.033
1.25				0.034	0.042

B. Metric module system (backlash in millimeters)

Module, m	Center distance, C (mm)				
	50	100	200	400	800
1.5	0.13	0.16			
2	0.14	0.17	0.22		
3	0.18	0.20	0.25	0.35	
5		0.26	0.31	0.41	
8		0.35	0.40	0.50	0.70
12			0.52	0.62	0.82
18				0.80	1.00

**FIGURE 8-15** Two spur gears in mesh showing the pressure angle, line of action, base circles, pitch diameters, and other features

the line of action. This length represents the path of the point of contact during the meshing of the two gears.

Also, the actual shape of the gear tooth depends on the pressure angle, as illustrated in Figure 8-16. The teeth in this figure were drawn according to the proportions for a 20-tooth, 5-pitch gear having a pitch diameter of 4.000 in.

All three teeth have the same tooth thickness because, as stated in Table 8-1 the thickness at the

pitch line depends only on the pitch. The difference between the three teeth shown is due to the different pressure angles because the pressure angle determines the size of the base circle. Remember that the base circle is the circle from which the involute is generated. The line of action is always tangent to the base circle. Therefore, the size of the base circle can be found from

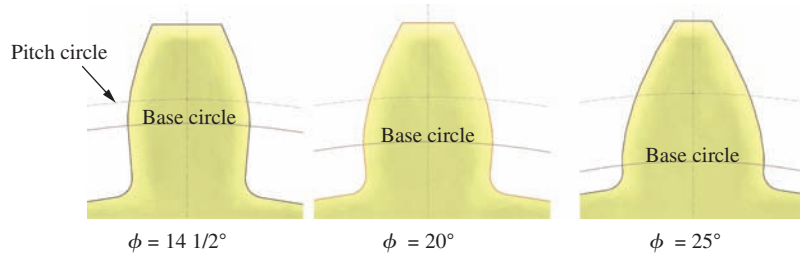


FIGURE 8-16 Illustration of how the shape of gear teeth change as the pressure angle, (ϕ), changes

Base Circle Diameter

$$D_b = D \cos \phi \quad (8-9)$$

As the pressure angle increases, the thickness at the bottom of the tooth increases. Therefore the larger the pressure angle, the stronger the tooth and the higher the load carrying capacity of the tooth. A higher pressure angle may not run as smoothly or quietly as the smaller pressure angle. As the pressure angle decreases, the top land of the tooth will increase. This can create interference problems when a small number of pinion teeth are required. The topic of gear interference will be covered in more detail in Section 8-5.

Standard values of the pressure angle are established by gear manufacturers, and the pressure angles of two gears in mesh must be the same. Current standard pressure angles are $14\frac{1}{2}^\circ$, 20° , and 25° as illustrated in Figure 8-16. Actually, the $14\frac{1}{2}^\circ$ tooth form is considered obsolete. Although it is still available, it should be avoided for new designs. The 20° tooth form is the most readily available at this time. The advantages and disadvantages of the different values of pressure angle relate to the strength of the teeth, the occurrence of interference, and the magnitude of forces exerted on the shaft. Interference is discussed in Section 8-5. The other points are discussed in a later chapter.

Contact Ratio

When two gears mesh, it is essential for smooth operation that a second tooth begin to make contact before a given tooth disengages. The term *contact ratio* is used to indicate the average number of teeth in contact during the transmission of power. A recommended minimum contact ratio is 1.2 and typical spur gear combinations often have values of 1.5 or higher.

The contact ratio is defined as the ratio of the length of the line-of-action to the base pitch for the gear. The line-of-action is the straight-line path of a tooth from where it encounters the outside diameter of the mating gear to the point where it leaves engagement. The base pitch is the diameter of the base circle divided by the number of teeth in the gear. A convenient formula for computing the contact ratio, m_f , is,

Contact Ratio

$$m_f = \frac{\sqrt{R_{oP}^2 - R_{bP}^2} + \sqrt{R_{oG}^2 - R_{bG}^2} - C \sin \phi}{p \cos \phi} \quad (8-10)$$

where,

ϕ = Pressure angle

$$R_{oP} = \text{Outside radius of the pinion} = D_{oP}/2 \\ = (N_P + 2)/(2P_d)$$

$$R_{bP} = \text{Radius of the base circle for the pinion} \\ = D_{bP}/2 = (D_P/2) \cos \phi = (N_P/2P_d) \cos \phi$$

$$R_{oG} = \text{Outside radius of the gear} = D_{oG}/2 \\ = (N_G + 2)/(2P_d)$$

$$R_{bG} = \text{Radius of the base circle for the gear} = D_{bG}/2 \\ = (D_G/2) \cos \phi = (N_G/2P_d) \cos \phi$$

$$C = \text{Center distance} = (N_P + N_G)/(2P_d)$$

$$p = \text{Circular pitch} = (\pi D_P/N_P) = \pi/P_d$$

For example, consider a pair of gears with the following data:

$$N_P = 18, N_G = 64, P_d = 8, \phi = 20^\circ$$

Then,

$$R_{oP} = (N_P + 2)/(2P_d) = (18 + 2)/[2(8)] = 1.2500 \text{ in}$$

$$R_{bP} = (N_P/2P_d) \cos \phi = 18/[2(8)] \cos 20^\circ = 1.05715 \text{ in}$$

$$R_{oG} = (N_G + 2)/(2P_d) = (64 + 2)/[2(8)] = 4.1250 \text{ in}$$

$$R_{bG} = (N_G/2P_d) \cos \phi = 64/[2(8)] \cos 20^\circ = 3.75877 \text{ in}$$

$$C = (N_P + N_G)/(2P_d) = (18 + 64)/[2(8)] = 5.1250 \text{ in}$$

$$p = \pi/P_d = \pi/8 = 0.392699 \text{ in}$$

Finally, the contact ratio is,

$$m_f = \frac{\sqrt{(1.250)^2 - (1.05715)^2} + \sqrt{(4.125)^2 - (3.75877)^2} - (5.125) \sin 20^\circ}{(0.392699) \cos 20^\circ}$$

$$m_f = 1.66$$

This states that for these two gears in mesh, there will always be one tooth in full contact and for 66% of the time there will be two teeth in contact. The value of $m_f = 1.66$ is comfortably above the recommended minimum value of 1.20.

Similar developments can be used to determine the factors needed to implement the contact ratio calculation in Equation (8-10) with terms expressed in the Metric Module System. Table 8-6 summarizes the relationships for both systems.

TABLE 8–6 Formulas for Use When Implementing Gear Pair Contact Ratio Calculation In U.S. and SI Systems in Terms of Diametral Pitch and Module

Factors		Diametral Pitch System	Metric Module System
ϕ	Pressure angle		
R_{oP}	Outside radius—pinion	$(N_p + 2)/(2P_d)$	$m(N_p + 2)/2$
R_{bP}	Base circle radius—pinion	$(N_p/2P_d) \cos \phi$	$(mN_p/2) \cos \phi$
R_{oG}	Outside radius—gear	$(N_G + 2)/(2P_d)$	$m(N_G + 2)/2$
R_{bG}	Base circle radius—gear	$(N_G/2P_d) \cos \phi$	$(mN_G/2) \cos \phi$
C	Center distance	$(N_p + N_G)/(2P_d)$	$m(N_p + N_G)/2$
p	Circular pitch	π/P_d	πm

Example Problem 8–1

For the pair of gears shown in Figure 8–1, compute all of the features of the gear teeth described in this section. The gears conform to the standard AGMA form and have a diametral pitch of 12 and a 20° pressure angle.

Solution

Given $P_d = 12$; $N_p = 11$; $N_G = 18$; $\phi = 20^\circ$.

Analysis Unless otherwise noted, we use equations from Table 8–1 to compute the features. Refer to the text in this section for explanation of terms.

Note that results are reported to four decimal places as is typical for precise mechanical devices like gears. A similar level of accuracy is expected for problems in this book.

Results

Pitch Diameters

For the pinion,

$$D_p = N_p/P_d = 11/12 = 0.9167 \text{ in}$$

For the gear,

$$D_G = N_G/P_d = 18/12 = 1.5000 \text{ in}$$

Circular Pitch

Three different approaches could be used.

$$p = \pi/P_d = \pi/12 = 0.2618 \text{ in}$$

Note that data for either the pinion or the gear data may also be used. For the pinion,

$$p = \pi D_p/N_p = \pi(0.9167 \text{ in})/11 = 0.2618 \text{ in}$$

For the gear,

$$p = \pi D_G/N_G = \pi(1.500 \text{ in})/18 = 0.2618 \text{ in}$$

Addendum

$$a = 1/P_d = 1/12 = 0.8333 \text{ in}$$

Dedendum

Note that the 12-pitch gear is considered to be coarse. Thus,

$$b = 1.25/P_d = 1.25/12 = 0.1042 \text{ in}$$

Clearance

$$c = 0.25/P_d = 0.25/12 = 0.0208 \text{ in}$$

Outside Diameters

For the pinion,

$$D_{oP} = (N_p + 2)/P_d = (11 + 2)/12 = 1.0833 \text{ in}$$

For the gear,

$$D_{oG} = (N_G + 2)/P_d = (18 + 2)/12 = 1.6667 \text{ in}$$

Root Diameters

First, for the pinion,

$$D_{RP} = D_p - 2b = 0.9167 \text{ in} - 2(0.1042 \text{ in}) = 0.7083 \text{ in}$$

For the gear,

$$D_{RG} = D_G - 2b = 1.500 \text{ in} - 2(0.1042 \text{ in}) = 1.2917 \text{ in}$$

Whole Depth

$$h_t = a + b = 0.0833 \text{ in} + 0.1042 \text{ in} = 0.1875 \text{ in}$$

Working Depth

$$h_k = 2a = 2(0.0833 \text{ in}) = 0.1667 \text{ in}$$

Tooth Thickness

$$t = \pi/[2(P_d)] = \pi/[2(12)] = 0.1309 \text{ in}$$

Center Distance

$$C = (N_G + N_P)/(2P_d) = (18 + 11)/[2(12)] = 1.2083 \text{ in}$$

Base Circle Diameter

$$D_{bP} = D_P \cos \phi = (0.9167 \text{ in}) \cos (20^\circ) = 0.8614 \text{ in}$$

$$D_{bG} = D_G \cos \phi = (1.5000 \text{ in}) \cos (20^\circ) = 1.4095 \text{ in}$$

8-5 INTERFERENCE BETWEEN MATING SPUR GEAR TEETH

For certain combinations of numbers of teeth in a gear pair, there is interference between the tip of the teeth on the pinion and the fillet or root of the teeth on the gear. Obviously this cannot be tolerated because the gears simply will not mesh. The probability that interference will occur is greatest when a small pinion drives a large gear, with the worst case being a small pinion driving a rack. A *rack* is a gear with a straight pitch line; it can be thought of as a gear with an infinite pitch diameter [see Figure 8-4(c)].

It is the designer's responsibility to ensure that interference does not occur in a given application. The surest way to do this is to control the minimum number of teeth in the pinion to the limiting values shown on the left side of Table 8-7. With this number of teeth or a greater number, there will be no interference with a rack or with any other gear. A designer who desires to use fewer than the listed number of teeth can use a graphical layout to test the combination of pinion and gear for interference.

Texts on kinematics provide the necessary procedure. The right side of Table 8-7 indicates the maximum number of 20° full depth gear teeth that you can use for a given number of pinion teeth to avoid interference. (See References 11 and 17.)

Using the information in Table 8-7, we can draw the following conclusions:

1. If a designer wants to be sure that there will not be interference between any two gears when using the $14\frac{1}{2}^\circ$, full-depth, involute system, the pinion of the gear pair must have no fewer than 32 teeth.
2. For the 20° , full-depth, involute system, using no fewer than 18 teeth will ensure that no interference occurs.
3. For the 25° , full-depth, involute system, using no fewer than 12 teeth will ensure that no interference occurs.
4. If a designer desires to use fewer than 18 teeth in a pinion having 20° , full-depth teeth, there is an upper limit to the number of teeth that can be used on

TABLE 8-7 Number of Pinion Teeth to Ensure No Interference

For a pinion meshing with a rack		For a 20° , full-depth pinion meshing with a gear		
Tooth form	Minimum number of teeth	Number of pinion teeth	Maximum number of gear teeth	Maximum ratio
$14\frac{1}{2}^\circ$, involute, full-depth	32	17	1309	77.00
20° , involute, full-depth	18	16	101	6.31
25° , involute, full-depth	12	15	45	3.00
		14	26	1.85
		13	16	1.23

the mating gear without interference. For 17 teeth in the pinion, any number of teeth on the gear can be used up to 1309, a very high number. Most gear drive systems use no more than about 200 teeth in any gear. But a 17-tooth pinion *would* have interference with a *rack* which is effectively a gear with an infinite number of teeth or an infinite pitch diameter. Similarly, the following requirements apply for 20° full-depth teeth:

- A 16-tooth pinion requires a gear having 101 or fewer teeth, producing a maximum velocity ratio of $N_G/N_P = 101/16 = 6.31$.
- A 15-tooth pinion requires a gear having 45 or fewer teeth, producing a maximum velocity ratio of $45/15 = 3.00$.
- A 14-tooth pinion requires a gear having 26 or fewer teeth, producing a maximum velocity ratio of $26/14 = 1.85$.
- A 13-tooth pinion requires a gear having 16 or fewer teeth, producing a maximum velocity ratio of $16/13 = 1.23$.

As noted earlier, the 14½° system is considered to be obsolete. The data in Table 8–7 indicate one of the main disadvantages with that system: its potential for causing interference.

Overcoming Interference

If a proposed design encounters interference, there are ways to make it work. But caution should be exercised because the tooth form or the alignment of the mating gears is changed, causing the stress and wear analysis to be inaccurate. With this in mind, the designer can provide for undercutting, modification of the addendum on the pinion or the gear, or modification of the center distance:

Undercutting is the process of cutting away the material at the fillet or root of the gear teeth, thus relieving the interference.

Figure 8–17 shows the result of undercutting. It should be obvious that this process weakens the tooth; this point is discussed further in Chapter 9 in the section on stresses in gear teeth.

To alleviate the problem of interference, increase the addendum of the pinion while decreasing the addendum of the gear. The center distance can remain the same as its theoretical value for the number of teeth in the pair. But the resulting gears are, of course, non-standard. (See Reference 12.) It is possible to make the pinion of a gear pair larger than standard while keeping the gear standard if the center distance for the pair is enlarged. (See Reference 11.)

Also shown in Figure 8–17 is tip relief that modifies the top of the tooth profile to provide clearance when a gear tooth bends under load.

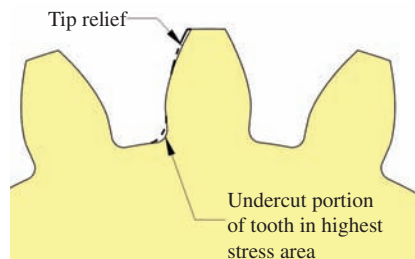


FIGURE 8–17 Undercutting and tip relief of gear teeth

8-6 INTERNAL GEAR GEOMETRY

An internal gear is one for which the teeth are machined on the inside of a ring instead of on the outside of a gear blank.

A small internal gear mating with a standard, external pinion is illustrated at the lower left in Figure 8–2, along with a variety of other kinds of gears.

Figure 8–18 is a drawing of an external pinion driving an internal gear. Note the following:

1. The gear rotates in the *same direction* as the pinion. This is different from the case when an external pinion drives an external gear.
2. The center distance is

Center Distance-Internal Gear

$$C = D_G/2 - D_P/2 = (D_G - D_P)/2$$

$$C = (N_G/P_d - N_P/P_d)/2 = (N_G - N_P)/(2P_d) \quad (8-10)$$

The last form is preferred because its factors are all integers for typical gear trains, giving maximum precision in calculations.

3. The descriptions of most other features of internal gears are the same as those for external gears presented earlier. Exceptions for an internal gear are as follows:

The addendum, a , is the radial distance from the pitch circle to the inside of a tooth.

The inside diameter, D_i , is

$$D_i = D - 2a$$

The root diameter, D_R , is

$$D_R = D + 2b$$

where b = dedendum.

Internal gears are used when it is desired to have the same direction of rotation for the input and the output. Also note that less space is taken for an internal gear mating with an external pinion compared with two external gears in mesh.

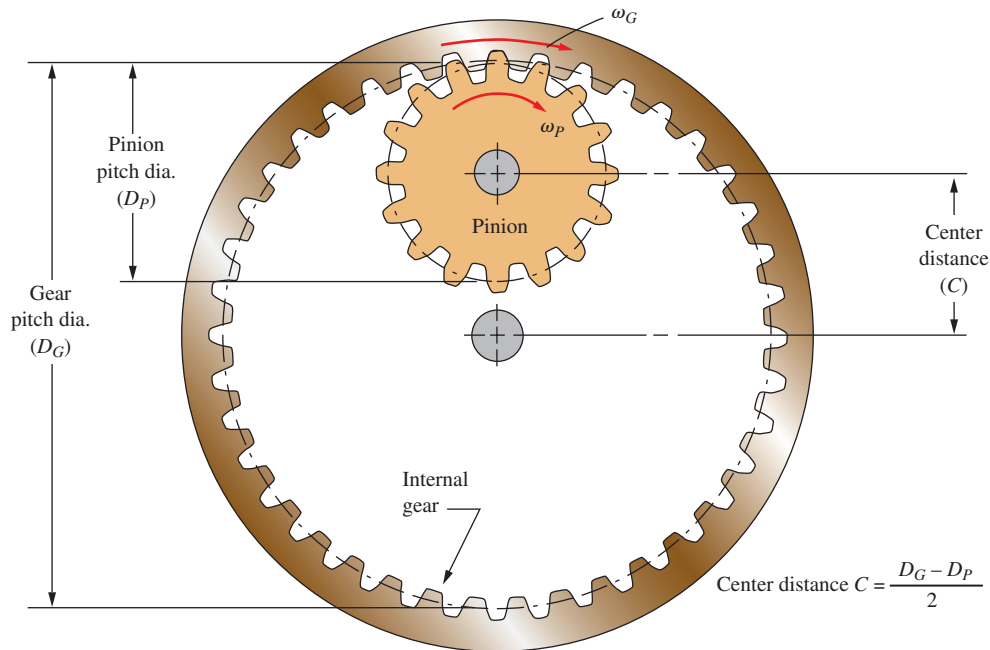


FIGURE 8-18 Internal gear driven by an external pinion

8-7 HELICAL GEAR GEOMETRY

Helical and spur gears are distinguished by the orientation of their teeth. On spur gears, the teeth are straight and are aligned with the axis of the gear. On helical gears, the teeth are inclined at an angle with the axis, that angle being called the *helix angle*. If the gear were very wide, it would appear that the teeth wind around the gear blank in a continuous, helical path. However, practical considerations limit the width of the gears so that the teeth normally appear to be merely inclined with respect to the axis. Figure 8-19 shows two examples of commercially available helical gears.

The forms of helical gear teeth are very similar to those discussed for spur gears. The basic task is to account for the effect of the helix angle.

Helix Angle

The helix for a given gear can be either *left-hand* or *right-hand*. The teeth of a right-hand helical gear would appear to lean to the right when the gear is lying on a flat surface. Conversely, the teeth of a left-hand helical gear would lean to the left. In normal installation, helical gears would be mounted on parallel shafts as shown in Figure 8-19(a). To achieve this arrangement, it is required that one gear be of the right-hand design and that the other be left-hand with an equal helix angle. If both gears in mesh are of the same hand, as shown in Figure 8-19(b), the shafts will be at 90° to each other. Such gears are called *crossed helical gears*.

The parallel shaft arrangement for helical gears is preferred because it results in a much higher powertransmitting capacity for a given size of gear than the crossed helical arrangement. In this book, we will assume that

the parallel shaft arrangement is being used unless otherwise stated.

The main advantage of helical gears over spur gears is smoother engagement because a given tooth assumes its load gradually instead of suddenly. Contact starts at one end of a tooth near the tip and progresses across the face in a path downward across the pitch line to the lower flank of the tooth, where it leaves engagement. Simultaneously, other teeth are coming into engagement before a given tooth leaves engagement, with the result that a larger average number of teeth are engaged and are sharing the applied loads compared with a spur gear. The lower average load per tooth allows a greater power transmission capacity for

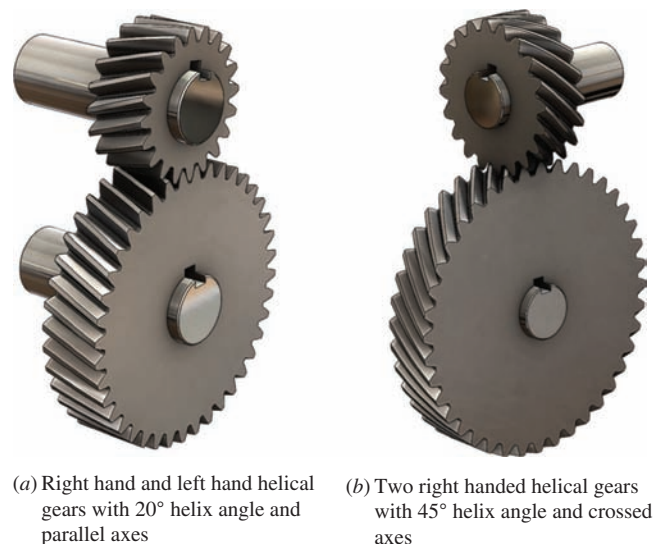


FIGURE 8-19 Two varieties of pairs of helical gears

a given size of gear, or a smaller gear can be designed to carry the same power. Helical gears also generate less noise and vibration than spur gears due to the smoother tooth engagement.

The main disadvantage of helical gears is that an *axial thrust load* is produced as a natural result of the inclined arrangement of the teeth. The bearings that hold the shaft carrying the helical gear must be capable of reacting against the thrust load.

The helix angle is specified for each given gear design. A balance should be sought to take advantage of the smoother engagement of the gear teeth when the helix angle is high while maintaining a reasonable value of the axial thrust load that increases with increasing helix angle. A typical range of values of helix angles is from 15° to 45° .

Primary Planes for Helical Gears

The three primary planes of a helical gear are illustrated in Figure 8–20(a) for a circular gear and 8–21(b) for a rack.

The *tangential plane* shown in Figure 8–20(b) is tangent to the pitch diameter (pitch circle) of the helical gear and is also called the pitch plane. The *transverse plane* shown in Figure 8–20(c) is perpendicular to the axis of the helical gear and the tangential plane. The transverse plane coincides with the plane of rotation. The *normal plane* shown in Figure 8–20(d) is perpendicular to the tooth and the tangential plane.

Pressure Angles for Helical Gears

In design of a helical gear, there are three angles of interest as illustrated in Figure 8–21(d): (1) the *helix angle*, ψ ; (2) the *normal pressure angle*, ϕ_n ; and (3) the *transverse pressure angle*, ϕ_t . Designers must specify the helix angle

and one of the two pressure angles. The other pressure angle can be computed from the following relationship:

$$\tan \phi_n = \tan \phi_t \cos \psi \quad (8-11)$$

For example, one manufacturer's catalog offers standard helical gears with a normal pressure angle of $14\frac{1}{2}^\circ$ and a 45° helix angle. Then the transverse pressure angle is found from

$$\tan \phi_n = \tan \phi_t \cos \psi$$

$$\tan \phi_t = \tan \phi_n / \cos \psi = \tan(14.5^\circ) / \cos(45^\circ) = 0.3657$$

$$\phi_t = \tan^{-1}(0.3657) = 20.09^\circ$$

Pitches for Helical Gears

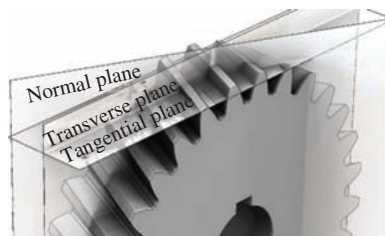
To obtain a clear picture of the geometry of helical gears, you must understand the following five different pitches. The helical rack shown in Figure 8–21 will be used to describe and explain the different pitches of a helical gear.

Transverse circular pitch, p_t . The transverse circular pitch is the distance from a point on one tooth to the corresponding point on the next adjacent tooth, measured at the pitch line in the transverse plane. The transverse circular pitch is the length of line AB shown on the transverse plane and the tangential plane in Figure 8–21(d). This is the same definition used for spur gears. Then

Transverse Circular Pitch

$$p_t = \pi D / N = \pi / P_d \quad (8-12)$$

Normal Circular Pitch, p_n . *Normal circular pitch* is the distance between corresponding points on adjacent teeth measured on the pitch surface in the normal direction. The normal circular pitch is the length of the line



(a) The three primary planes defining helical gears



(b) The tangential plane only



(c) The transverse plane only



(d) The normal plane only

FIGURE 8–20 Identities of the three primary planes for helical gears

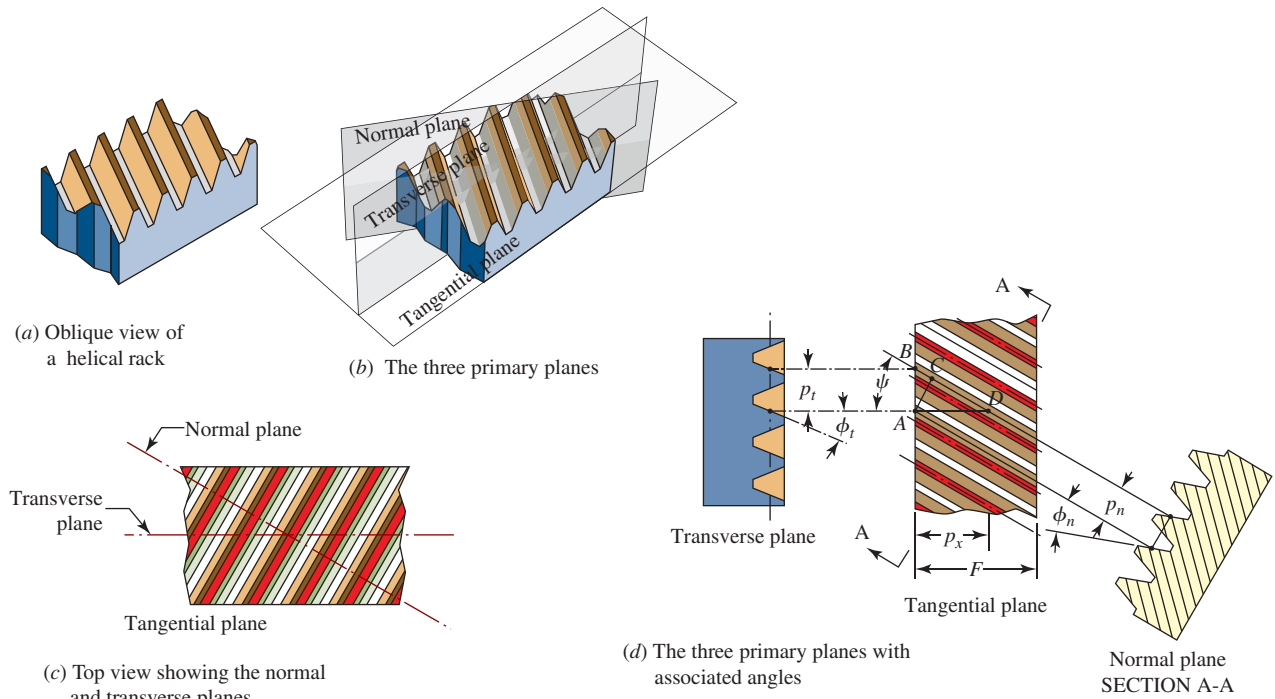


FIGURE 8-21 Identities of the three primary planes and associated angles shown on a helical rack

AC shown on the normal plane in Figure 8-21(d). The triangle ABC is a right triangle with the angle BAC equal to the helix angle. Pitches p_t and p_n are related by the following equation:

⇒ **Normal Circular Pitch**

$$p_n = p_t \cos \psi \quad (8-13)$$

Axial Pitch, p_x . Axial pitch is the distance AD between corresponding points on adjacent teeth, measured on the pitch surface in the axial direction.

⇒ **Axial Pitch**

$$p_x = p_t / \tan \psi = \pi (P_d / \tan \psi) = \pi m \tan \psi \quad (8-14)$$

It is necessary to have at least two axial pitches in the face width to have the benefit of full helical action and its smooth transfer of the load from tooth to tooth. A convenient method of checking this concept is to compute the *face contact ratio* defined as:

$$\text{Face Contact Ratio} = F/P_x$$

The face contact ratio is also called the *helical overlap*. Then we can check a given design to ensure that

$$F/P_x > 2.0$$

Diametral Pitch, P_d . Diametral pitch is the ratio of the number of teeth in the gear to the pitch diameter. This is the same definition as the one for spur gears; it applies in considerations of the form of the teeth in the diametral or transverse plane. Thus, this pitch is sometimes called the *transverse diametral pitch*:

⇒ **Diametral Pitch**

$$P_d = N/D \quad (8-15)$$

Normal Diametral Pitch, P_{nd} . Normal diametral pitch is the equivalent diametral pitch in the plane normal to the teeth:

⇒ **Normal Diametral Pitch**

$$P_{nd} = P_d / \cos \psi \quad (8-16)$$

It is helpful to remember these relationships:

$$P_d p = \pi \quad (8-17)$$

$$P_{nd} p_n = \pi \quad (8-18)$$

Metric Module, m . As stated for spur gears, the metric module is essentially the inverse of the diametral pitch with the value reported in mm.

⇒ **Metric Module**

$$m = D/N$$

This applies in the transverse plane of the gear.

Normal Metric Module, m_n . This is the inverse of the normal diametral pitch with the value reported in mm. It is the module in the plane normal to the gear tooth.

⇒ **Normal Metric Module**

$$m_n = 1/P_{nd} = 1/(P_d / \cos \psi) = \cos \psi / P_d$$

$$m_n = \cos \psi / (N/D) = D \cos \psi / N = m \cos \psi \quad (8-19)$$

The use of Equations (8-11) through (8-16) is now illustrated in the following example problem.

Example Problem 8-2

A helical gear has a transverse diametral pitch of 12, a transverse pressure angle of $14\frac{1}{2}^\circ$, 28 teeth, a face width of 1.25 in, and a helix angle of 30° . Compute the transverse circular pitch, normal circular pitch, normal diametral pitch, axial pitch, pitch diameter, and the normal pressure angle. Compute the number of axial pitches in the face width.

Solution *Transverse Circular Pitch*
Use Equation (8-12):

$$p_t = \pi/P_d = \pi/12 = 0.262 \text{ in}$$

Normal Circular Pitch
Use Equation (8-13):

$$p_n = p_t \cos \psi = (0.262) \cos(30) = 0.227 \text{ in}$$

Normal Diametral Pitch
Use Equation (8-16)

$$P_{nd} = P_d / \cos \psi = 12 / \cos(30) = 13.856$$

Axial Pitch
Use Equation (8-14):

$$p_x = p_t / \tan \psi = 0.262 / \tan(30) = 0.453 \text{ in}$$

Pitch Diameter
Use Equation (8-15):

$$D = N/P_d = 28/12 = 2.333 \text{ in}$$

Normal Pressure Angle
Use Equation (8-11):

$$\begin{aligned}\phi_n &= \tan^{-1}(\tan \phi_t \cos \psi) \\ \phi_n &= \tan^{-1}[\tan(14\frac{1}{2}) \cos(30)] = 12.62^\circ\end{aligned}$$

Number of Axial Pitches in the Face Width

$$F/p_x = 1.25/0.453 = 2.76 \text{ pitches}$$

Since this is greater than 2.0, there will be full helical action.

8-8 BEVEL GEAR GEOMETRY

Bevel gears are used to transfer motion between non-parallel shafts, usually at 90° to one another. The four primary styles of bevel gears are straight bevel, spiral bevel, zero spiral bevel, and hypoid. Figure 8-22 shows the general appearance of these four types of bevel gear sets. The surface on which bevel gear teeth are machined is inherently a part of a cone. The differences occur in the specific shape of the teeth and in the orientation of the pinion relative to the gear. (See References 3, 5, 14, and 16.)

Straight Bevel Gears

The teeth of a straight bevel gear are straight and lie along an element of the conical surface. See Figure 8-22 (a), (e) and (f). Lines along the face of the teeth through the pitch circle meet at the apex of the pitch cone. As shown in Figure 8-22(f), the centerlines of both the

pinion and the gear also meet at this apex. In the standard configuration, the teeth are tapered toward the center of the cone.

Key dimensions are specified either at the outer end of the teeth or at the mean, midface position. The relationships that control some of these dimensions are listed in Table 8-8 for the case when the shafts are at the 90° angle. The pitch cone angles for the pinion and the gear are determined by the ratio of the number of teeth, as shown in the table. Note that their sum is 90° . Also, for a pair of bevel gears having a ratio of unity, each has a pitch cone angle of 45° . Such gears, called *miter gears*, are used simply to change the direction of the shafts in a machine drive without affecting the speed of rotation.

You should understand that many more features need to be specified before the gears can be produced. Furthermore, many successful, commercially available gears are made in some nonstandard form. For example,

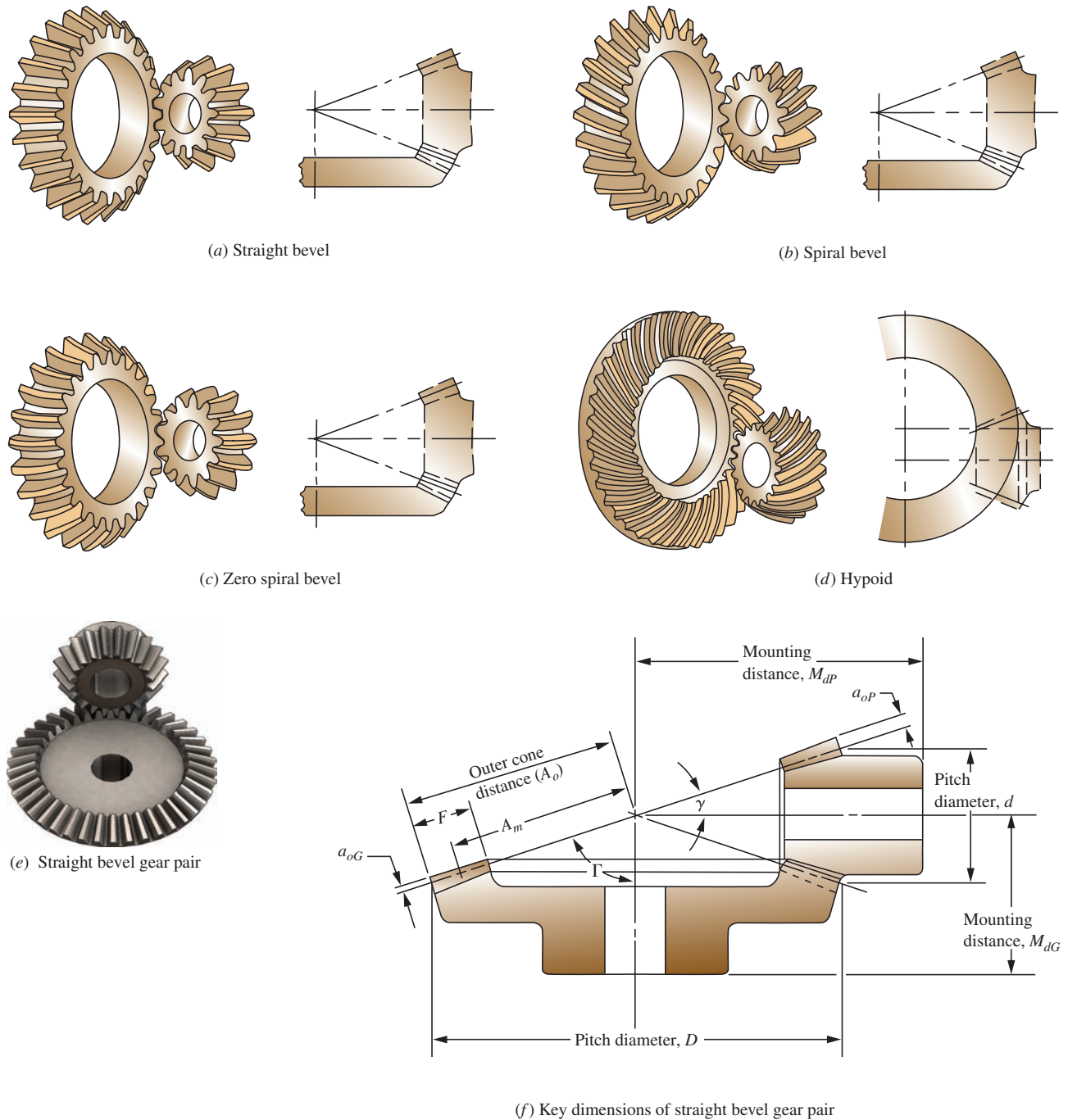


FIGURE 8-22 Types of bevel gears [Parts (a) through (d) extracted from ANSI/AGMA 2005-C96, *Design Manual for Bevel Gears*, with the permission of the publisher]

the addendum of the pinion is often made longer than that of the gear. Some manufacturers modify the slope of the root of the teeth to produce a uniform depth, rather than using the standard, tapered form. Reference 5 gives many more data.

The pressure angle, ϕ , is typically 20° , but 22.5° and 25° are often used to avoid interference. The minimum number of teeth for straight bevel gears is typically 12. More is said about the design of straight bevel gears in Chapter 10.

The mounting of bevel gears is critical if satisfactory performance is to be achieved. Most commercial gears have a defined mounting distance similar to that shown in Figure 8-22(f). It is the distance from some reference surface, typically the back of the hub of the gear, to the apex of the pitch cone. Because the pitch cones of the mating gears have coincident apexes, the mounting distance also locates the axis of the mating gear. If the gear is mounted at a distance smaller than the recommended mounting distance, the teeth will

TABLE 8–8 Geometrical Features of Straight Bevel Gears

Given Diametral pitch = $P_d = N_P/d = N_G/D$ or $m = d/N_P = D/N_G$

where N_P = number of teeth in pinion

N_G = number of teeth in gear

Dimension	Formula
Gear ratio	$m_G = N_G/N_P$
Pitch diameters:	
Pinion	$d = N_P/P_d$ or $d = mN_P$
Gear	$D = N_G/P_d$ or $D = mN_G$
Pitch cone angles:	
Pinion	$\gamma = \tan^{-1}(N_P/N_G)$ (lowercase Greek <i>gamma</i>)
Gear	$\Gamma = \tan^{-1}(N_G/N_P)$ (uppercase Greek <i>gamma</i>)
Outer cone distance	$A_o = 0.5D/\sin(\Gamma)$
Face width must be specified:	$F =$
Nominal face width	$F_{\text{nom}} = 0.30A_o$
Maximum face width	$F_{\text{max}} = A_o/3$ or $F_{\text{max}} = 10/P_d$ or $m/2.54$ (whichever is less)
Mean cone distance	$A_m = A_o - 0.5F$ (Note: A_m is defined for the gear, also called A_{mG} .)
Mean circular pitch	$p_m = (\pi/P_d)(A_m/A_o)$ or $\pi m(A_m/A_o)$
Mean working depth	$h = (2.00/P_d)(A_m/A_o)$ or $2.00 m(A_m/A_o)$
Clearance	$c = 0.125h$
Mean whole depth	$h_m = h + c$
Mean addendum factor	$c_1 = 0.210 + 0.290/(m_G)^2$
Gear mean addendum	$a_G = c_1h$
Pinion mean addendum	$a_P = h - a_G$
Gear mean dedendum	$b_G = h_m - a_G$
Pinion mean dedendum	$b_P = h_m - a_P$
Gear dedendum angle	$\delta_G = \tan^{-1}(b_G/A_{mG})$
Pinion dedendum angle	$\delta_P = \tan^{-1}(b_P/A_{mG})$
Gear outer addendum	$a_{oG} = a_G + 0.5F \tan \delta_P$
Pinion outer addendum	$a_{oP} = a_P + 0.5F \tan \delta_G$
Gear outside diameter	$D_o = D + 2a_{oG} \cos \Gamma$
Pinion outside diameter	$d_o = d + 2a_{oP} \cos \gamma$

likely bind. If it is mounted at a greater distance, there will be excessive backlash, causing noisy and rough operation.

Spiral Bevel Gears

The teeth of a spiral bevel gear are curved and sloped with respect to the surface of the pitch cone. See Figure 8–22(b). Spiral angles, ψ , of 20° to 45° are used, with 35° being

typical. Contact starts at one end of the teeth and moves along the tooth to its end. For a given tooth form and number of teeth, more teeth are in contact for spiral bevel gears than for straight bevel gears. The gradual transfer of loads and the greater average number of teeth in contact make spiral bevel gears smoother and allow smaller designs than for typical straight bevel gears. Recall that similar advantages were described for a helical gear relative to a spur gear.

The pressure angle, ϕ , is typically 20° for spiral bevel gears, and the minimum number of teeth is typically 12 to avoid interference. But nonstandard spiral gears allow as few as five teeth in the pinion of high-ratio sets if the tips of the teeth are trimmed to avoid interference. The rather high average number of teeth in contact (high contact ratio) for spiral gears makes this approach acceptable and can result in a very compact design. Reference 5 gives the relationships for computing the geometric features of spiral bevel gears that are extensions of those given in Table 8–8.

Zero Spiral Bevel Gears

The teeth of a zero spiral bevel gear are curved somewhat as in a spiral bevel gear, but the spiral angle is zero. See Figure 8–22(c). These gears can be used in the same mountings as straight bevel gears, but they operate more smoothly. They are sometimes called ZEROL[®] bevel gears.

Hypoid Gears

The major difference between hypoid gears and the others just described is that the centerline of the pinion for a set of hypoid gears is offset either above or below the centerline of the gear. See Figure 8–22(d). The teeth are designed specially for each combination of offset distance and spiral angle of the teeth. A major advantage is the more compact design that results, particularly when applied to vehicle drive trains and machine tools. (See References 5, 14, and 16 for more data.)

The hypoid gear geometry is the most general form of bevel gearing, and the others are special cases. The hypoid gear has an offset axis for the pinion, and its curved teeth are cut at a spiral angle. Then the spiral bevel gear is a hypoid gear with a zero offset distance. A ZEROL[®] bevel gear is a hypoid gear with a zero offset and a zero spiral angle. A straight bevel gear is a hypoid gear with a zero offset, a zero spiral angle, and straight teeth.

Example Problem 8–3

Compute the values for the geometrical features listed in Table 8–8 for a pair of straight bevel gears having a diametral pitch of 8, a 20° pressure angle, 16 teeth in the pinion, and 48 teeth in the gear. Specify a suitable face width. The shafts are at 90° .

Solution

Given $P_d = 8$; $N_p = 16$; $N_G = 48$.

Computed Values *Gear Ratio*

$$m_G = N_G/N_p = 48/16 = 3.000$$

Pitch Diameter
For the pinion,

$$d = N_p/P_d = 16/8 = 2.000 \text{ in}$$

For the gear,

$$D = N_G/P_d = 48/8 = 6.000 \text{ in}$$

Pitch Cone Angles
For the pinion,

$$\gamma = \tan^{-1}(N_p/N_G) = \tan^{-1}(16/48) = 18.43^\circ$$

For the gear,

$$\Gamma = \tan^{-1}(N_G/N_p) = \tan^{-1}(48/16) = 71.57^\circ$$

Outer Cone Distance

$$A_o = 0.5 D/\sin(\Gamma) = 0.5(6.00 \text{ in})/\sin(71.57^\circ) = 3.162 \text{ in}$$

Face Width

The face width must be specified based on the following guidelines:

Nominal face width:

$$F_{\text{nom}} = 0.30A_o = 0.30(3.162 \text{ in}) = 0.949 \text{ in}$$

Maximum face width:

$$F_{\text{max}} = A_o/3 = (3.162 \text{ in})/3 = 1.054 \text{ in}$$

or

$$F_{\text{max}} = 10/P_d = 10/8 = 1.25 \text{ in}$$

Therefore the face width should be in the range from 0.949 in to 1.054 in. Let's specify $F = 1.000$ in.

Mean Cone Distance

$$A_m = A_{mG} = A_o - 0.5F = 3.162 \text{ in} - 0.5(1.00 \text{ in}) = 2.662 \text{ in}$$

Ratio $A_m/A_o = 2.662/3.162 = 0.842$ (This ratio occurs in several following calculations.)

Mean Circular Pitch

$$p_m = (\pi/P_d)(A_m/A_o) = (\pi/8)(0.842) = 0.331 \text{ in}$$

Mean Working Depth

$$h = (2.00/P_d)(A_m/A_o) = (2.00/8)(0.842) = 0.210 \text{ in}$$

Clearance

$$c = 0.125h = 0.125(0.210 \text{ in}) = 0.026 \text{ in}$$

Mean Whole Depth

$$h_m = h + c = 0.210 \text{ in} + 0.026 \text{ in} = 0.236 \text{ in}$$

Mean Addendum Factor

$$c_1 = 0.210 + 0.290/(m_G)^2 = 0.210 + 0.290/(3.00)^2 = 0.242$$

Gear Mean Addendum

$$a_G = c_1 h = (0.242)(0.210 \text{ in}) = 0.051 \text{ in}$$

Pinion Mean Addendum

$$a_P = h - a_G = 0.210 \text{ in} - 0.051 \text{ in} = 0.159 \text{ in}$$

Gear Mean Dedendum

$$b_G = h_m - a_G = 0.236 \text{ in} - 0.051 \text{ in} = 0.185 \text{ in}$$

Pinion Mean Dedendum

$$b_P = h_m - a_P = 0.236 \text{ in} - 0.159 \text{ in} = 0.077 \text{ in}$$

Gear Dedendum Angle

$$\delta_G = \tan^{-1}(b_G/A_{mG}) = \tan^{-1}(0.185/2.662) = 3.975^\circ$$

Pinion Dedendum Angle

$$\delta_P = \tan^{-1}(b_P/A_{mG}) = \tan^{-1}(0.077/2.662) = 1.657^\circ$$

Gear Outer Addendum

$$\begin{aligned} a_{oG} &= a_G + 0.5F \tan \delta_P \\ a_{oG} &= (0.051 \text{ in}) + (0.5)(1.00 \text{ in}) \tan(1.657^\circ) = 0.0655 \text{ in} \end{aligned}$$

Pinion Outer Addendum

$$\begin{aligned} a_{oP} &= a_P + 0.5F \tan \delta_G \\ a_{oP} &= (0.159 \text{ in}) + (0.5)(1.00 \text{ in}) \tan(3.975^\circ) = 0.1937 \text{ in} \end{aligned}$$

Gear Outside Diameter

$$\begin{aligned} D_o &= D + 2a_{oG} \cos \Gamma \\ D_o &= 6.000 \text{ in} + 2(0.0655 \text{ in}) \cos(71.57^\circ) = 6.041 \text{ in} \end{aligned}$$

Pinion Outside Diameter

$$\begin{aligned} d_o &= d + 2a_{oP} \cos \gamma \\ d_o &= 2.000 \text{ in} + 2(0.1937 \text{ in}) \cos(18.43^\circ) = 2.368 \text{ in} \end{aligned}$$

8-9 TYPES OF WORMGEARING

Wormgearing is used to transmit motion and power between nonintersecting shafts, usually at 90° to each other. The drive consists of a worm on the high-speed shaft which has the general appearance of a power screw

thread: a cylindrical, helical thread. The worm drives a wormgear, which has an appearance similar to that of a helical gear. Figures 8-23 shows a typical worm and wormgear set. Sometimes the wormgear is referred to as a *worm wheel* or simply a *wheel* or *gear*. (See Reference 6.)

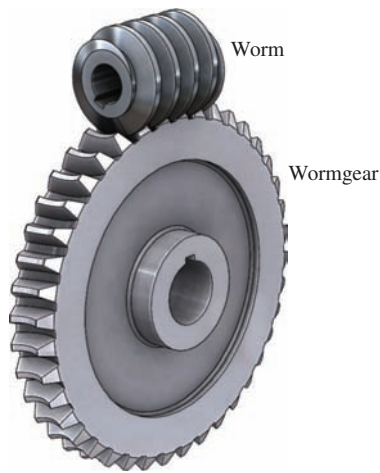


FIGURE 8–23 Worm and wormgear with a single-threaded worm

Worms and wormgears can be provided with either right hand or left hand threads on the worm and correspondingly designed teeth on the wormgear affecting the rotational direction of the wormgear.

Several variations of the geometry of wormgear drives are available. The most common one, shown in Figures 8–23 and 8–24, employs a cylindrical worm mating with a wormgear having teeth that are throated, wrapping partially around the worm. This is called a

single-enveloping type of wormgear drive. The contact between the threads of the worm and wormgear teeth is along a line, and the power transmission capacity is quite good. Many manufacturers offer this type of wormgear set as a stocked item. Installation of the worm is relatively easy because axial alignment is not very critical. However, the wormgear must be carefully aligned radially in order to achieve the benefit of the enveloping action. Figure 8–25 shows a cutaway of a commercial wormgear reducer.

A simpler form of wormgear drive allows a special cylindrical worm to be used with a standard spur gear or helical gear. Neither the worm nor the gear must be aligned with great accuracy, and the center distance is not critical. However, the contact between the worm threads and the wormgear teeth is theoretically a point, drastically reducing the power transmission capacity of the set. Thus, this type is used mostly for nonprecision positioning applications at low speeds and low power levels.

A third type of wormgear set is the *double-enveloping type* in which the worm is made in an hourglass shape and mates with an enveloping type of wormgear. This results in area contact rather than line or point contact and allows a much smaller system to transmit a given power at a given reduction ratio. However, the worm is more difficult to manufacture, and the alignment of both the worm and the wormgear is very critical.

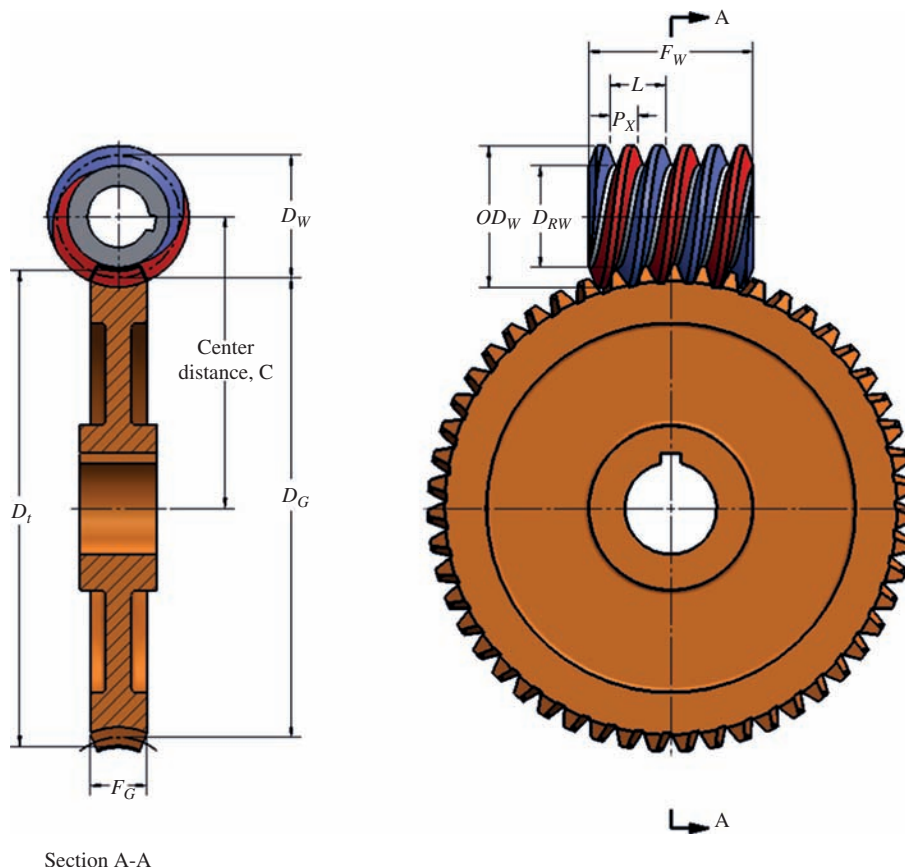


FIGURE 8–24 Single-enveloping wormgear set with a double-threaded worm

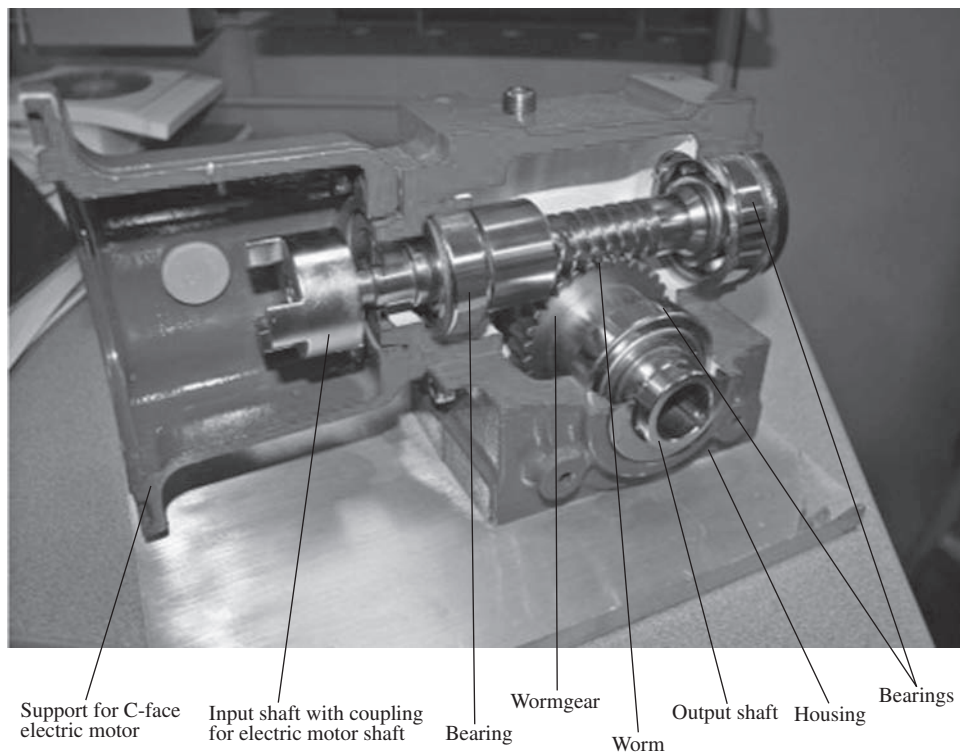


FIGURE 8-25 Cutaway view of a wormgear reducer
(Baldor/Dodge, Greenville, SC)

8-10 GEOMETRY OF WORMS AND WORMGEARS

Pitches, p , P_d , and m

A basic requirement of the worm and wormgear set is that the *axial pitch* of the worm must be equal to the *circular pitch* of the wormgear in order for them to mesh. Figure 8-24 shows the basic geometric features of a single-enveloping worm and wormgear set. *Axial pitch*, P_x , is defined as the distance from a point on the worm thread to the corresponding point on the next adjacent thread, measured axially on the pitch cylinder. As before, the circular pitch is defined for the wormgear as the distance from a point on a tooth on the pitch circle of the gear to the corresponding point on the next adjacent tooth, measured along the pitch circle. Thus, the circular pitch is an arc distance that can be calculated from

◇ Circular Pitch

$$p = \pi D_G / N_G \quad (8-20)$$

where D_G = pitch diameter of the gear

N_G = number of teeth in the gear

Some wormgears are made according to the circular pitch convention. But, as noted with spur gears, commercially available wormgear sets are usually made to a diametral pitch convention with the following pitches readily available: 48, 32, 24, 16, 12, 10, 8, 6, 5, 4, and

3. See Internet sites 2, 3, 10, and 11. The diametral pitch is defined for the gear as

◇ Diametral Pitch

$$P_d = N_G / D_G \quad (8-21)$$

The conversion from diametral pitch to circular pitch can be made from the following equation:

$$P_d p = \pi \quad (8-22)$$

Metric module worms and wormgears are available commercially with modules from 0.50 to 6.00. See Internet sites 9 and 10. As for other types of gears, the module is defined for the wormgear as

$$m = D / N$$

Then the pitch of the wormgear is $p = \pi m$ and this is also the axial pitch of the worm.

Number of Worm Threads, N_W

Worms can have a single thread, as in a typical screw, or multiple threads, usually 2 or 4, but sometimes 3, 5, 6, 8, or more. It is common to refer to the number of threads as N_W and then to treat that number as if it were the number of teeth in the worm. The number of threads in the worm is frequently referred to as the number of *starts*; this is convenient because if you look at the end of a worm, you can count the number of threads that start at the end and wind down the cylindrical worm.

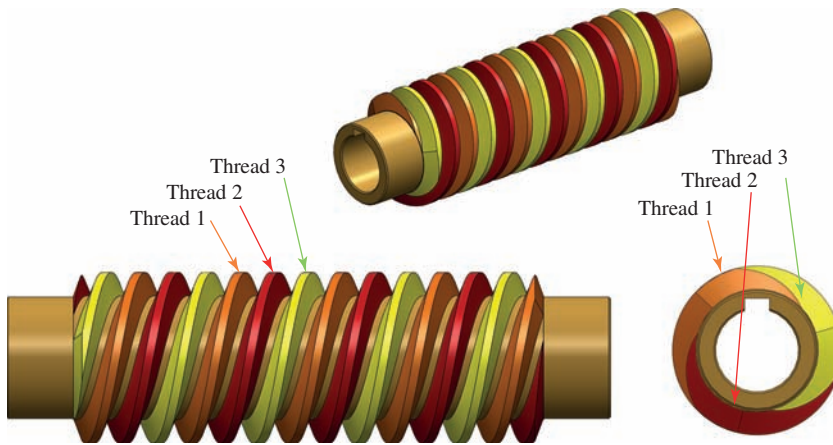


FIGURE 8-26 Worm with triple thread

Figure 8-26 shows a worm with a triple thread, with each thread shown as a different color.

Lead, L

The *lead* of a worm is the axial distance that a point on the worm would move as the worm is rotated one revolution. Lead is related to the axial pitch, P_x , by

◇ **Lead**

$$L = N_W P_x \quad (8-23)$$

See Figure 8-24 that shows the axial pitch and lead on a double-threaded worm.

Lead Angle, λ

The *lead angle* is the angle between the tangent to the worm thread and the line perpendicular to the axis of the worm. To visualize the method of calculating the lead angle, refer to Figure 8-27, which shows a simple triangle that would be formed if one thread of the worm were unwrapped from the pitch cylinder and laid flat

on the paper. The length of the hypotenuse is the length of the thread itself. The horizontal side is the lead, L . The vertical side is the circumference of the pitch cylinder, πD_W , where D_W is the pitch diameter of the worm. Then

◇ **Lead Angle**

$$\tan \lambda = L / \pi D_W \quad (8-24)$$

Pitch Line Speed, v_t

As before, the pitch line speed is the linear velocity of a point on the pitch line for the worm or the wormgear. For the worm having a pitch diameter D_W in, rotating at n_W rpm,

◇ **Pitch Line Speed for worm**

$$v_{tW} = \frac{\pi D_W n_W}{12} \text{ ft/min} \quad \text{or} \quad v_{tW} = \frac{\pi D_W n_W}{60000} \text{ m/s}$$

For the wormgear having a pitch diameter D_G in, rotating at n_G rpm,

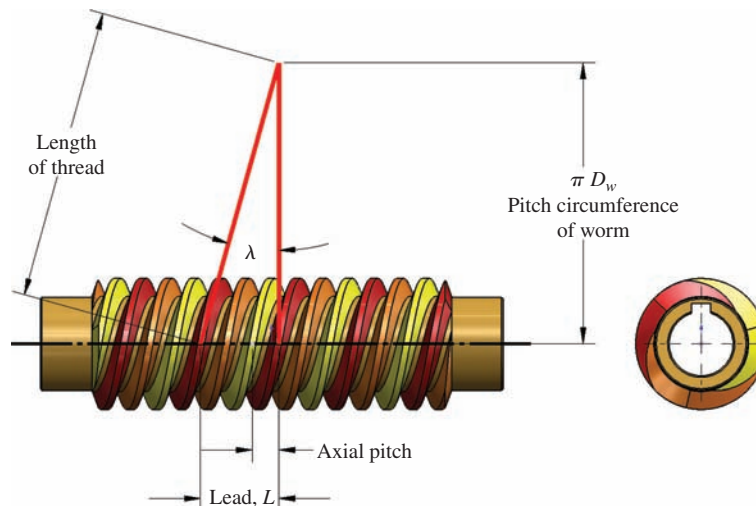


FIGURE 8-27 Lead angle

⇒ Pitch Line Speed for Gear

$$v_{tG} = \frac{\pi D_G n_G}{12} \text{ ft/min} \quad \text{or} \quad v_{tG} = \frac{\pi D_G n_G}{60\,000} \text{ m/s}$$

Note that these two values for pitch line speed are *not* equal.

Velocity Ratio, VR

It is most convenient to calculate the velocity ratio of a worm and wormgear set from the ratio of the input rotational speed to the output rotational speed:

⇒ Velocity Ratio for Worm/Wormgear Set

$$VR = \frac{\text{speed of worm}}{\text{speed of gear}} = \frac{n_W}{n_G} = \frac{N_G}{N_W} \quad (8-25)$$

Typical commercially available wormgear drives have ratios of 5, 7.5, 10, 12.5, 15, 20, 25, 30, 35, 40, 45, 50, 60, and 70. Drives may have a single worm/wormgear pair or two or more pairs in series. Of course, specially designed drives can have a virtually unlimited array of ratios within limits of size and practicality.

Example Problem 8-4

A wormgear has 52 teeth and a diametral pitch of 6. It mates with a triple-threaded worm that rotates at 1750 rpm. The pitch diameter of the worm is 2.000 in. Compute the circular pitch, the axial pitch, the lead, the lead angle, the pitch diameter of the wormgear, the center distance, the velocity ratio, and the rotational speed of the wormgear. See Figure 8-28.

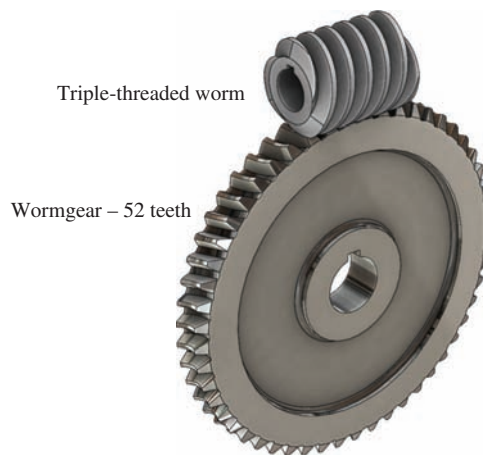


FIGURE 8-28 Worm and wormgear set for Example Problem 8-4

Solution Circular Pitch

$$p = \pi/P_d = \pi/6 = 0.5236 \text{ in}$$

Axial Pitch

$$P_x = p = 0.5236 \text{ in}$$

Lead

$$L = N_W P_x = (3)(0.5236) = 1.5708 \text{ in}$$

Lead Angle

$$\lambda = \tan^{-1}(L/\pi D_W) = \tan^{-1}(1.5708/\pi 2.000)$$

$$\lambda = 14.04^\circ$$

Gear Pitch Diameter

$$D_G = N_G/P_d = 52/6 = 8.667 \text{ in}$$

Center Distance

$$C = (D_W + D_G)/2 = (2.000 + 8.667)/2 = 5.333 \text{ in}$$

Velocity Ratio

$$VR = N_G/N_W = 52/3 = 17.333$$

Gear rpm

$$n_G = n_W/VR = 1750/17.333 = 101 \text{ rpm}$$

Pressure Angle

Most commercially available wormgears are made with pressure angles of $14\frac{1}{2}^\circ$, 20° , 25° , or 30° . The low pressure angles are used with worms having a low lead angle and/or a low diametral pitch. For example, a $14\frac{1}{2}^\circ$ pressure angle may be used for lead angles up to about 17° . For higher lead angles and with higher diametral pitches (smaller teeth), the 20° or 25° pressure angle is used to eliminate interference without excessive undercutting. The 20° pressure angle is the preferred value for lead angles up to 30° . From 30° to 45° of lead angle, the 25° pressure angle is recommended. Either the normal pressure angle, ϕ_n , or the transverse pressure angle, ϕ_t , may be specified. These are related by

Pressure Angle

$$\tan \phi_n = \tan \phi_t \cos \lambda \quad (8-26)$$

Self-Locking Wormgear Sets

Self-locking is the condition in which the worm drives the wormgear, but if torque is applied to the gear shaft,

the worm does not turn. It is locked! This means the wormgear cannot back-drive the worm. The locking action is produced by the friction force between the worm threads and the wormgear teeth, and this is highly dependent on the lead angle. It is recommended that a lead angle no higher than about 5.0° be used in order to ensure that self-locking will occur. This lead angle usually requires the use of a single-threaded worm. Note that the triple-threaded worm in Example Problem 8-4 has a lead angle of 14.04° . It is *not* likely to be self-locking.

Although a worm gear set may be self-locking, they are not meant to be used as a holding device. If vibration and shock are present in the system, the worm gear set could back-drive. A brake should be designed in the system if a wormgear set is required not to backdrive, especially if safety is a concern.

Typical Designs of Wormgear Sets

Considerable latitude is permissible in the design of wormgear sets because the worm and wormgear combination is designed as a unit. However, there are some guidelines as shown next.

GENERAL GUIDELINES FOR WORM AND WORMGEAR DIMENSIONS ▼

Typical Tooth Dimensions

Table 8-9 shows typical values used for the dimensions of worm threads and gear teeth.

TABLE 8-9 Typical Tooth Dimensions for Worms and Wormgears

Dimension	Formula
Addendum	$a = 0.3183P_x = 1/P_d$
Whole depth	$h_t = 0.6866P_x = 2.157/P_d$
Working depth	$h_k = 2a = 0.6366P_x = 2/P_d$
Dedendum	$b = h_t - a = 0.3683P_x = 1.157/P_d$
Root diameter of worm	$D_{rW} = D_W - 2b$
Outside diameter of worm	$D_{oW} = D_W + 2a = D_W + h_k$
Root diameter of gear	$D_{rG} = D_G - 2b$
Throat diameter of gear	$D_t = D_G + 2a$

Worm Diameter

The diameter of the worm affects the lead angle, which in turn affects the efficiency of the set. For this reason, small diameters are desirable. But for practical reasons and proper

proportion with respect to the wormgear, it is recommended that the worm diameter be approximately $C^{0.875}/2.2$, where C is the center distance in inches between the worm and the wormgear. Variation of about 30% is allowed. (See Reference 6.) Thus, the worm diameter should fall in the range

Wormgear Drive Center Distance-in

$$1.6 < \frac{C^{0.875}}{D_W} < 3.0 \quad (8-27)$$

For metric designs with dimensions in mm,

Wormgear Drive Center Distance, mm

$$1.07 < \frac{C^{0.875}}{D_W} < 2.0 \quad (8-27M)$$

The recommended nominal worm diameter is approximately $C^{0.875}/1.54$.

But some commercially available wormgear sets fall outside this range, especially in the smaller sizes. Also, those worms designed to have a through-hole bored in them for installation on a shaft are typically larger than you would find from Equation (8-27). Proper proportion and efficient use of material should be the guide. The worm shaft must also be checked for deflection under operating loads. For worms machined integral with the shaft, the root of the worm threads determines the minimum shaft diameter. For worms having bored holes, sometimes called *shell worms*, care must be exercised to leave sufficient material between the thread root and the keyway in the bore. Figure 8-29 shows the recommended thickness above the keyway to be a minimum of one-half the whole depth of the threads.

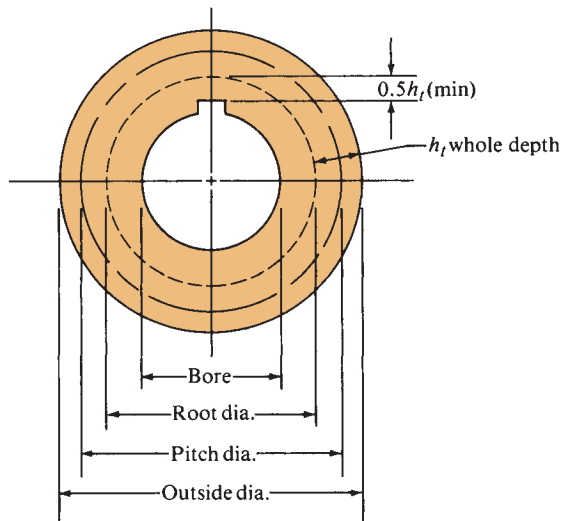


FIGURE 8-29 Shell worm

Wormgear Dimensions

We are concerned here with the single-enveloping type of wormgear, as shown in Figures 8-23 and 8-30. Its addendum, dedendum, and depth dimensions are assumed to be the same as those listed in Table 8-9, measured at the throat of the wormgear teeth. The throat is in line with the vertical centerline of the worm. The recommended face width for the wormgear is

Face Width of Wormgear

$$F_G = (D_{oW}^2 - D_W^2)^{1/2} \quad (8-28)$$

This corresponds to the length of the line tangent to the pitch circle of the worm and limited by the outside diameter of the worm. Any face width beyond this value would not be effective in resisting stress or wear, but a convenient value slightly greater than the minimum should be used. The outer edges of the wormgear teeth should be chamfered approximately as shown in Figure 8-30.

Another recommendation, which is convenient for initial design, is that the face width of the gear should be approximately 2.0 times the circular pitch. Because we are working in the diametral pitch system, we will use

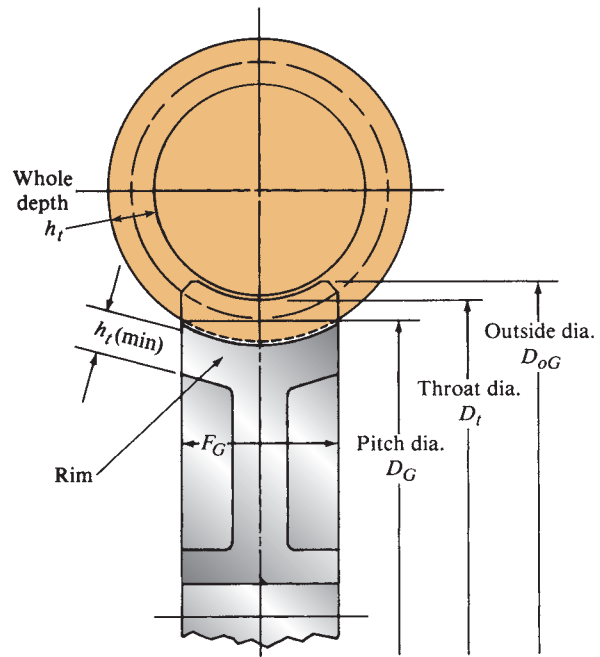


FIGURE 8-30 Wormgear details

$$F_G = 2p = 2\pi/P_d \quad (8-29)$$

However, since this is only approximate and 2π is approximately 6, we will use

$$F_G = 6/P_d \quad (8-30)$$

If the gear web is thinned, a rim thickness at least equal to the whole depth of the teeth should be left.

Face Length of the Worm

For maximum load sharing, the worm face length should extend to at least the point where the outside diameter of the worm intersects the throat diameter of the wormgear. This length is

Face Length of Worm

$$F_W = 2[(D_t/2)^2 - (D_G/2 - a)^2]^{1/2} \quad (8-31)$$

Example Problem 8-5

A worm and wormgear set is to be designed to produce a velocity ratio of 40. It has been proposed that the diametral pitch of the wormgear be 8, based on the torque that must be transmitted. (This will be discussed in Chapter 10.) Using the relationships presented in this section, specify the following:

- Worm diameter, D_W
- Number of threads in the worm, N_W
- Number of teeth in the gear, N_G
- Actual center distance, C
- Face width of the gear, F_G
- Face length of the worm, F_W
- Minimum thickness of the rim of the gear

Solution Many design decisions need to be made, and multiple solutions could satisfy the requirements. Presented here is one solution, along with comparisons with the various guidelines discussed in this section. This

type of analysis precedes the stress analysis and the determination of the power-transmitting capacity of the worm and wormgear drive which is discussed in Chapter 10.

Trial Design: Let's specify a double-threaded worm: $N_W = 2$. Then there must be 80 teeth in the wormgear to achieve a velocity ratio of 40. That is,

$$VR = N_G/N_W = 80/2 = 40$$

With the known diametral pitch, $P_d = 8$, the pitch diameter of the wormgear is

$$D_G = N_G/P_d = 80/8 = 10.000 \text{ in}$$

An initial estimate for the magnitude of the center distance is approximately $C = 6.50$ in. We know that it will be greater than 5.00 in, the radius of the wormgear. Using Equation (8-27), the recommended minimum size of the worm is

$$D_W = C^{0.875}/3.0 = 1.71 \text{ in}$$

Similarly, the maximum diameter should be

$$D_W = C^{0.875}/1.6 = 3.21 \text{ in}$$

A small worm diameter is desirable. Let's specify $D_W = 2.25$ in. The actual center distance is

$$C = (D_W + D_G)/2 = 6.125 \text{ in}$$

Worm Outside Diameter

$$D_{oW} = D_W + 2a = 2.25 + 2(1/P_d) = 2.25 + 2(1/8) = 2.50 \text{ in}$$

Whole Depth

$$h_t = 2.157/P_d = 2.157/8 = 0.270 \text{ in}$$

Face Width for Gear

Let's use Equation (8-28):

$$F_G = (D_{oW}^2 - D_W^2)^{1/2} = (2.50^2 - 2.25^2)^{1/2} = 1.090 \text{ in}$$

Let's specify $F_G = 1.25$ in.

Addendum

$$a = 1/P_d = 1/8 = 0.125 \text{ in}$$

Throat Diameter of Wormgear

$$D_t = D_G + 2a = 10.000 + 2(0.125) = 10.250 \text{ in}$$

Recommended Minimum Face Length of Worm

$$F_W = 2[(D_t/2)^2 - (D_G/2 - a)^2]^{1/2} = 3.16 \text{ in}$$

Let's specify $F_W = 3.25$ in.

Minimum Thickness of the Rim of the Gear

The rim thickness should be greater than the whole depth:

$$h_t > 0.270 \text{ in}$$

8-11 GEAR MANUFACTURE

The discussion of gear manufacture will begin with the method of producing the gear blank. Small gears are frequently made from wrought plate or bar, with the hub, web, spokes, and rim machined to final or near-final dimensions before the gear teeth are produced. The

face width and the outside diameter of the gear teeth are also produced at this stage. Other gear blanks may be forged, sand cast, or die cast to achieve the basic form prior to machining. A few gears in which only moderate precision is required may be die cast with the teeth in virtually final form.

Large gears are frequently fabricated from components. The rim and the portion into which the teeth are machined may be rolled into a ring shape from a flat bar and then welded. The web or spokes and the hub are then welded inside the ring. Very large gears may be made in segments with the final assembly of the segments by welding or by mechanical fasteners.

The popular methods of machining the gear teeth are form milling, shaping, and hobbing. (See References 11, 12 and 19.)

In *form milling* [Figure 8–31(a)], a milling cutter that has the shape of the tooth space is used, and each space is cut completely before the gear blank is indexed to the position of the next adjacent space. This method is used mostly for large gears, and great care is required to achieve accurate results.

Shaping [Figures 8–31(b) and 8–32] is a process in which the cutter reciprocates, usually on a vertical spindle. The shaping cutter rotates as it reciprocates and is fed into the gear blank. Thus, the involute-tooth form is

generated gradually. This process is frequently used for internal gears.

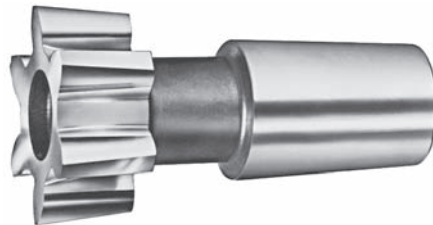
Scudding is a process similar to shaping but it is much faster, and closer to hobbing in production rate. The workpiece and the scudding cutter rotate synchronously and the cutter feeds directly through the workpiece. Both internal and external gears can be produced by scudding. See Internet site 12.

Hobbing [Figures 8–31(c) and (d) and 8–33] is a process similar to milling except that the workpiece (the gear blank) and the cutter (the hob) rotate in a coordinated fashion. Here also, the tooth form is generated gradually as the hob is fed into the blank.

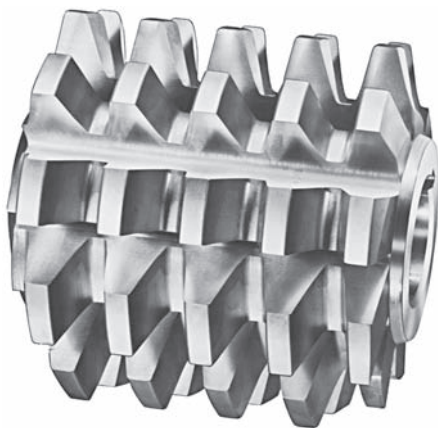
The gear teeth are finished to greater precision after form milling, shaping, or hobbing by the processes of grinding, shaving, and honing. Being products of secondary processes, they are expensive and should be used only where the operation requires high accuracy in the tooth form and spacing. Figure 8–34 shows a gear grinding machine.



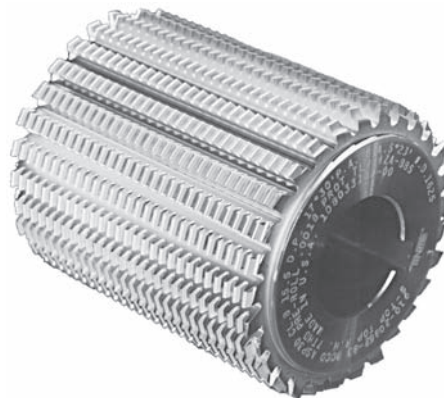
(a) Form milling cutter



(b) Spur gear shaper cutter

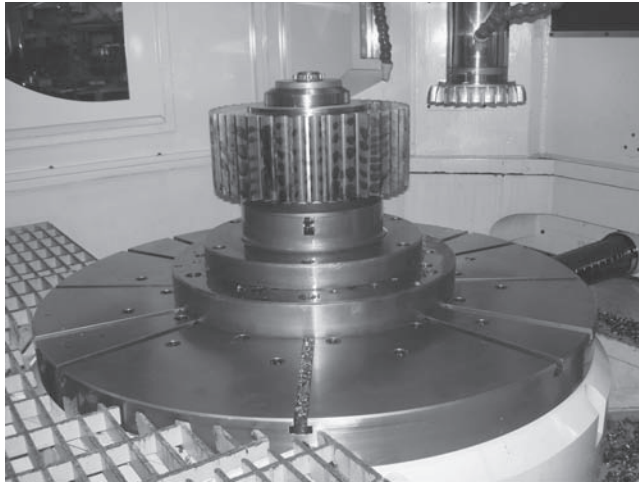


(c) Hob for small pitch gears having large teeth



(d) Hob for high pitch gears having small teeth

FIGURE 8–31 A variety of gear cutting tools (Courtesy of Gleason Cutting Tools Corporation)

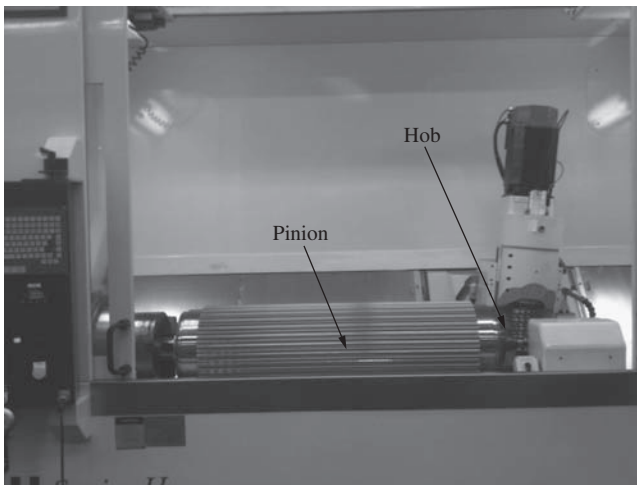


(a) Gear being shaped with shaping cutter



(b) Gear shaping machine

FIGURE 8-32 Gear shaping operation and shaping machine (Supplied by Bourn & Koch Inc)

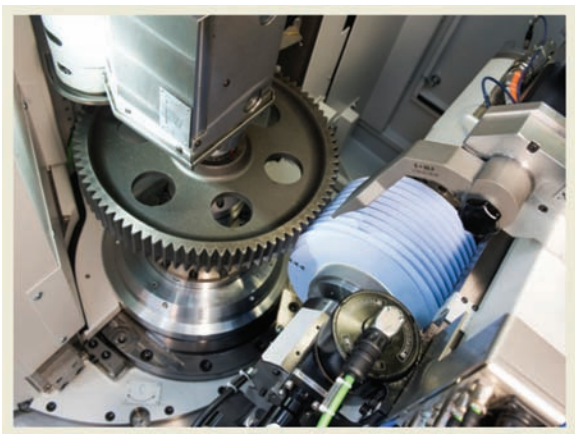


(a) Long pinion and a hob in a hobbing machine



(b) Large hobbing machine

FIGURE 8-33 Gear hobbing operation and hobbing machine (Supplied by Bourn & Koch Inc)



(a) Gear with grinding wheel



(b) Gear grinding machine

FIGURE 8-34 Gear grinding operation and grinding machine (Supplied by Bourn & Koch Inc)

8-12 GEAR QUALITY

Ensuring proper dimensional accuracy of power-transmission gears is essential to their suitability for use in machinery. Stresses in gears, smoothness of operation, life, and noise are affected by the degree of precision obtained in manufacturing. References 2, 8, and 12 provide comprehensive treatments of gear quality. Here, we give an overview of the fundamentals and equipment used to measure gear quality.

Quality in gearing is indicated by either of two methods: (1) the composite variation of a product test gear rotating in mesh with a precise master gear, called *functional measurement* or (2) the precision of specific features of a single gear, called *analytical measurement*. These two methods are described next.

Functional Measurement

Functional measurement employs a device called a *double flank roll tester* to measure the *total radial composite deviation* and the *tooth-to-tooth radial composite*

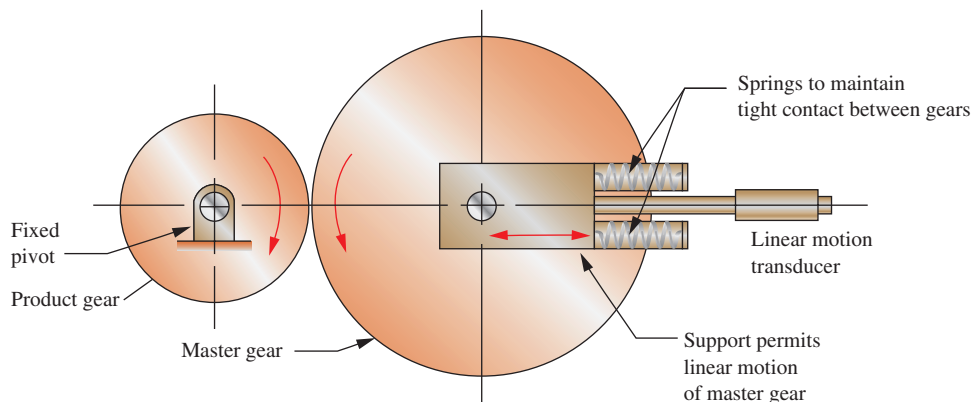
variation. Figure 8-35 shows a photograph (a) of one commercially available roll tester and a schematic diagram of its construction (b). The primary features of the physical testing device are:

1. Fixed mounting of the axis for the product gear to be tested
2. Mounting for an accurate master gear on a linearly movable slide with means for maintaining full double flank contact between the two gears
3. Measurement device for indicating the movement of the axis for the master gear as the product gear rotates one complete revolution
4. Recorder to display the total excursion of the master gear during the cycle

Figure 8-36 shows an example of a test report for a gear generated by the double-flank roll tester shown in part (a). Shown are the *Total composite error*, the *Tooth to tooth error*, and the *Runout*. These data are sensitive to the amount of radial force exerted on the test gear by the master gear and that force is indicated on the test report; 13 oz in this example.



(a) Double flank roll testing device (PECO - Process Equipment Company)



(b) Schematic diagram of double flank roll tester

FIGURE 8-35 Radial composite deviation testing of gears “Penta Gear Metrology, Dayton, Ohio”

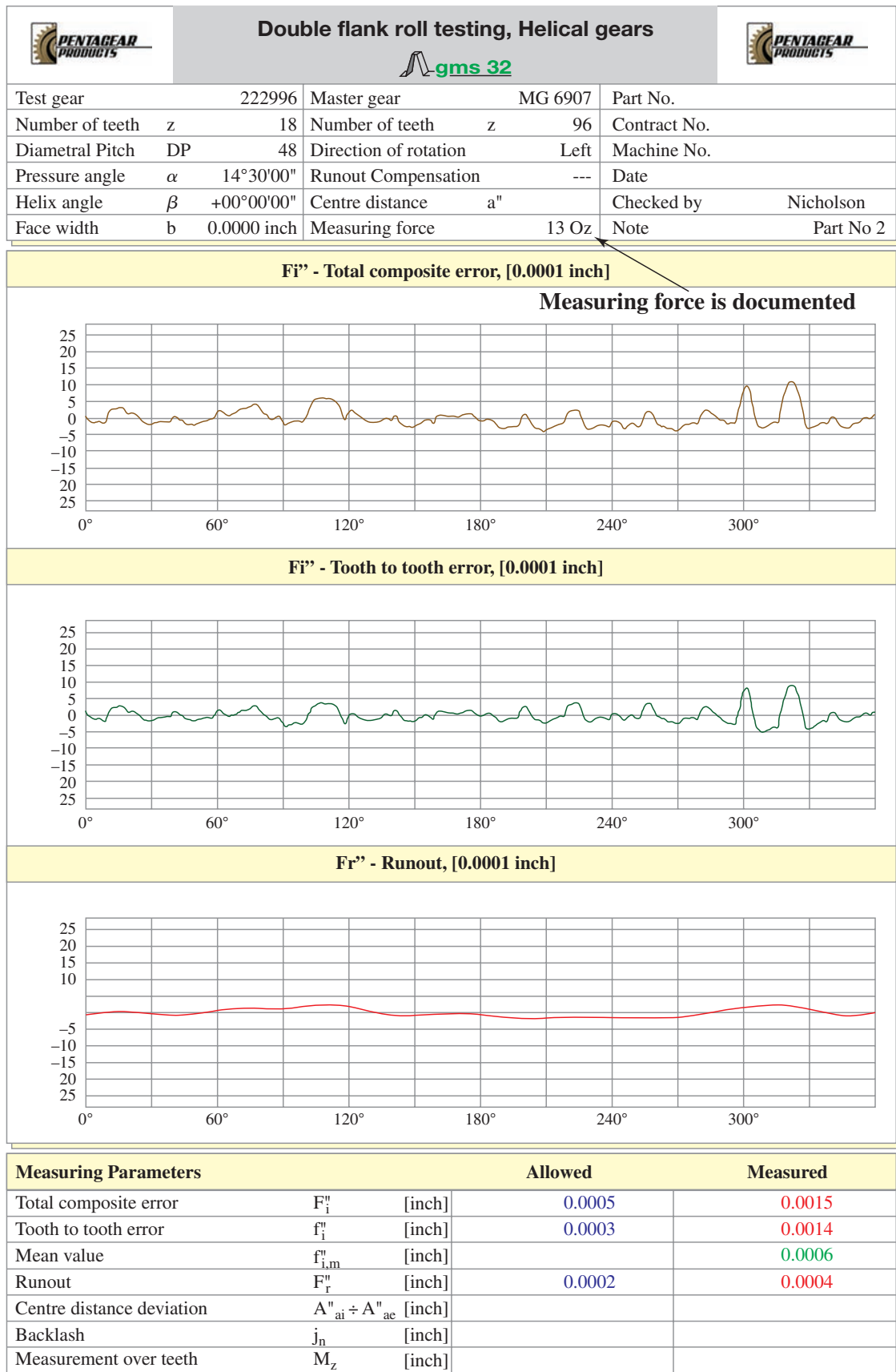


FIGURE 8-36 Example of a report from a double flank roll test of a gear.

Analytical Measurement

Analytical measurement measures numerous features of individual teeth using a highly sensitive probe moving over the surface of the tooth under the control of a specially designed coordinate measurement machine (CMM). The computer in the complete system guides the probe over specified trajectories while accurately measuring its position. Figure 8–37(a) shows a commercially available system. Figure 8–37(b) shows a test gear in position for making the measurements using sensitive probes that can be seen at the right of the gear. The probe is guided over a prescribed path, making a complete survey of the geometry of the tooth features described next.

Figure 8–38 shows an example report giving extensive detailed information about the test gear provided by an analytical measurement system. On the left side are graphs for the *Profile* and *Helix* variations on both the right and left sides for selected teeth on the gear.

On the right side, data are reported on *Index Variation*, *Pitch Variation*, *Pitch Line Runout*, and *Tooth Thickness*.

Profile: The measurement of the actual profile of the surface of a gear tooth from the point of the start of the active profile to the tip of the tooth. The theoretical profile is a true involute curve. Variations of the actual profile from the theoretical profile cause variations in the instantaneous velocity ratio between the two gears in

mesh, affecting the smoothness of the motion and noise.

Helix: The deviation of the actual line on the gear tooth surface at the pitch circle from the theoretical line. Measurements are made across the face from one end to the other. For a spur gear, the theoretical line is straight. For a helical gear, it is a part of a helix. Helix measurement is sometimes called the *tooth alignment* measurement. It is important because excessive misalignment causes nonuniform loading on the gear teeth.

Index variation: The difference between the actual location of a point on the face of a gear tooth at the pitch circle and the corresponding point of a reference tooth measured on the pitch circle. The variation causes inaccuracy in the action of mating gear teeth.

Pitch variation: The measurement of the variation of the pitch between gear teeth through a complete revolution.

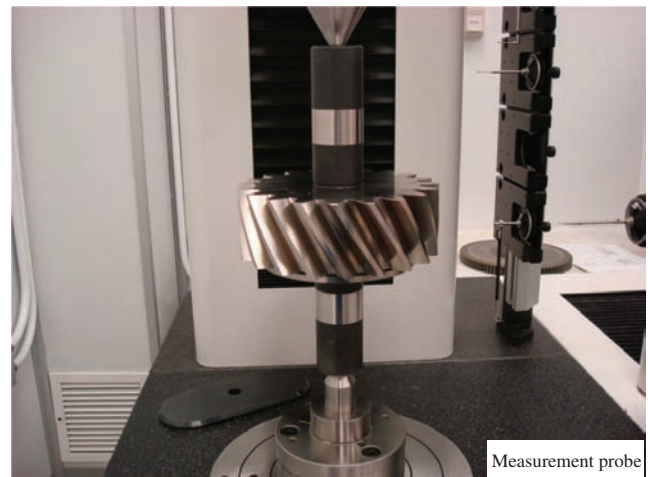
Pitch line runout: A measure of the eccentricity and out of roundness of a gear. Excessive runout causes the contact point on mating gear teeth to move radially during each revolution.

Tooth thickness: A measure of the thickness of each tooth on the gear.

As the analytical measurement system completes its tests, comparisons are automatically made with the theoretical tooth forms and with tolerance values from the



(a) Analytical gear measurement system



(b) Gear installed in an analytical gear measurement system. Note probe at right.

FIGURE 8–37 Analytical measurement system for gear quality “Penta Gear Metrology, Dayton, Ohio” (PECO- Process Equipment Company)

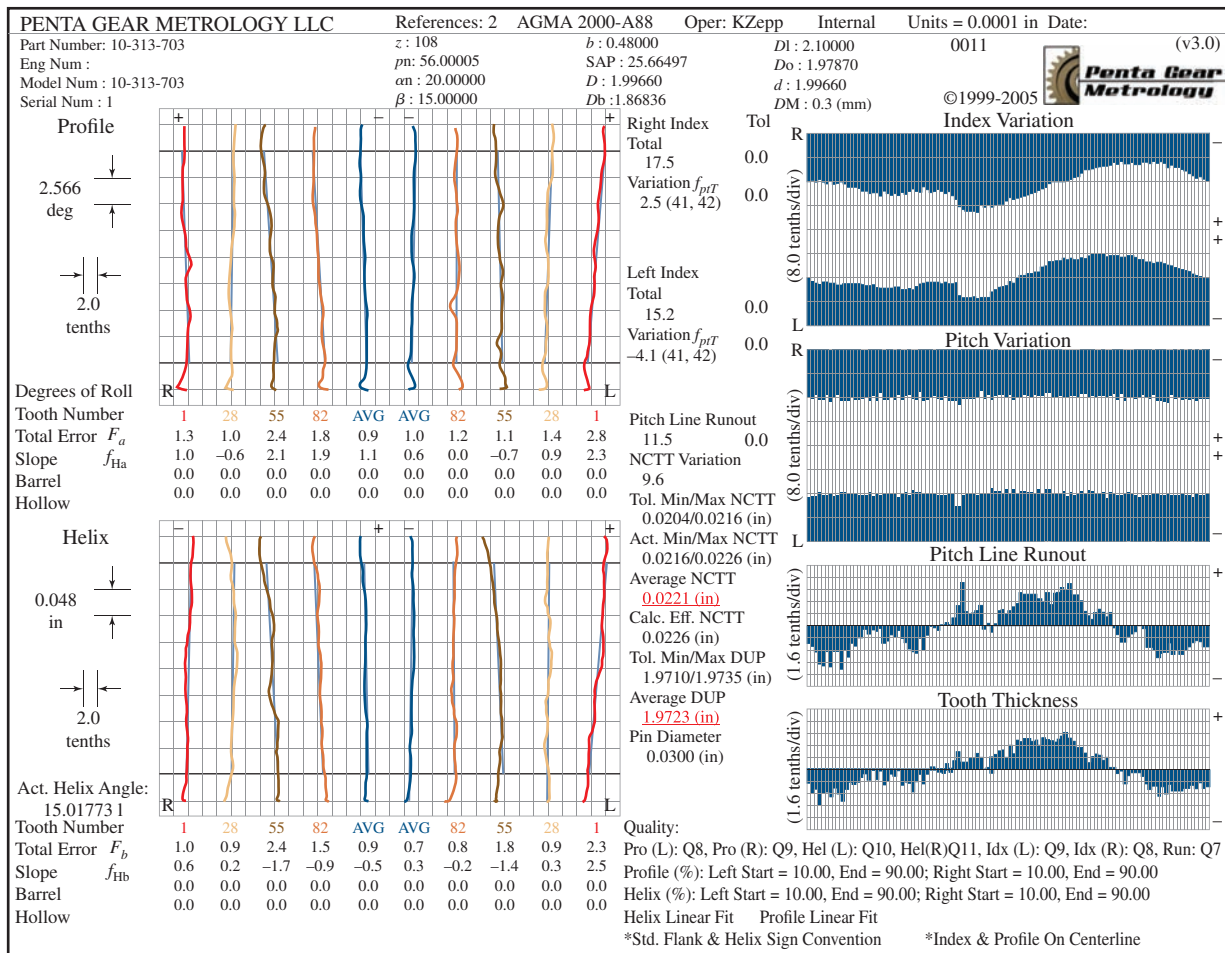


FIGURE 8-38 Example test data report for an analytical measurement of a helical gear “Penta Gear Metrology, Dayton, Ohio”

applicable standard (described in Chapter 9.) to report the resulting *quality number*. The data are also useful to manufacturing staff for making adjustments to gear cutting and grinding machine control settings to improve the accuracy of the total process.

Using the general capabilities of the analytical measurement system, dimensions of features other than those of the gear teeth may also be determined while the gear is in its fixture. For example, when a gear is machined onto a shaft, diameters, keyseats, shoulder fillets, and other geometric features may be checked for dimensions, perpendicularity, parallelism, and concentricity. Gear segments, composite gears having two or more gears on the same shaft, splines, and cam surfaces can also be measured.

8-13 VELOCITY RATIO AND GEAR TRAINS

A gear train is one or more pairs of gears operating together to transmit power.

Normally there is a speed change from one gear to the next due to the different sizes of the gears in mesh. The fundamental building block of the total speed change

ratio in a gear train is the *velocity ratio* between two gears in a single pair.

Velocity Ratio

The velocity ratio (VR) is defined as the ratio of the rotational speed of the input gear to that of the output gear for a single pair of gears.

To develop the equation for computing the velocity ratio, it is helpful to view the action of two gears in mesh, as shown in Figure 8-39. The action is equivalent to the action of two smooth wheels rolling on each other without slipping, with the diameters of the two wheels equal to the pitch diameters of the two gears. Remember that when two gears are in mesh, their pitch circles are tangent, obviously, the gear teeth prohibit any slipping.

As shown in Figure 8-39, without slipping there is no relative motion between the two pitch circles at the pitch point, and therefore the tangential linear velocity of a point on either pitch circle is the same. We will use the symbol v_t for this velocity. The linear velocity of a point that is in rotation at a distance R from its center